PROBABILITY & STATISTICS · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

54 question sets • 1435 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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Represent data with dot plots

MAIN	Probability & Statistics main question	**HSS-ID.A.1**
BACKWARD 1	Count how often each value occurs	**6.SP.B.5.a**
BACKWARD 2	Locate values on a number-line scale	**6.NS.C.6.c**
BACKWARD 3	Read counts from a dot plot	**6.SP.B.4**
FORWARD 1	Find the median and mean from a dot plot	**HSS-ID.A.2**
FORWARD 2	Choose the display that keeps the information you need	**HSS-ID.A.1**

SET 1 • HSS-ID.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

The numbers of books 15 students read over the summer are 2, 4, 3, 2, 5, 3, 2, 6, 3, 4, 2, 3, 1, 3, 4. A dot plot shows one dot for each student above the number of books. How many dots are above 2, above 3, and above 4?

Dots above 2: _____

Dots above 3: _____

Dots above 4: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

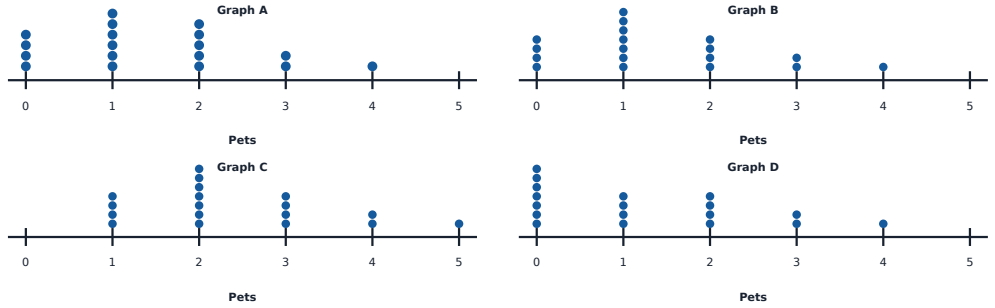
4; 5; 3

GRADING NOTE Form required: all three. Counts: 1 → 1, 2 → 4, 3 → 5, 4 → 3, 5 → 1, 6 → 1 (15 dots in all).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q1/5

The numbers of pets in 18 households are 0, 1, 1, 2, 0, 3, 1, 2, 1, 0, 4, 1, 2, 1, 0, 2, 1, 3. Which dot plot shows the data?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Counts: 0 → 4, 1 → 7, 2 → 4, 3 → 2, 4 → 1. The others move one dot from 1 to 2, shift every value up 1, or swap the counts of 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q2/5

Fourteen quiz scores (out of 10) are 7, 9, 8, 10, 7, 8, 8, 6, 9, 8, 7, 10, 8, 9. In a dot plot of the scores, which score has the tallest stack of dots, and how many dots are in that stack?

Score: _____

Dots in that stack: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

8; 5

GRADING NOTE Form required: both. Counts: 6 → 1, 7 → 3, 8 → 5, 9 → 3, 10 → 2.

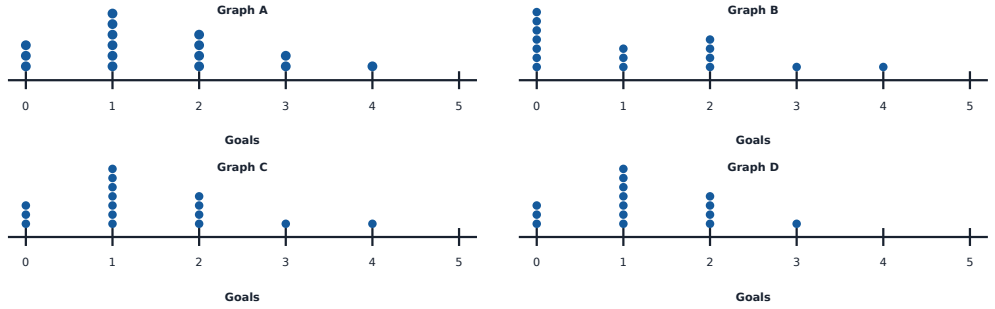
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

A team scored these numbers of goals in 16 games: 1, 0, 2, 1, 3, 1, 0, 2, 2, 1, 4, 1, 0, 1, 2, 1. Which dot plot shows the data?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE

Counts: 0 → 3, 1 → 7, 2 → 4, 3 → 1, 4 → 1 (16 dots). The others move a dot from 1 to 3, swap the counts of 0 and 1, or leave out the 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q4/5

Twenty students reported their numbers of siblings: 1, 2, 0, 1, 3, 2, 1, 1, 0, 2, 5, 1, 2, 0, 1, 2, 3, 1, 1, 2. In a dot plot of the data, how many dots are at values of 2 or more, and how many dots are there in all?

Dots at 2 or more: _____

Dots in all: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

9; 20

GRADING NOTE Form required: both. Counts: 0 → 3, 1 → 8, 2 → 6, 3 → 2, 5 → 1; 6 + 2 + 1 = 9 at 2 or more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSS-ID.A.1

BACKWARD 1

Count how often each value occurs • Grade 6 • Statistics & Probability • 6.SP.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

In the list 3, 5, 3, 4, 3, 5, how many times does 3 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 3 appears in positions 1, 3 and 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q1/5

In the list 1, 2, 2, 4, 2, 1, 2, how many times does 2 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Four of the seven values are 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q2/5

True or false?

The list 6, 6, 7, 8, 6 has three values equal to 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 6 appears three times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q3/5

How many values are in the list 2, 4, 4, 5, 2, 4, 4?

A. 3

B. 5

C. 4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE Count every value, repeats included. 4 is how often the value 4 appears; 3 and 5 are other miscounts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the list 0, 1, 0, 2, 1, 0, how many values are 0 or 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Three 0s and two 1s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSS-ID.A.1

BACKWARD 2

Locate values on a number-line scale • Grade 6 • The Number System • 6.NS.C.6.c

A number line has marks labeled 0, 1, 2, 3, 4, 5, 6.
Where does a dot for the value 4 go?

A. at the right end of the number line

B. above the fourth mark from the left, labeled 3

C. above the mark labeled 4

D. above the mark labeled 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. above the mark labeled 4

GRADING NOTE Place the dot at the label, not by counting marks; the fourth mark from the left is 3 because the first is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q1/5

True or false?

On a number line with marks at 0, 2, 4, 6, 8, the value 5 is halfway between the marks 4 and 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 is 1 more than 4 and 1 less than 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q2/5

A number line has marks at 10, 15, 20, 25, 30. What value is halfway between the marks 15 and 20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Short answer

CORRECT ANSWER

17.5

GRADING NOTE (15 + 20)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q3/5

A number line runs from 0 to 10 with a mark at every whole number. How many marks are there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE 0, 1, ..., 10 is eleven marks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q4/5

True or false?

On a horizontal number line, a larger value is placed to the left of a smaller value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Values increase to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

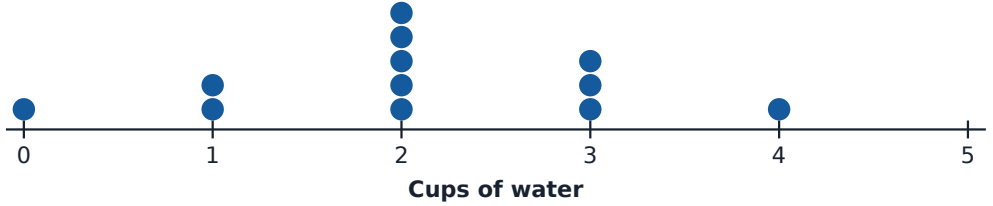
Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSS-ID.A.1

BACKWARD 3

Read counts from a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank 3 cups?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q1/5

26

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

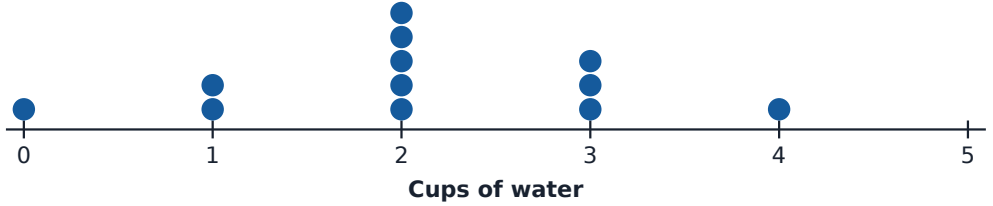
3

GRADING NOTE The stack above 3 has 3 dots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q1/5

The dot plot shows the cups of water some workers drank before noon. How many workers are shown in all?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q2/5

27

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12

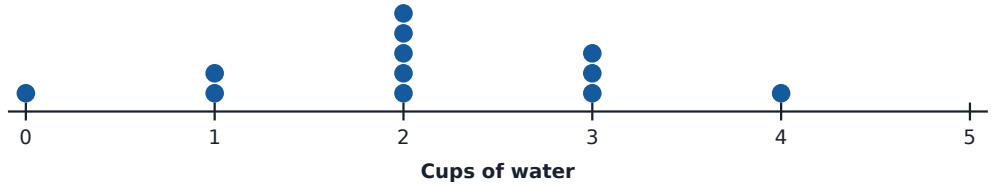
GRADING NOTE 1 + 2 + 5 + 3 + 1 = 12 dots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q2/5

True or false?

The dot plot shows the cups of water 12 workers drank before noon. No worker drank 5 cups.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q3/5

28

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

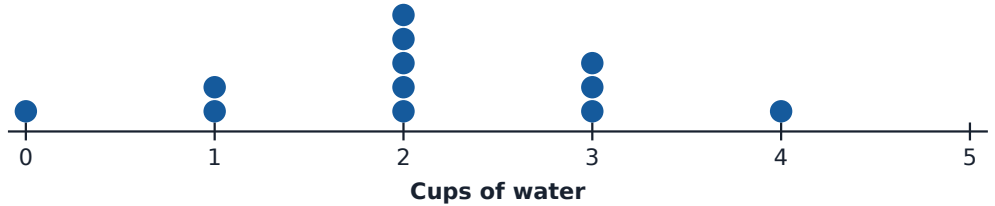
A. True

GRADING NOTE There are no dots above 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q3/5

The dot plot shows the cups of water 12 workers drank before noon. Which number of cups occurs most often?



A. 3 **B.** 12 **C.** 2 **D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q4/5

29

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

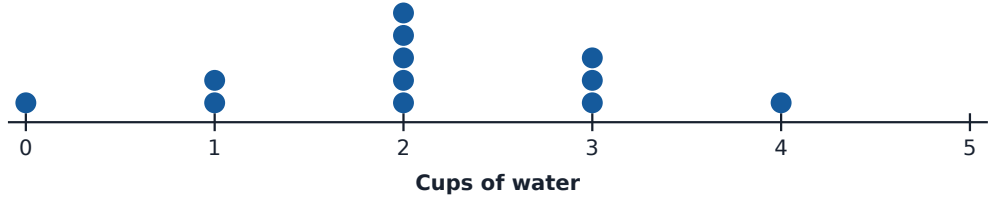
C. 2

GRADING NOTE The tallest stack (5 dots) is above 2. 12 is the number of workers; 5 is the height of the stack.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q4/5

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank fewer than 2 cups?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q5/5

30

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 1 at 0 and 2 at 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSS-ID.A.1

FORWARD 1

Find the median and mean from a dot plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The dot plot shows rainy days in 10 two-week periods.
Find the median number of rainy days.



Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q1/5

32

QUESTION ID
S1-F1-Q1

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE Ordered: 3, 3, 4, 4, 4, 4, 5, 5, 6, 7; the middle two values are 4 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q1/5

The dot plot shows rainy days in 10 two-week periods. Find the mean number of rainy days. Give a decimal.



Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q2/5

33

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5

GRADING NOTE Sum 45 over 10 values = 4.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q2/5

True or false?

A dot plot shows the values 3, 3, 4, 4, 4, 4, 5, 5, 6, 7. Adding one more dot at 20 changes the median more than it changes the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q3/5

34

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median stays 4; the mean moves from 4.5 to 65/11 ≈ 5.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q3/5

The dot plot shows the cups of water 12 workers drank before noon. Which statement is correct?



A. The mean is 12 cups, because 12 workers are shown in the plot.

B. The median is 2.5 cups, halfway between the smallest and largest values.

C. The median is 5 cups, the height of the tallest stack in the plot.

D. The median is 2 cups and the mean is 2.08 cups (to the nearest hundredth).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q4/5

35

QUESTION ID

S1-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

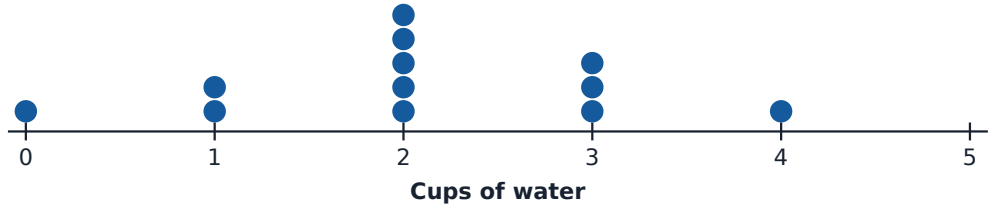
D. The median is 2 cups and the mean is 2.08 cups (to the nearest hundredth).

GRADING NOTE Ordered values: 0, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 4; median (2 + 2)/2 = 2; mean 25/12 ≈ 2.08. The others use the midrange, a stack height, or the number of workers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q4/5

The dot plot shows the cups of water 12 workers drank before noon. Find the total number of cups drunk.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q5/5

36

QUESTION ID
S1-F1-Q5

TYPE
Short answer

CORRECT ANSWER
25

GRADING NOTE $0\cdot1 + 1\cdot2 + 2\cdot5 + 3\cdot3 + 4\cdot1 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSS-ID.A.1

FORWARD 2

Choose the display that keeps the information you need • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Which display always shows every individual value of a small data set?

A. a box plot

B. a histogram with bins of width 10

C. a dot plot

D. a box plot with outliers marked

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q1/5

38

QUESTION ID

S1-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. a dot plot

GRADING NOTE A dot plot keeps every value; box plots keep five summary values; a histogram keeps only counts per bin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q1/5

From a box plot alone, which of these can always be found?

A. the median

B. the number of values

C. the most common value

D. the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q2/5

39

QUESTION ID

S1-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. the median

GRADING NOTE The box plot shows the five-number summary only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q2/5

True or false?

A histogram with bins 0-10, 10-20, 20-30 shows how many values are exactly 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q3/5

40

QUESTION ID

S1-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows only how many values are in the bin 10-20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q3/5

A school records the number of people in each of 400 students' households (whole numbers from 1 to 9). Which display shows how many households have each size and stays readable?

A. a histogram with one bin for each whole number

B. a list of all 400 household sizes

C. a box plot of the household sizes

D. a dot plot with one dot per household

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q4/5

41

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. a histogram with one bin for each whole number

GRADING NOTE A dot plot of 400 values has stacks too tall to read; a box plot loses the counts; a list is not a display.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

Two different data sets can have the same box plot but different dot plots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q5/5

42

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1, 2, 3, 4, 5, 6, 7 and 1, 2, 3.5, 4, 4.5, 6, 7 both have the five-number summary 1, 2, 4, 6, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q5/5

Represent data with histograms

MAIN	Probability & Statistics main question	**HSS-ID.A.1**
BACKWARD 1	Assign values to bins with an endpoint rule	**6.SP.B.4**
BACKWARD 2	Count the values in an interval	**6.SP.B.5.a**
BACKWARD 3	Read a histogram	**6.SP.B.4**
FORWARD 1	Regroup a histogram with wider bins	**HSS-ID.A.1**
FORWARD 2	Locate the median in a histogram	**HSS-ID.A.2**

SET 2 • HSS-ID.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Commute times (minutes) of 20 employees are 12, 25, 8, 31, 17, 22, 15, 40, 19, 27, 10, 33, 21, 14, 26, 18, 35, 23, 29, 16. A histogram uses the bins 0-10, 10-20, 20-30, 30-40 and 40-50. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 10-20, 30-40 and 40-50?

10-20: _____

30-40: _____

40-50: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q1/5

45

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

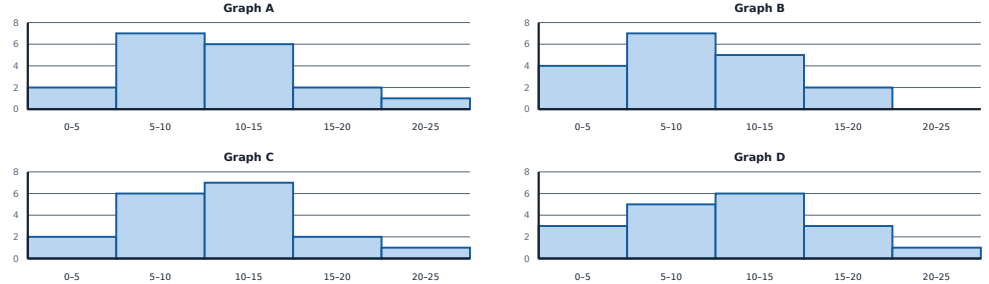
8; 3; 1

GRADING NOTE Form required: all three. Counts 0-10: 1, 10-20: 8 (10 is included), 20-30: 7, 30-40: 3, 40-50: 1 (40 goes in 40-50).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q1/5

Waiting times (minutes) of 18 patients are 4, 11, 7, 15, 9, 5, 12, 20, 10, 6, 14, 8, 17, 10, 3, 13, 5, 9. Which histogram shows the data with bins 0-5, 5-10, 10-15, 15-20, 20-25? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows minutes and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q2/5

46

QUESTION ID
S2-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Graph A

GRADING NOTE Counts 0-5: 2, 5-10: 7, 10-15: 6, 15-20: 2, 20-25: 1 (5, 10, 15 and 20 go in the bin they start). The others put each endpoint value in the lower bin (4, 7, 5, 2, 0), move one value from 5-10 into 10-15 (2, 6, 7, 2, 1), or miscount three bins (3, 5, 6, 3, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q2/5

Standing long-jump distances (cm) of 16 students are 152, 168, 145, 175, 160, 158, 171, 149, 163, 180, 155, 166, 150, 172, 161, 157. A histogram uses the bins 140-150, 150-160, 160-170, 170-180 and 180-190. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 150-160 and 170-180?

150-160: _____

170-180: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q3/5

47

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

5; 3

GRADING NOTE Form required: both. Counts 140-150: 2, 150-160: 5, 160-170: 5, 170-180: 3, 180-190: 1.

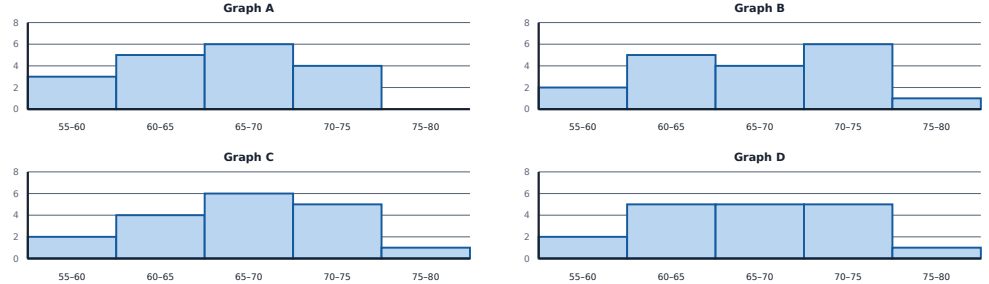
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Daily high temperatures (°F) over 18 days are 61, 58, 70, 66, 72, 59, 64, 75, 68, 63, 71, 60, 67, 65, 74, 69, 62, 70. Which histogram shows the data with bins 55-60, 60-65, 65-70, 70-75, 75-80? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows °F and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q4/5

48

QUESTION ID

S2-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Counts 55-60: 2, 60-65: 5, 65-70: 5, 70-75: 5, 75-80: 1. The others put endpoints such as 60, 65, 70 and 75 in the lower bin (3, 5, 6, 4, 0), move one value from 65-70 into 70-75 (2, 5, 4, 6, 1), or move one value from 60-65 into 65-70 (2, 4, 6, 5, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q4/5

Page counts of 14 novels are 120, 255, 188, 310, 96, 205, 150, 275, 230, 182, 140, 200, 260, 175. A histogram uses bins of width 50: 50-100, 100-150, ..., 300-350. Each bin includes its left endpoint and excludes its right endpoint. Which bin has the most values (give its left endpoint), and how many values does it have?

Left endpoint of the tallest bin: _____

Values in that bin: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q5/5

49

QUESTION ID

S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

150; 4

GRADING NOTE Form required: both. Counts 50-100: 1, 100-150: 2, 150-200: 4, 200-250: 3, 250-300: 3, 300-350: 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSS-ID.A.1

BACKWARD 1

Assign values to bins with an endpoint rule • Grade 6 • Statistics & Probability • 6.SP.B.4

Bins 10-20 and 20-30 each include their left endpoint and exclude their right endpoint. Which bin holds 20?

A. 20-30

B. 10-20

C. neither bin

D. both bins

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q1/5

51

QUESTION ID

S2-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 20-30

GRADING NOTE 20 is the left endpoint of 20-30 and the excluded right end of 10-20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q1/5

True or false?

With bins that include the left endpoint and exclude the right endpoint, the value 30 goes in the bin 20-30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q2/5

52

QUESTION ID

S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 30 is excluded from 20-30; it goes in 30-40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q2/5

Bins 0-5, 5-10 and 10-15 include their left endpoint and exclude their right endpoint. Which bin holds 9.9? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q3/5

53

QUESTION ID

S2-B1-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE $5 \le 9.9 < 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q3/5

How many of the values 4, 5, 9, 10, 12 fall in the bin 5-10, which includes 5 and excludes 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q4/5

54

QUESTION ID

S2-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 5 and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q4/5

True or false?

If every bin included both of its endpoints, a value equal to an endpoint shared by two bins would be counted twice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q5/5

55

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is why a rule must give each endpoint to exactly one bin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSS-ID.A.1

BACKWARD 2

Count the values in an interval • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many of the values 12, 18, 25, 31, 19, 22 are at least 15 and less than 25?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q1/5

57

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 18, 19, 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q1/5

How many of the values 3.5, 4, 6.2, 7, 8 are at least 4 and less than 7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q2/5

58

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 4 and 6.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q2/5

True or false?

Exactly two of the values 40, 45, 50, 55 are at least 45 and less than 55.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q3/5

59

QUESTION ID

S2-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 45 and 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q3/5

How many of the values 2, 9, 11, 14, 20, 21 are at least 10 and less than 20?

A. 3

B. 4

C. 1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q4/5

60

QUESTION ID

S2-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2

GRADING NOTE 11 and 14; 20 is not less than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q4/5

How many of the values 100, 150, 199, 200, 250 are at least 100 and less than 200?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q5/5

61

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 100, 150, 199.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

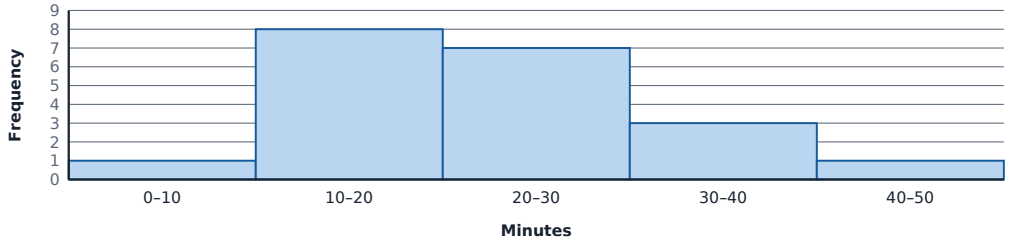
Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSS-ID.A.1

BACKWARD 3

Read a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

The histogram shows the commute times of some employees. How many employees have commutes in the bin 20-30?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q1/5

63

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

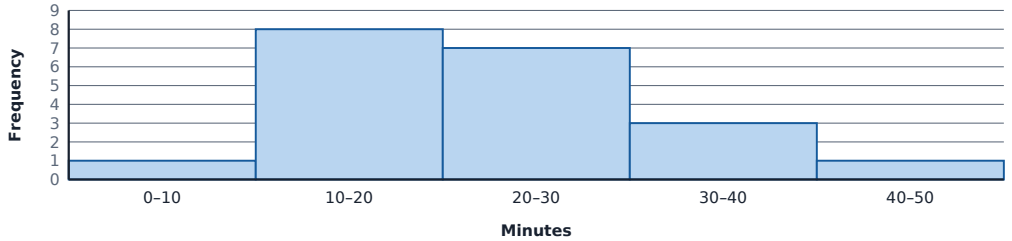
7

GRADING NOTE The 20-30 bar reaches 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q1/5

The histogram shows the commute times of some employees. How many employees are shown in all?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q2/5

64

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

20

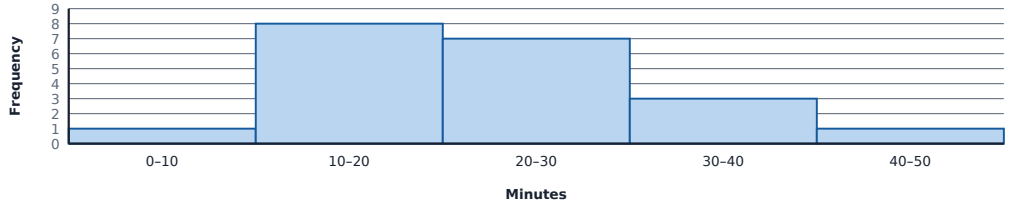
GRADING NOTE 1 + 8 + 7 + 3 + 1 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q2/5

True or false?

The histogram shows the commute times of some employees. It shows that one employee's commute was exactly 15 minutes.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q3/5

65

QUESTION ID

S2-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

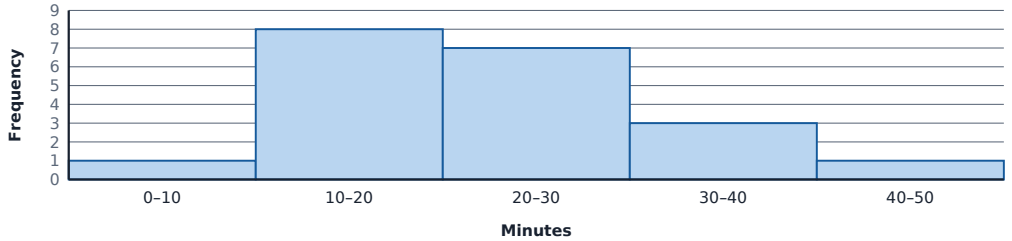
GRADING NOTE A histogram shows counts per bin, not individual values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows the commute times of some employees. Which bin has the most values?



A. 10-20 **B.** 0-10 **C.** 20-30 **D.** 30-40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q4/5

66

QUESTION ID

S2-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10-20

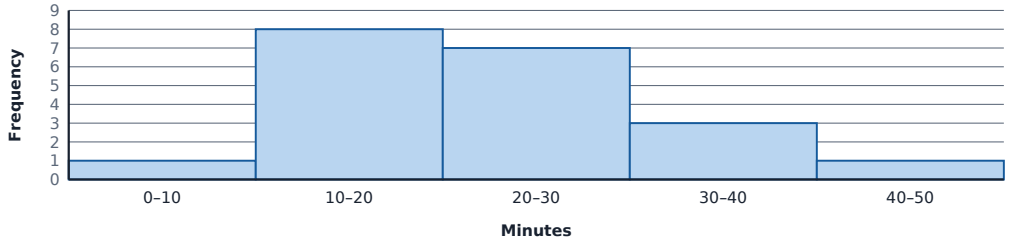
GRADING NOTE Its bar reaches 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows the commute times of some employees. How many employees have commutes of at least 30 minutes?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q5/5

67

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 3 in 30-40 and 1 in 40-50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSS-ID.A.1

FORWARD 1

Regroup a histogram with wider bins • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

A histogram with bins 0-5, 5-10, 10-15, 15-20 has counts 3, 6, 8, 2. What are the counts for the wider bins 0-10 and 10-20?

0-10: _____

10-20: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q1/5

69

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9; 10

GRADING NOTE Form required: both. 3 + 6 and 8 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q1/5

A histogram with bins 60-65, 65-70, 70-75, 75-80 has counts 2, 5, 6, 4. What are the counts for 60-70 and 70-80?

60-70: _____

70-80: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q2/5

70

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7; 10

GRADING NOTE Form required: both. 2 + 5 and 6 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q2/5

True or false?

Combining each pair of neighboring bins into one bin never changes the total number of values shown.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q3/5

71

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every value is still counted exactly once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q3/5

A histogram with bins of width 1 has counts 1, 0, 2, 0, 1, 0, 3, 0, 2, so it looks jagged. Why might bins of width 2 help?

A. Wider bins always make a distribution look symmetric.

B. Wider bins change the data values, so the shape becomes smoother.

C. Wider bins add values to the empty bins, so no bin is 0.

D. Wider bins group nearby values, so the overall shape is easier to see.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q4/5

72

QUESTION ID

S2-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Wider bins group nearby values, so the overall shape is easier to see.

GRADING NOTE Regrouping changes only the summary, not the data. The others claim the data change, values are added, or symmetry always appears.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q4/5

A histogram with bins of width 10 has counts 4, 9, 5. Can the counts for bins of width 5 be found from it?

A. Yes: halve each count, because each narrower bin is half as wide.

B. Yes: keep the same counts, because the data have not changed.

C. No: each count could be split between the two narrower bins in many ways.

D. No: a histogram with bins of width 10 cannot be redrawn at all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q5/5

73

QUESTION ID

S2-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. No: each count could be split between the two narrower bins in many ways.

GRADING NOTE Combining bins loses information that cannot be recovered. Halving assumes an even split; the others are false.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSS-ID.A.1

FORWARD 2

Locate the median in a histogram • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

A histogram of 21 values has counts 0-10: 4, 10-20: 6, 20-30: 7, 30-40: 4. The median is the 11th value in order. Which bin holds it? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q1/5

75

QUESTION ID
S2-F2-Q1

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE 4 + 6 = 10 values come before 20, so the 11th is in 20-30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q1/5

A histogram of 20 values has counts 50-60: 3, 60-70: 5, 70-80: 8, 80-90: 4. The median is the mean of the 10th and 11th values. Which bin holds both? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q2/5

76

QUESTION ID
S2-F2-Q2

TYPE
Short answer

CORRECT ANSWER
70

GRADING NOTE 3 + 5 = 8 values lie below 70; the 9th through 16th are in 70-80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q2/5

True or false?

The bin with the tallest bar always holds the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q3/5

77

QUESTION ID
S2-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Counts 10, 2, 3, 1, 9 (25 values): the tallest bar is the first, but the median (13th value) is in the third bin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q3/5

A histogram of 15 values has counts 0-5: 10, 5-10: 2, 10-15: 2, 15-20: 1. Which bin holds the median (the 8th value)?

A. 15-20

B. 10-15

C. 5-10

D. 0-5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q4/5

78

QUESTION ID
S2-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 0-5

GRADING NOTE The first 10 values are in 0-5. The others use the middle bin or later bins.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q4/5

A histogram of 15 values has counts 0-10: 2, 10-20: 3, 20-30: 3, 30-40: 7. Which bin holds the median (the 8th value)? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q5/5

79

QUESTION ID
S2-F2-Q5

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE 2 + 3 = 5 below 20, 8 below 30; the 8th is in 20-30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q5/5

Represent data with box plots

MAIN	Probability & Statistics main question	**HSS-ID.A.1**
BACKWARD 1	Order data and find the median	**6.SP.B.5.c**
BACKWARD 2	Find quartiles as medians of the halves	**6.SP.B.5.c**
BACKWARD 3	Read a box plot	**6.SP.B.4**
FORWARD 1	Mark outliers on a box plot	**HSS-ID.A.3**
FORWARD 2	Compare two box plots	**HSS-ID.A.2**

SET 3 • HSS-ID.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Laps swum by 11 swimmers: 12, 15, 9, 18, 14, 11, 16, 20, 13, 10, 17. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q1/5

82

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

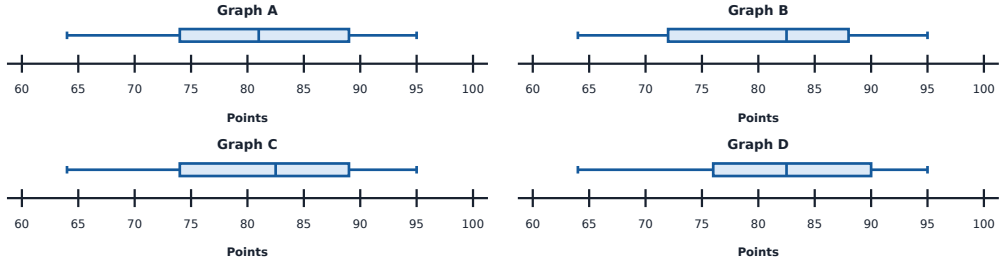
9; 11; 14; 17; 20

GRADING NOTE Form required: all five. Ordered: 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20; median 14; lower half 9-13 gives Q1 = 11; upper half 15-20 gives Q3 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q1/5

Test scores of 12 students: 72, 85, 90, 64, 78, 88, 95, 70, 81, 76, 84, 92. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q2/5

83

QUESTION ID

S3-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Ordered: 64, 70, 72, 76, 78, 81, 84, 85, 88, 90, 92, 95; median (81 + 84)/2 = 82.5; Q1 = (72 + 76)/2 = 74; Q3 = (88 + 90)/2 = 89. The others use 81 as the median, the 3rd and 9th values as quartiles, or the wrong middle pair of each half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Hours of sleep of 14 teenagers: 7.5, 6, 8, 6.5, 9, 7, 5.5, 8.5, 7, 6.5, 7.5, 8, 6, 7. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q3/5

84

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

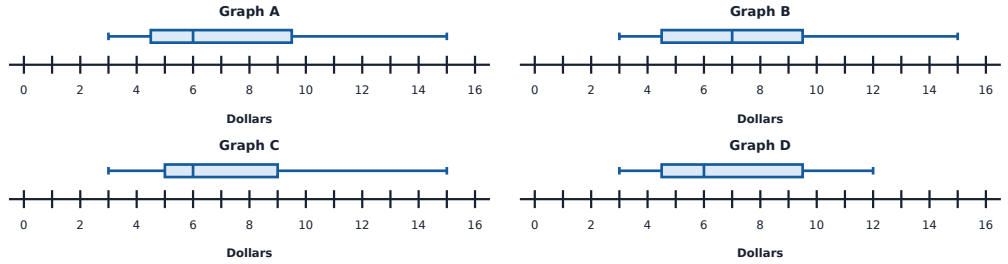
5.5; 6.5; 7; 8; 9

GRADING NOTE Form required: all five. Ordered: 5.5, 6, 6, 6.5, 6.5, 7, 7 | 7, 7.5, 7.5, 8, 8, 8.5, 9; median 7; Q1 = 6.5; Q3 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q3/5

Prices (dollars) of 13 menu items: 3, 8, 5, 12, 6, 4, 9, 7, 10, 5, 6, 15, 4. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q4/5

85

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Ordered: 3, 4, 4, 5, 5, 6, 6, 7, 8, 9, 10, 12, 15; median 6 (7th value); Q1 = (4 + 5)/2 = 4.5; Q3 = (9 + 10)/2 = 9.5. The others use 7 as the median, keep the median in both halves (Q1 = 5, Q3 = 9), or end the whisker at 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q4/5

Distances (km) 10 students live from school: 2.1, 3.4, 1.8, 5, 2.7, 4.2, 3, 2.4, 3.9, 1.5. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q5/5

86

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

1.5; 2.1; 2.85; 3.9; 5

GRADING NOTE Form required: all five. Ordered: 1.5, 1.8, 2.1, 2.4, 2.7 | 3.0, 3.4, 3.9, 4.2, 5.0; median (2.7 + 3.0)/2 = 2.85; Q1 = 2.1; Q3 = 3.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q5/5

SET 3 • HSS-ID.A.1

BACKWARD 1

Order data and find the median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 9, 3, 7, 5, 11.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q1/5

88

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Ordered: 3, 5, 7, 9, 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q1/5

Find the median of 4, 10, 6, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q2/5

89

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Ordered: 4, 6, 8, 10; (6 + 8)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q2/5

True or false?

The median of 2, 9, 4 is 9, the middle value of the list as written.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q3/5

90

QUESTION ID

S3-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Order first: 2, 4, 9; the median is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q3/5

What is the median of 1, 3, 3, 6, 7, 8?

A. 3

B. 6

C. 4.67

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q4/5

91

QUESTION ID

S3-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4.5

GRADING NOTE (3 + 6)/2 = 4.5. 3 is the mode, 6 a middle value alone, 4.67 the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q4/5

Find the median of 12, 15, 11, 14, 13, 16, 10.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q5/5

92

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Ordered: 10, 11, 12, 13, 14, 15, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSS-ID.A.1

BACKWARD 2

Find quartiles as medians of the halves • Grade 6 • Statistics & Probability • 6.SP.B.5.c

For 2, 4, 6, 8, 10, 12, 14, 16, find Q1 and Q3 (each the median of a half).

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q1/5

94

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5; 13

GRADING NOTE Form required: both. Halves 2, 4, 6, 8 and 10, 12, 14, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q1/5

For 1, 3, 5, 7, 9, 11, 13 (an odd count), leave out the median and find Q1 and Q3.

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q2/5

95

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3; 11

GRADING NOTE Form required: both. Halves 1, 3, 5 and 9, 11, 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q2/5

True or false?

For 1, 2, 3, 4, 5, 6, 7, 8, 9, Q1 is 2.5 when the median 5 is left out of both halves.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q3/5

96

QUESTION ID

S3-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lower half 1, 2, 3, 4; its median is 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q3/5

For 10, 20, 30, 40, 50, 60, what is the interquartile range Q3 − Q1 (Q1 and Q3 are the medians of the halves)?

A. 30

B. 50

C. 20

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q4/5

97

QUESTION ID

S3-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 30

GRADING NOTE Q1 = 20, Q3 = 50. 50 is the range; 20 and 35 come from wrong halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q4/5

For 5, 6, 8, 9, 12, 15, find Q1 (the median of the lower half).

Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q5/5

98

QUESTION ID

S3-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Lower half 5, 6, 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

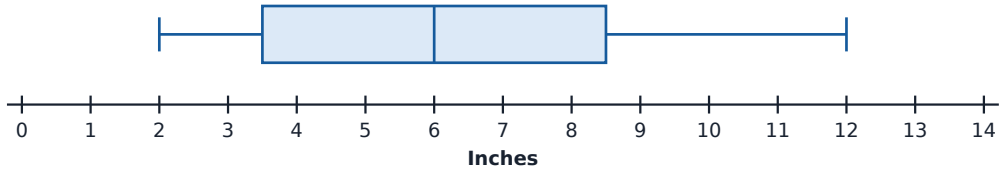
Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSS-ID.A.1

BACKWARD 3

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The box plot shows the snowfall in some storms. What is the median snowfall?



Median (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q1/5

100

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

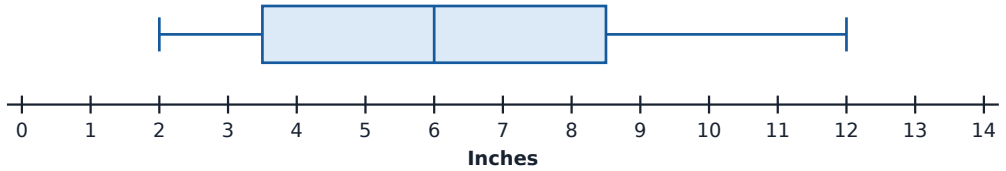
6

GRADING NOTE The line inside the box is at 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q1/5

The box plot shows the snowfall in some storms. What is the interquartile range?



IQR (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q2/5

101

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

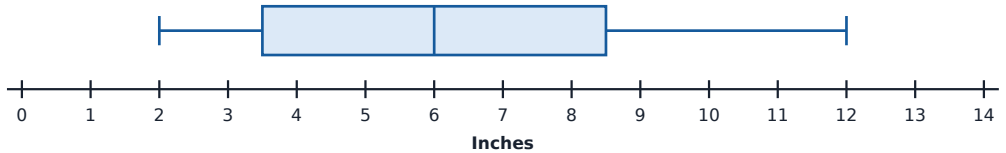
GRADING NOTE Q3 − Q1 = 8.5 − 3.5 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q2/5

True or false?

The box plot shows the snowfall in some storms. About half of the storms had at least 6 inches.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q3/5

102

QUESTION ID

S3-B3-Q3

TYPE

True or false

CORRECT ANSWER

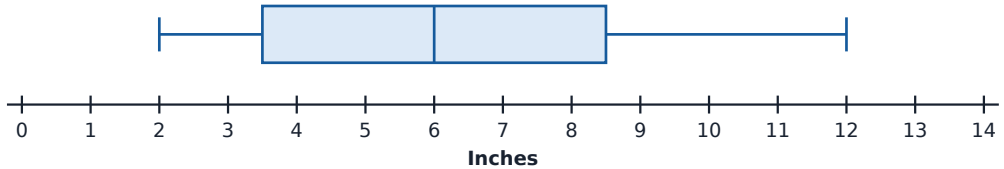
A. True

GRADING NOTE The median is 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q3/5

The box plot shows the snowfall in some storms. What is the range?



A. 6 inches **B.** 10 inches **C.** 12 inches **D.** 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q4/5

103

QUESTION ID

S3-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

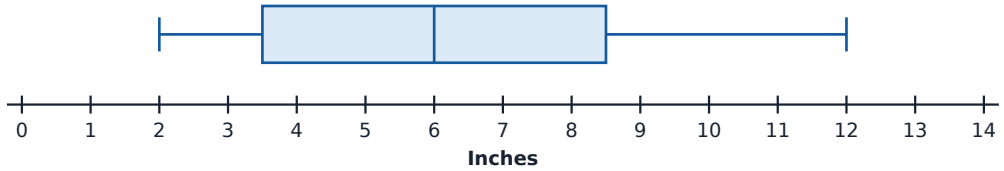
B. 10 inches

GRADING NOTE 12 − 2 = 10. 5 is the IQR, 12 the maximum, 6 the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q4/5

The box plot shows the snowfall in some storms. What is Q1?



Q1 (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q5/5

104

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE The left edge of the box is at 3.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSS-ID.A.1

FORWARD 1

Mark outliers on a box plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set has Q1 = 20 and Q3 = 30. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. What are the lower and upper fences?

Lower fence: _____

Upper fence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q1/5

106

QUESTION ID
S3-F1-Q1

TYPE
Short answer

CORRECT ANSWER
5; 45

GRADING NOTE Form required: both. IQR 10; 20 − 15 and 30 + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q1/5

Prices of 10 used bikes (dollars): 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. How many outliers are there, and where does the upper whisker end?

Number of outliers: _____

Upper whisker ends at: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q2/5

107

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1; 220

GRADING NOTE Form required: both. Ordered 180, 185, 190, 195, 200, 205, 210, 215, 220, 600; Q1 = 190, Q3 = 215, IQR 25; fences 152.5 and 252.5; only 600 is an outlier; the whisker ends at the largest non-outlier, 220.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q2/5

True or false?

In a box plot that marks outliers, a whisker ends at the fence value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q3/5

108

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It ends at the most extreme data value inside the fence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q3/5

Race times (seconds) of 9 swimmers: 52, 55, 54, 31, 56, 53, 57, 55, 54. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Which value is an outlier?

A. 31

B. 57

C. both 31 and 57

D. none

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q4/5

109

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 31

GRADING NOTE Ordered 31, 52, 53, 54, 54, 55, 55, 56, 57; Q1 = (52 + 53)/2 = 52.5, Q3 = (55 + 56)/2 = 55.5; IQR 3; fences 48 and 60; only 31 is outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q4/5

A box plot that marks outliers has Q1 = 14 and Q3 = 17, and the largest data values are 18 and 41. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Where does the upper whisker end?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q5/5

110

QUESTION ID
S3-F1-Q5

TYPE
Short answer

CORRECT ANSWER
18

GRADING NOTE Upper fence 17 + 4.5 = 21.5; 41 is an outlier, 18 is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q5**

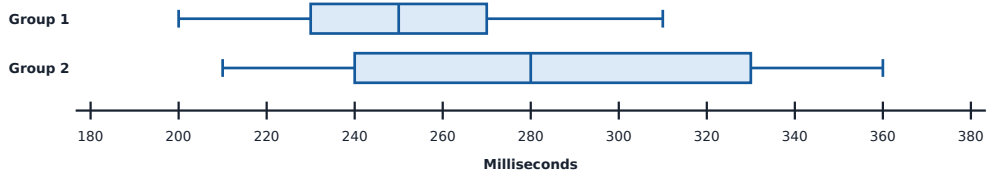
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSS-ID.A.1

FORWARD 2

Compare two box plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The box plots show the reaction times of two groups. Which comparison is correct?



A. Group 1 has the greater median and the greater IQR.

B. Group 1 has the greater range and the greater median.

C. The groups have the same median but different IQRs.

D. Group 2 has the greater median and the greater IQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q1/5

112

QUESTION ID

S3-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Group 2 has the greater median and the greater IQR.

GRADING NOTE Medians 250 and 280; IQRs 40 and 90. The others reverse or misread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q1/5

Box plot 1 has median 40 and IQR 12; box plot 2 has median 46 and IQR 8. By how much do the medians differ, and which has the smaller IQR (enter 1 or 2)?

Difference of medians: _____

Smaller IQR (1 or 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q2/5

113

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6; 2

GRADING NOTE Form required: both. $46 - 40 = 6$; $8 < 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

If two box plots have the same median, the two data sets have the same mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q3/5

114

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

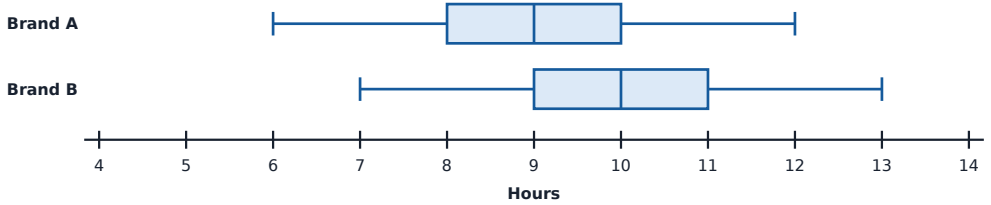
B. False

GRADING NOTE The mean is not shown by a box plot and can differ (for example with different skew).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q3/5

The box plots show the battery lives of two brands. About what fraction of brand B batteries last longer than the median life of brand A?



A. about 1/2 **B.** about 3/4 **C.** none **D.** about 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q4/5

115

QUESTION ID
S3-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. about 3/4

GRADING NOTE Brand A's median is 9, which is Brand B's Q1, so about 3/4 of Brand B lies above it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q4/5

Two box plots: class A has Q1 = 70 and Q3 = 85; class B has Q1 = 72 and Q3 = 80. Give each IQR.

IQR of A: _____

IQR of B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q5/5

116

QUESTION ID

S3-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15; 8

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSS-ID.A.2 • INTERPRETING CATEGORICAL AND QUANTITATIVE DATA

Compare centers with the measure suited to the shape

MAIN	Probability & Statistics main question	**HSS-ID.A.2**
BACKWARD 1	Compute a mean	**6.SP.B.5.c**
BACKWARD 2	Compute a median	**6.SP.B.5.c**
BACKWARD 3	Recognize skew and extreme values	**6.SP.A.2**
FORWARD 1	Find a missing value from a target mean or median	**HSS-ID.A.2**
FORWARD 2	Predict how changes in data move the mean and median	**HSS-ID.A.3**

SET 4 • HSS-ID.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Sale prices (thousands of dollars) of 8 homes in each of two towns: Town A: 210, 225, 230, 240, 245, 250, 260, 900; Town B: 190, 220, 235, 255, 270, 280, 300, 310. Compare the typical prices with one measure of center suited to the shapes of both data sets. Give that measure for each town.

Town A: _____

Town B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q1/5

119

QUESTION ID
S4-M-Q1

TYPE
Short answer

CORRECT ANSWER
242.5; 262.5

GRADING NOTE Form required: both medians. Town A has an extreme value (900), so the median is suited; the same measure is used for both towns. Town A: (240 + 245)/2 = 242.5; Town B: (255 + 270)/2 = 262.5. (Town A's mean, 320, is pulled up by 900.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q1/5

Scores of 7 students in each of two classes: class P: 70, 74, 76, 78, 80, 82, 86; class Q: 66, 70, 72, 74, 76, 78, 82. Which comparison of centers is appropriate and correct?

A. Compare medians: P's median 76 is 2 points above Q's median 74, so P is 2 points higher.

B. Compare the highest scores: P's 86 is 4 points above Q's 82, so P is 4 points higher.

C. Both are roughly symmetric with no outliers, so compare means: P's mean 78 is 4 points above Q's mean 74.

D. Both are roughly symmetric, so compare means: P's mean 76 is 2 points above Q's mean 74.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q2/5

120

QUESTION ID
S4-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Both are roughly symmetric with no outliers, so compare means: P's mean 78 is 4 points above Q's mean 74.

GRADING NOTE Means: 546/7 = 78 and 518/7 = 74 (the medians are also 78 and 74). The others miscompute the mean or median of P, or compare maximums, which do not describe the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q2/5

Delivery times (days) of 9 orders from each of two stores: store X: 2, 2, 3, 3, 3, 4, 4, 5, 14; store Y: 2, 3, 3, 4, 4, 5, 5, 6, 6. Compare the typical delivery times with one measure of center suited to the shapes of both data sets. Give that measure for each store.

Store X: _____

Store Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q3/5

121

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

3; 4

GRADING NOTE Form required: both medians. Store X has an extreme value (14), so medians are used: 3 and 4 (5th values).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q3/5

Heights (cm) of 7 seedlings of each of two plants: plant A: 5, 6, 6, 7, 7, 8, 30; plant B: 8, 8, 9, 9, 9, 10, 10. Which comparison of typical heights is appropriate?

A. Compare means: A's mean (about 9.9 cm) is greater, so A's seedlings are typically taller.

B. Compare means: both are about 9 cm, so the seedlings are typically the same height.

C. Compare maximums: A's tallest seedling (30 cm) is taller, so A's seedlings are typically taller.

D. A is skewed by the value 30, so compare medians: B's median 9 cm is greater than A's median 7 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q4/5

122

QUESTION ID

S4-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A is skewed by the value 30, so compare medians: B's median 9 cm is greater than A's median 7 cm.

GRADING NOTE A's mean 69/7 ≈ 9.86 is pulled up by 30; medians 7 and 9 describe typical seedlings. The others use the mean despite the outlier or use the maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Masses (grams) of 6 bags filled by each of two machines: machine 1: 498, 499, 500, 500, 501, 502; machine 2: 503, 504, 505, 505, 506, 507. Compare the typical masses with one measure of center suited to the shapes of both data sets. Give that measure for each machine.

Machine 1: _____

Machine 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q5/5

123

QUESTION ID
S4-M-Q5

TYPE
Short answer

CORRECT ANSWER
500; 505

GRADING NOTE Form required: both. Both sets are symmetric with no outliers, so means suit: 3000/6 = 500 and 3030/6 = 505. (The medians are also 500 and 505, so either measure gives the same values.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSS-ID.A.2

BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 4, 7, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q1/5

125

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 21/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 2, 3, 5, 6. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q2/5

126

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 16/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q2/5

True or false?

The mean of 1, 2, 3, 10 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q3/5

127

QUESTION ID

S4-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 16/4 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q3/5

What is the mean of 5, 5, 6, 8?

A. 24

B. 5

C. 6

D. 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q4/5

128

QUESTION ID

S4-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE 24/4. 5 is the mode, 5.5 the median, 24 the sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q4/5

Find the mean of 1.5, 2.5, 3.5. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q5/5

129

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE 7.5/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSS-ID.A.2

BACKWARD 2

Compute a median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 8, 2, 6.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q1/5

131

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Ordered 2, 6, 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q1/5

Find the median of 3, 9, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q2/5

132

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Ordered 3, 5, 7, 9; (5 + 7)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q2/5

True or false?

The median of 1, 2, 3, 100 is 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q3/5

133

QUESTION ID

S4-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (2 + 3)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q3/5

What is the median of 10, 12, 12, 15, 30?

A. 15.8

B. 12

C. 30

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q4/5

134

QUESTION ID

S4-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 12

GRADING NOTE Third value. 15.8 is the mean; 15 and 30 are other values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q4/5

Find the median of 4.5, 6, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q5/5

135

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5.5

GRADING NOTE Ordered 4.5, 5, 6, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSS-ID.A.2

BACKWARD 3

Recognize skew and extreme values • Grade 6 • Statistics & Probability • 6.SP.A.2

Which data set is skewed to the right?

A. 5, 5, 5, 5, 5

B. 1, 12, 13, 13, 14, 14, 14, 15

C. 1, 2, 2, 3, 3, 3, 4, 15

D. 1, 2, 3, 4, 5, 6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q1/5

137

QUESTION ID

S4-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 1, 2, 2, 3, 3, 3, 4, 15

GRADING NOTE A long tail of high values (15). The others are constant, skewed left (a low tail at 1), or symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q1/5

True or false?

A data set with a single value far below all the others is skewed to the left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q2/5

138

QUESTION ID

S4-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The tail points toward the low value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q2/5

Which value is extreme in 20, 22, 21, 23, 22, 60?

A. 60

B. 23

C. none

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q3/5

139

QUESTION ID

S4-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60

GRADING NOTE 60 is far from the rest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q3/5

True or false?

In 4, 5, 5, 6, 6, 6, 7, 7, 8 the values are spread evenly on both sides of 6, so the data are roughly symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q4/5

140

QUESTION ID

S4-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mirror image around 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q4/5

Incomes in a town include a few very large incomes. The distribution is most likely

A. skewed to the right

B. skewed to the left

C. uniform

D. symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q5/5

141

QUESTION ID

S4-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. skewed to the right

GRADING NOTE A few large values make a long right tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSS-ID.A.2

FORWARD 1

Find a missing value from a target mean or median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Four quiz scores are 80, 85, 90, 75. What fifth score makes the mean of all five scores 84?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q1/5

143

QUESTION ID
S4-F1-Q1

TYPE
Short answer

CORRECT ANSWER
90

GRADING NOTE $5 \cdot 84 = 420$; $420 - 330 = 90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q1/5

The values 3, 5, 8 and x have mean 7. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q2/5

144

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE $4 \cdot 7 = 28$; $28 - 16 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q2/5

True or false?

In a data set of 7 values, replacing the largest value with a larger number can change the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q3/5

145

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median is the 4th value, which does not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q3/5

The values 4, 6, 9, 10, 12 and one more value x have a median of 8. What is x?

A. 16

B. 7

C. 8

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q4/5

146

QUESTION ID

S4-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

GRADING NOTE With x = 7: 4, 6, 7, 9, 10, 12 has median (7 + 9)/2 = 8. x = 8 gives 8.5, x = 16 gives 9.5, x = 6 gives 7.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q4/5

A team's mean score over 9 games is 70. What score in a 10th game makes the mean of all 10 games 72?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q5/5

147

QUESTION ID
S4-F1-Q5

TYPE
Short answer

CORRECT ANSWER
90

GRADING NOTE 720 − 630.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSS-ID.A.2

FORWARD 2

Predict how changes in data move the mean and median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The values are 10, 12, 13, 15, 40. If 40 is replaced by 20, what happens?

A. The mean decreases by 20 and the median stays 13.

B. The mean and the median both decrease by 4.

C. The mean decreases by 4 and the median stays 13.

D. The mean stays the same and the median decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q1/5

149

QUESTION ID

S4-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The mean decreases by 4 and the median stays 13.

GRADING NOTE The sum drops by 20, so the mean drops by 20/5 = 4; the middle value 13 is unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q1/5

True or false?

Adding 5 to every value in a data set increases both the mean and the median by 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q2/5

150

QUESTION ID

S4-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every value, and so every summary of center, moves up by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q2/5

Find the mean and the median of 2, 4, 6, 8, 100.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q3/5

151

QUESTION ID

S4-F2-Q3

TYPE

Short answer

CORRECT ANSWER

24; 6

GRADING NOTE Form required: both. 120/5 = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q3/5

Salaries at a company include a few very high salaries. Which statement is most likely true?

A. The median is pulled up more than the mean.

B. The mean salary is greater than the median salary.

C. The median salary is greater than the mean salary.

D. The mean and the median are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q4/5

152

QUESTION ID
S4-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. The mean salary is greater than the median salary.

GRADING NOTE High values pull the mean up; the median resists them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q4/5

True or false?

Doubling every value in a data set doubles the median but leaves the mean unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q5/5

153

QUESTION ID
S4-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q5/5

Compare spreads with the IQR or the standard deviation

MAIN	Probability & Statistics main question	**HSS-ID.A.2**
BACKWARD 1	Find the range and the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Measure distances from the mean	**6.SP.B.5.c**
BACKWARD 3	Square, average, and take square roots	**8.EE.A.2**
FORWARD 1	Compare population and sample standard deviations	**HSS-ID.A.2**
FORWARD 2	Predict how adding or multiplying changes center and spread	**HSS-ID.A.2**

SET 5 • HSS-ID.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Ages of 9 players on each of two teams: team A: 14, 15, 15, 16, 16, 17, 18, 18, 19; team B: 13, 14, 16, 16, 17, 18, 20, 22, 25. Give the interquartile range of each team. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of team A: _____

IQR of team B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q1/5

156

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

3; 6

GRADING NOTE Form required: both. A: Q1 = 15, Q3 = 18, IQR 3. B: Q1 = 15, Q3 = 21, IQR 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q1/5

Bus delays (minutes) on 5 days: route 1: 4, 6, 8, 10, 12; route 2: 7, 8, 8, 8, 9. Both are symmetric. Give the standard deviation of each route, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Route 1: _____

Route 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q2/5

157

QUESTION ID
S5-M-Q2

TYPE
Short answer

CORRECT ANSWER
2.8; 0.6

GRADING NOTE Form required: both, to the nearest tenth. Route 1: deviations −4, −2, 0, 2, 4; variance 40/5 = 8; √8 ≈ 2.83. Route 2: deviations −1, 0, 0, 0, 1; variance 2/5 = 0.4; √0.4 ≈ 0.63.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q2/5

Monthly rents (dollars) of 7 apartments in each of two buildings: A: 800, 850, 900, 950, 1000, 1050, 2400; B: 700, 800, 900, 1000, 1100, 1200, 1300. Which comparison of spread is appropriate and correct?

A. Compare standard deviations: A's is greater, so A's typical rents vary more than B's.

B. A has an outlier, so compare IQRs: B's IQR ($400) is twice A's ($200), so B's typical rents vary more.

C. Compare IQRs: A's IQR ($400) is twice B's ($200), so A's typical rents vary more.

D. Compare ranges: A's range ($1,600) is greater than B's ($600), so A's typical rents vary more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q3/5

158

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A has an outlier, so compare IQRs: B's IQR ($400) is twice A's ($200), so B's typical rents vary more.

GRADING NOTE Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. A: Q1 850, Q3 1050, IQR 200; B: Q1 800, Q3 1200, IQR 400. The range and standard deviation of A are inflated by the outlier 2400; the last choice swaps the IQRs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q3/5

Lengths (cm) of 8 fish from each of two lakes: lake A: 20, 22, 23, 25, 26, 28, 30, 31; lake B: 18, 24, 24, 25, 25, 26, 26, 40. Lake B has outliers. Give the interquartile range of each lake. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of lake A: _____

IQR of lake B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q4/5

159

QUESTION ID
S5-M-Q4

TYPE
Short answer

CORRECT ANSWER
6.5; 2

GRADING NOTE Form required: both. A: Q1 = 22.5, Q3 = 29, IQR 6.5. B: Q1 = 24, Q3 = 26, IQR 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q4/5

Weekly savings (dollars) of 5 students in each of two groups: group 1: 10, 20, 30, 40, 50; group 2: 25, 28, 30, 32, 35. Both are symmetric. Give the standard deviation of each group, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Group 1: _____

Group 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q5/5

160

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

14.1; 3.4

GRADING NOTE Form required: both, to the nearest tenth. Group 1: variance (400 + 100 + 0 + 100 + 400)/5 = 200; √200 ≈ 14.14. Group 2: deviations −5, −2, 0, 2, 5; variance 58/5 = 11.6; √11.6 ≈ 3.41.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSS-ID.A.2

BACKWARD 1

Find the range and the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the range of 7, 3, 12, 5, 9.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q1/5

162

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 12 − 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q1/5

Q1 = 18 and Q3 = 27. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q2/5

163

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 27 − 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q2/5

True or false?

The IQR is the spread of the middle half of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q3/5

164

QUESTION ID

S5-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE From Q1 to Q3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q3/5

For 2, 4, 6, 8, 10, 12, the IQR (with Q1 and Q3 the medians of the halves) is

A. 10

B. 6

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q4/5

165

QUESTION ID

S5-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE Q1 = 4, Q3 = 10. 10 is the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q4/5

A box plot has minimum 5, Q1 9, median 12, Q3 16, maximum 25. Give the IQR and the range.

IQR: _____

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q5/5

166

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

7; 20

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSS-ID.A.2

BACKWARD 2

Measure distances from the mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

The mean of 2, 6, 10 is 6. What are the distances of 2 and of 10 from the mean?

Distance of 2: _____

Distance of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q1/5

168

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4; 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q1/5

Find the mean absolute deviation of 1, 3, 5, 7.

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q2/5

169

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Mean 4; distances 3, 1, 1, 3; mean 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q2/5

True or false?

A data set whose values are all far from the mean has a large spread.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q3/5

170

QUESTION ID

S5-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Spread measures distance from center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q3/5

Which set has the larger spread around its mean?

A. 0, 10, 20

B. cannot tell

C. 9, 10, 11

D. they are the same

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q4/5

171

QUESTION ID

S5-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0, 10, 20

GRADING NOTE Distances 10, 0, 10 versus 1, 0, 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q4/5

The values 5, 5, 5, 5 have mean 5. What is their mean absolute deviation?

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q5/5

172

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Every distance is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSS-ID.A.2

BACKWARD 3

Square, average, and take square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find √16 + √9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q1/5

174

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q1/5

The squared distances from a mean are 4, 0, 4. Find their average.

Average: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q2/5

175

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

8/3

GRADING NOTE 8/3 (about 2.67).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q3/5

176

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$, but $3 + 4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q3/5

Which is √0.25?

A. 0.5

B. 0.125

C. 0.0625

D. 0.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q4/5

177

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.5

GRADING NOTE $0.5^2 = 0.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q4/5

Find √8 to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q5/5

178

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2.8

GRADING NOTE 2.828....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSS-ID.A.2

FORWARD 1

Compare population and sample standard deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

For the data 4, 6, 8, 10, 12, give the sample standard deviation (divide by n − 1), to the nearest hundredth. A calculator or spreadsheet may be used.

s = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q1/5

180

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.16

GRADING NOTE Sum of squared deviations 40; 40/4 = 10; √10 ≈ 3.162.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q1/5

True or false?

For the same data, the sample standard deviation (divide by n − 1) is never smaller than the population standard deviation (divide by n).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q2/5

181

QUESTION ID

S5-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dividing by the smaller number n − 1 gives a larger or equal value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For 2, 4, 6, give the population and the sample standard deviations, to the nearest hundredth. A calculator or spreadsheet may be used.

Population: _____

Sample: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q3/5

182

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

1.63; 2.00

GRADING NOTE Form required: both, to the nearest hundredth. Squared deviations 4, 0, 4: $\sqrt{(8/3)} \approx 1.633$ and $\sqrt{(8/2)} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q3/5

A spreadsheet gives 5.2 for STDEV.S and 4.9 for STDEV.P of the same data. Which statement is correct?

A. STDEV.S divides by n and treats the data as the whole population; STDEV.P divides by n − 1 for a sample.

B. The two functions measure different spreads: STDEV.S the range of the data and STDEV.P their IQR.

C. One of the two values must be an error, because a data set has only one standard deviation.

D. STDEV.S divides by n − 1 and treats the data as a sample; STDEV.P divides by n and treats them as the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q4/5

183

QUESTION ID
S5-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. STDEV.S divides by n − 1 and treats the data as a sample; STDEV.P divides by n and treats them as the population.

GRADING NOTE The sample formula divides by n − 1, so it is the larger value. The others swap the formulas or deny the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q4/5

Ten values have sum of squared deviations from their mean equal to 90. Give the population and sample variances.

Population variance: _____

Sample variance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q5/5

184

QUESTION ID

S5-F1-Q5

TYPE

Short answer

CORRECT ANSWER

9; 10

GRADING NOTE Form required: both. 90/10 and 90/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSS-ID.A.2

FORWARD 2

Predict how adding or multiplying changes center and spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Every value in a data set is increased by 10. What happens to the standard deviation and the IQR?

A. Both stay the same.

B. The SD stays the same and the IQR increases by 10.

C. Both increase by 10.

D. Both are multiplied by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q1/5

186

QUESTION ID
S5-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. Both stay the same.

GRADING NOTE Shifting every value keeps every distance the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q1/5

A data set has mean 20 and standard deviation 4. Every value is multiplied by 3. Give the new mean and standard deviation.

Mean: _____

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q2/5

187

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

60; 12

GRADING NOTE Form required: both. Both are multiplied by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q2/5

True or false?

Converting temperatures from degrees to tenths of a degree (multiplying by 10) multiplies the IQR by 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q3/5

188

QUESTION ID
S5-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Q1 and Q3 are both multiplied by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q3/5

A data set has median 50 and IQR 8. Each value is changed to 2·value + 5. Give the new median and IQR.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q4/5

189

QUESTION ID

S5-F2-Q4

TYPE

Short answer

CORRECT ANSWER

105; 16

GRADING NOTE Form required: both. 2·50 + 5; the + 5 does not change the IQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q4/5

Test scores have SD 6. The teacher adds 3 bonus points to every score. The new SD is

A. 18

B. 9

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q5/5

190

QUESTION ID

S5-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE Adding a constant does not change spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q5/5

Interpret differences in shape, center and spread in context

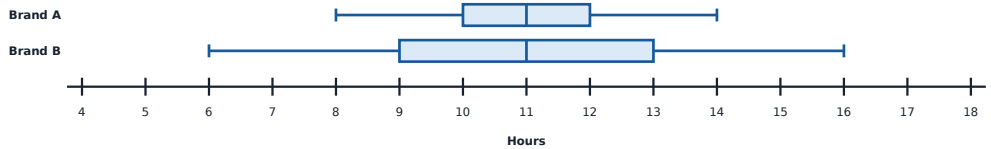
MAIN	Probability & Statistics main question	**HSS-ID.A.3**
BACKWARD 1	Describe the shape of a distribution	**6.SP.A.2**
BACKWARD 2	What center and spread tell about data	**6.SP.A.3**
BACKWARD 3	Compare two distributions informally	**7.SP.B.4**
FORWARD 1	Measure separation in units of spread	**HSS-ID.A.3**
FORWARD 2	Judge whether a difference supports a claim	**HSS-IC.B.6**

SET 6 • HSS-ID.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The box plots show the battery lives (hours) of two brands. Which interpretation is correct?



A. Brand A's lives vary more, because its box is narrower than brand B's box is.

B. Typical lives are the same (both medians 11 hours), but brand B's lives vary more (IQR 4 versus 2 hours).

C. Brand B lasts longer, because its longest-lasting battery runs longer than brand A's does.

D. Brand B is better in every way, because its whiskers and its box are both longer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q1/5

193

QUESTION ID

S6-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Typical lives are the same (both medians 11 hours), but brand B's lives vary more (IQR 4 versus 2 hours).

GRADING NOTE Same median; IQRs 2 and 4. The others use the maximum as the center, read a narrow box as more spread, or call more spread better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q1/5

Waiting times at two stores: store 1 is roughly symmetric with mean 6 minutes and standard deviation 1.5 minutes; store 2 is skewed right with median 5 minutes, IQR 6 minutes, and a few waits over 20 minutes. Which interpretation is correct?

A. Store 2 often has a shorter wait, but its waits are much less predictable, with occasional very long waits.

B. Store 2 always has shorter waits than store 1, because its median is 1 minute lower than the mean of store 1.

C. The stores cannot be compared at all, because they are summarized with different measures.

D. Store 1's waits vary more, because its waits are described with a mean and a standard deviation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q2/5

194

QUESTION ID
S6-M-Q2

TYPE
Multiple choice

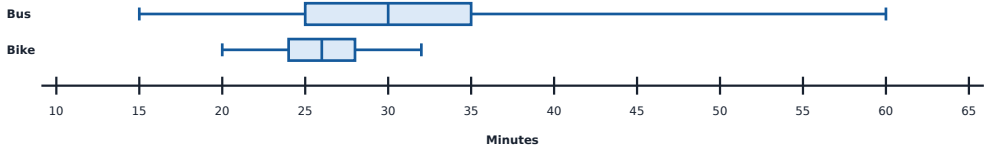
CORRECT ANSWER
A. Store 2 often has a shorter wait, but its waits are much less predictable, with occasional very long waits.

GRADING NOTE Median 5 < mean 6 for a typical wait; IQR 6 and long waits show more variability. The others overstate, invert the spread, or refuse a reasonable comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q2/5

The box plots show commute times (minutes) by bus and by bike for the same group of workers. Which interpretation is correct?



A. Bus trips are shorter in general, because the bus box plot starts at a lower time than the bike box plot.

B. Bike trips vary more, because the bike box plot is shorter than the bus box plot on the scale.

C. Both ways take the same time, because both box plots contain the times from 25 to 28 minutes.

D. Bike trips are more consistent; bus trips take a little longer in the middle and vary much more, including long trips.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q3/5

195

QUESTION ID
S6-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Bike trips are more consistent; bus trips take a little longer in the middle and vary much more, including long trips.

GRADING NOTE Medians 30 and 26; IQRs 10 and 4; the bus range is 45. The others use the minimum, overlap, or read a short box as more spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q3/5

Two classes both have mean score 70 on a test. Class X's scores form two clusters, around 55 and around 85; class Y's form one mound around 70. Which interpretation is correct?

A. Class X did better, because some of its students scored around 85, which is above every score in class Y.

B. Class Y has more spread, because its scores are all close to the middle of the range of scores.

C. The equal means hide a difference: class X seems to contain two groups doing very differently; class Y is more alike.

D. The classes performed the same way, because their means are equal and the mean summarizes the whole class.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q4/5

196

QUESTION ID

S6-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The equal means hide a difference: class X seems to contain two groups doing very differently; class Y is more alike.

GRADING NOTE A bimodal shape suggests two subgroups; the mean alone hides it. The others ignore shape or invert spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q4/5

Apples from two farms both have mean weight 150 g; farm A's weights have standard deviation 5 g and farm B's have standard deviation 20 g. Both distributions are roughly symmetric. Which interpretation is correct?

A. Farm B's apples are heavier, because a standard deviation of 20 g is larger than one of 5 g.

B. Farm A's apples are more uniform in size; on average, apples from both farms weigh about the same.

C. Farm A's apples are lighter, because a smaller standard deviation means a smaller mean weight.

D. The farms' apples are the same in every way, because their mean weights are both 150 g.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q5/5

197

QUESTION ID
S6-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Farm A's apples are more uniform in size; on average, apples from both farms weigh about the same.

GRADING NOTE Same center, different spread. The others read spread as center or ignore spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSS-ID.A.3

BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

A histogram has one peak in the middle and falls off evenly on both sides. Its shape is

A. symmetric

B. bimodal

C. skewed left

D. skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q1/5

199

QUESTION ID

S6-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. symmetric

GRADING NOTE Mirror image about the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q1/5

True or false?

A histogram with two separate peaks is called bimodal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q2/5

200

QUESTION ID

S6-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two modes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q2/5

A histogram has a long tail toward small values. Its shape is

A. uniform

B. symmetric

C. skewed right

D. skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q3/5

201

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. skewed left

GRADING NOTE The tail names the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q3/5

True or false?

A distribution whose bars are all about the same height is skewed right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q4/5

202

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is uniform.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q4/5

Waiting times at a busy clinic are mostly short with a few very long waits. The shape is most likely

A. skewed left

B. symmetric

C. bimodal

D. skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q5/5

203

QUESTION ID

S6-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. skewed right

GRADING NOTE Long tail of large values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSS-ID.A.3

BACKWARD 2

What center and spread tell about data • Grade 6 • Statistics & Probability • 6.SP.A.3

True or false?

A measure of center summarizes all the values of a data set with a single number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q1/5

205

GRADING NOTE Definition.

QUESTION ID

S6-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A measure of spread tells how much the values vary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q2/5

206

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q2/5

Two data sets have the same median but different IQRs. Which is true?

A. Their typical values are similar, but one varies more.

B. One of them has a larger typical value than the other.

C. They are identical data sets with the same values.

D. They have the same range and the same maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q3/5

207

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Their typical values are similar, but one varies more.

GRADING NOTE Center versus spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which describes spread?

A. the median

B. the mean

C. the IQR

D. the mode

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q4/5

208

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the IQR

GRADING NOTE The others are centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A data set with a larger mean must have a larger standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q5/5

209

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5, 5, 5 has mean 5 and SD 0; 1, 2, 3 has mean 2 and SD about 0.82.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSS-ID.A.3

BACKWARD 3

Compare two distributions informally • Grade 7 • Statistics & Probability • 7.SP.B.4

Group 1 has median 12 and IQR 3; group 2 has median 15 and IQR 3. Which is true?

A. Group 2 varies more, since its median is higher.

B. The groups are the same, since the IQRs are equal.

C. Group 2's typical value is 3 higher, with similar spread.

D. Group 1's values are higher, since its IQR is the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q1/5

211

QUESTION ID

S6-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Group 2's typical value is 3 higher, with similar spread.

GRADING NOTE Same IQR, higher median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q1/5

True or false?

Group 1 has median 20 and IQR 2; group 2 has median 40 and IQR 2. The middle halves (Q1 to Q3) of the two groups cannot overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q2/5

212

QUESTION ID

S6-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each box contains its median and is 2 wide, so box 1 lies within 18 to 22 and box 2 within 38 to 42. The summaries say nothing about the tails, which could still overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q2/5

Group A has mean 40 and group B has mean 46. By how much are the means different?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q3/5

213

QUESTION ID

S6-B3-Q3

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 46 − 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q3/5

Which pair of groups has the smallest gap between medians measured in IQRs (median gap divided by the common IQR)?

A. medians 20 and 30, both IQRs 1

B. medians 20 and 21, both IQRs 10

C. medians 20 and 40, both IQRs 2

D. medians 10 and 50, both IQRs 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q4/5

214

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. medians 20 and 21, both IQRs 10

GRADING NOTE 1/10 = 0.1. The others give 20/2 = 10, 10/1 = 10 and 40/5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q4/5

True or false?

Two groups with the same center must have the same spread.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q5/5

215

QUESTION ID

S6-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Spread is separate from center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSS-ID.A.3

FORWARD 1

Measure separation in units of spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Two roughly symmetric groups have means 70 and 76; both have standard deviation 4. How many standard deviations apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q1/5

217

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 6/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q1/5

Group medians are 30 and 45; both IQRs are 10. How many IQRs apart are the medians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q2/5

218

QUESTION ID
S6-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1.5

GRADING NOTE 15/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q2/5

Pair 1: means 50 and 55, both SDs 10. Pair 2: means 50 and 53, both SDs 1. Which pair is more clearly separated?

A. pair 1, 5 points apart

B. pair 1, 0.5 SDs apart

C. pair 2, 3 SDs apart

D. they are equally separated

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q3/5

219

QUESTION ID
S6-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. pair 2, 3 SDs apart

GRADING NOTE Pair 1 is 0.5 SDs apart, pair 2 is 3 SDs apart. A larger raw gap is not the larger separation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q3/5

True or false?

Two approximately normal groups have the same standard deviation, and their means are 0.2 standard deviations apart. Their distributions overlap a great deal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q4/5

220

QUESTION ID
S6-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Under the stated normal model with equal spread, a 0.2-SD gap leaves most values of each group in the range of the other. Summaries alone, without a shape, would not decide this.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q4/5

Why measure a difference in units of spread?

A. Units of spread show which group has more values in it.

B. Units of spread remove the need to compare the centers of the groups.

C. A gap matters only relative to how much values vary within each group.

D. Units of spread make every difference look larger than it really is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q5/5

221

QUESTION ID
S6-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. A gap matters only relative to how much values vary within each group.

GRADING NOTE The others are false claims about standardizing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSS-ID.A.3

FORWARD 2

Judge whether a difference supports a claim • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says that brand X bulbs "last much longer" than brand Y. Lifetimes of both brands are approximately normal with means 1,010 and 1,000 hours and the same standard deviation, 80 hours. Evaluate.

A. Weak: the claim would be supported only if the standard deviations were different.

B. Strong: both brands have the same standard deviation, so the difference in means must be large.

C. Weak: the means differ by about 0.13 standard deviations, so the lives overlap almost completely.

D. Strong: 1,010 is greater than 1,000, so every brand X bulb lasts longer than any brand Y bulb.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q1/5

223

QUESTION ID

S6-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Weak: the means differ by about 0.13 standard deviations, so the lives overlap almost completely.

GRADING NOTE 10/80 = 0.125 SD; for two normal distributions with equal spread, that gap means near-total overlap. The others overstate or misuse the spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q1/5

True or false?

A difference of 5 points between two group means is 2.5 standard deviations when each group's standard deviation is 2, but only 0.25 standard deviations when it is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q2/5

224

QUESTION ID
S6-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 5/2 and 5/20. How strong the evidence is would also depend on sample sizes and design, which are not given.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q2/5

After new training, a team's median race time dropped from 52 to 50 seconds; the IQR was 8 seconds before and after. Which statement correctly describes the size of the drop relative to the spread?

A. It is a quarter of an IQR (2/8), small relative to the athlete-to-athlete spread.

B. It cannot be measured, because the two IQRs are equal to each other.

C. It shows that the athletes slowed down by two seconds on average.

D. It is four IQRs (8/2), large relative to the athlete-to-athlete spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q3/5

225

QUESTION ID
S6-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. It is a quarter of an IQR (2/8), small relative to the athlete-to-athlete spread.

GRADING NOTE 2/8 = 0.25 IQR. Whether 2 seconds matters in a race is a separate judgment that the ratio does not settle. The others invert the ratio, read a lower time as slower, or treat equal IQRs as a problem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q3/5

Two groups have means 120 and 135 and the same standard deviation 10. How many standard deviations apart are they?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q4/5

226

QUESTION ID

S6-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 15/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

If two groups' means differ by less than one standard deviation, the groups cannot differ in any important way.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q5/5

227

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A small standardized difference can still matter in context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q5/5

Identify outliers and account for their effects

MAIN	Probability & Statistics main question	**HSS-ID.A.3**
BACKWARD 1	Find the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Compute the fences	**6.NS.B.3**
BACKWARD 3	How an extreme value affects the mean and the median	**6.SP.B.5.d**
FORWARD 1	Decide whether an outlier is an error or a real value	**HSS-ID.A.3**
FORWARD 2	Report resistant summaries after checking outliers	**HSS-ID.A.3**

SET 7 • HSS-ID.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q1/5

230

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

9.5; 21.5; 1

GRADING NOTE Form required: all three. Ordered 12, 13, 14, 14, 15, 15, 16, 16, 17, 18, 41; Q1 = 14, Q3 = 17, IQR 3; fences 14 − 4.5 = 9.5 and 17 + 4.5 = 21.5; only 41 is an outlier.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q1/5

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. The value 41 is an outlier. Find the mean with and without it, each to the nearest tenth.

Mean with 41: _____

Mean without 41: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q2/5

231

QUESTION ID
S7-M-Q2

TYPE
Short answer

CORRECT ANSWER
17.4; 15.0

GRADING NOTE Form required: both. 191/11 ≈ 17.36; 150/10 = 15.0. (The median is 15 both ways.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q2/5

Hours of screen time 11 students reported for one weekend: 6, 8, 7, 9, 7, 8, 30, 6, 7, 8, 1. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q3/5

232

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

3; 11; 2

GRADING NOTE Form required: all three. Ordered 1, 6, 6, 7, 7, 7, 8, 8, 8, 9, 30; Q1 = 6, Q3 = 8, IQR 2; fences 3 and 11; 1 and 30 are outliers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q3/5

Prices (dollars) of 10 used bikes: 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. The $600 bike is an outlier. Which describes the effect of removing it?

A. The mean and the median both drop by $40.

B. The median drops by $40, while the mean drops by only $2.50.

C. The mean drops by $40, while the median drops by only $2.50.

D. Neither the mean nor the median changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q4/5

233

QUESTION ID
S7-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. The mean drops by $40, while the median drops by only $2.50.

GRADING NOTE Mean 2400/10 = 240 becomes 1800/9 = 200; median (200 + 205)/2 = 202.5 becomes 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q4/5

Support calls received in each of 10 hours: 3, 5, 4, 6, 5, 4, 22, 5, 3, 4. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the upper fence and the number of outliers.

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q5/5

234

QUESTION ID

S7-M-Q5

TYPE

Short answer

CORRECT ANSWER

6.5; 1

GRADING NOTE Form required: both. Ordered 3, 3, 4, 4, 4, 5, 5, 5, 6, 22; Q1 = 4, Q3 = 5, IQR 1; fences 2.5 and 6.5; only 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q5/5

SET 7 • HSS-ID.A.3

BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Q1 = 12 and Q3 = 20. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q1/5

236

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE 20 − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q1/5

For 1, 3, 4, 6, 8, 9, find the IQR (Q1 and Q3 are the medians of the halves).

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q2/5

237

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Q1 = 3, Q3 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q2/5

True or false?

The IQR is the difference between the maximum and the minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q3/5

238

QUESTION ID

S7-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q3/5

A box plot has Q1 = 30 and Q3 = 42. The IQR is

A. 36

B. 42

C. 72

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q4/5

239

QUESTION ID

S7-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE 42 − 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q4/5

For 2, 4, 5, 7, 9, 10, 12 (odd count; leave out the median), find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q5/5

240

QUESTION ID

S7-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Q1 = 4, Q3 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSS-ID.A.3

BACKWARD 2

Compute the fences • Grade 6 • The Number System • 6.NS.B.3

Find 1.5 × 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q1/5

242

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 6 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 1.5 × 2.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q2/5

243

GRADING NOTE 2.5 + 1.25.

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q2/5

True or false?

If Q1 = 10 and the IQR is 4, the lower fence is 10 − 6 = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q3/5

244

QUESTION ID

S7-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.5 · 4 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q3/5

If Q3 = 25 and the IQR is 10, the upper fence is

A. 15

B. 35

C. 40

D. 37.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q4/5

245

QUESTION ID

S7-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 40

GRADING NOTE 25 + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q4/5

Q1 = 7.5 and the IQR is 3. Find the lower fence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q5/5

246

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 7.5 − 4.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSS-ID.A.3

BACKWARD 3

How an extreme value affects the mean and the median • Grade 6 • Statistics & Probability • 6.SP.B.5.d

In 3, 4, 5, 6, 50, which is pulled more by the value 50?

A. the mean

B. neither

C. the median

D. both equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q1/5

248

QUESTION ID

S7-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. the mean

GRADING NOTE Mean 13.6; median 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q1/5

True or false?

The median of 2, 3, 4, 5, 100 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q2/5

249

QUESTION ID

S7-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Middle value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 2, 3, 4, 5, 100.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q3/5

250

QUESTION ID

S7-B3-Q3

TYPE

Short answer

CORRECT ANSWER

22.8

GRADING NOTE 114/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q3/5

True or false?

Removing an extreme value always changes the median more than the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q4/5

251

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Usually the reverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q4/5

For skewed data with outliers, the better summary of a typical value is usually

A. the maximum

B. the range

C. the median

D. the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q5/5

252

QUESTION ID

S7-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. the median

GRADING NOTE Resistant to extremes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSS-ID.A.3

FORWARD 1

Decide whether an outlier is an error or a real value • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Heights of 9 adults in a study are listed in centimeters; one value is 17.5. Which is the most reasonable judgment?

A. Outliers are always errors, so every outlier in the data should be removed.

B. It is a real height, so it must be kept exactly as it is in the data.

C. It is probably a recording error (likely 175 cm), so check the record before using it.

D. It is an outlier, so it should be deleted without any further checking.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q1/5

254

QUESTION ID

S7-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. It is probably a recording error (likely 175 cm), so check the record before using it.

GRADING NOTE An adult height of 17.5 cm is impossible; an outlier is checked, not removed automatically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q1/5

True or false?

A value that the 1.5·IQR rule flags as an outlier must be removed from the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q2/5

255

QUESTION ID
S7-F1-Q2

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The rule flags values to examine; a legitimate unusual value stays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q2/5

Among 30 home prices, one mansion sells for 10 times the median. Which is correct?

A. It must be removed, because the outlier rule requires it to be deleted.

B. It is a legitimate unusual value; report resistant summaries such as the median and the IQR.

C. It is a recording error, because no real home could cost ten times the median price.

D. It should replace the median, because it is the most informative value in the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q3/5

256

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. It is a legitimate unusual value; report resistant summaries such as the median and the IQR.

GRADING NOTE Real but extreme values are kept; resistant summaries describe the typical home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q3/5

True or false?

An outlier can be a correct measurement of an unusual case.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q4/5

257

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outliers are not automatically errors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q4/5

A thermometer log shows 18, 19, 21, 20, 210, 19 °C in a classroom. What is the best next step?

A. Check the 210 reading, since a decimal point was likely lost (21.0), before computing summaries.

B. Report the mean including 210, because every reading is data and must be used as recorded.

C. Delete the 19s, because they are the most common values and do not add information.

D. Report the 210 reading as the median, because it is the most unusual value in the list.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q5/5

258

QUESTION ID
S7-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. Check the 210 reading, since a decimal point was likely lost (21.0), before computing summaries.

GRADING NOTE 210 °C in a classroom is impossible; check it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSS-ID.A.3

FORWARD 2

Report resistant summaries after checking outliers • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

After checking, the $600 bike price is real. Which summaries best describe a typical used bike and the spread?

A. the median and the IQR

B. the mean and the range

C. the maximum and the minimum

D. the mean and the standard deviation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q1/5

260

QUESTION ID
S7-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. the median and the IQR

GRADING NOTE Resistant to the outlier.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q1/5

After a recording error is corrected, the data are 12, 15, 14, 18, 16, 13, 17, 15, 14, 14, 16 (the 41 was really 14). Find the new mean to the nearest tenth and the new median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q2/5

261

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

14.9; 15

GRADING NOTE Form required: both. Sum 164/11 ≈ 14.9; ordered middle value 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q2/5

True or false?

In most large data sets, removing one extreme value changes the IQR less than it changes the standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q3/5

262

QUESTION ID
S7-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Quartiles depend on the middle of the data, so the IQR is usually less sensitive; it can still move when the removal shifts a quartile across a gap in the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q3/5

Which summary is most changed by one extreme value?

A. Q1

B. the standard deviation

C. the median

D. the IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q4/5

263

QUESTION ID
S7-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. the standard deviation

GRADING NOTE Squared distances make the SD very sensitive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q4/5

Data 5, 6, 6, 7, 8, 30: find the median and the IQR (Q1 and Q3 are the medians of the halves).

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q5/5

264

QUESTION ID
S7-F2-Q5

TYPE
Short answer

CORRECT ANSWER
6.5; 2

GRADING NOTE Form required: both. Q1 = 6, Q3 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q5/5

Estimate percentages with the 68-95-99.7 rule

MAIN	Probability & Statistics main question	**HSS-ID.A.4**
BACKWARD 1	Count standard deviations from the mean	**HSS-ID.A.2**
BACKWARD 2	Find a percent of a population	**6.RP.A.3.c**
BACKWARD 3	Use the symmetry of the normal curve	**6.SP.A.2**
FORWARD 1	Find a cutoff from a percent	**HSS-ID.A.4**
FORWARD 2	Compare positions in two populations with standardized values	**HSS-ID.A.4**

SET 8 • HSS-ID.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights of adult women in a population are approximately normal with mean 64 inches and standard deviation 2.5 inches. Use the 68-95-99.7 rule. What percent are between 61.5 and 66.5 inches, and what percent are between 59 and 69 inches?

61.5 to 66.5: _____

59 to 69: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q1/5

267

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

68%; 95%

GRADING NOTE Form required: both, as percents. ±1 SD and ±2 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q1/5

Battery lives are approximately normal with mean 40 hours and standard deviation 4 hours. Use the 68-95-99.7 rule. What percent of batteries last more than 48 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q2/5

268

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

2.5%

GRADING NOTE 48 is 2 SD above; (100 − 95)/2 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q2/5

Commute times are approximately normal with mean 30 minutes and standard deviation 5 minutes. Use the 68-95-99.7 rule. What percent are between 25 and 40 minutes?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q3/5

269

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

81.5%

GRADING NOTE 25 to 30 is 34%; 30 to 40 is 47.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q3/5

Egg masses are approximately normal with mean 58 g and standard deviation 3 g. Use the 68-95-99.7 rule. What percent are less than 52 g, and what percent are more than 61 g?

Less than 52 g: _____

More than 61 g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q4/5

270

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5%; 16%

GRADING NOTE Form required: both. 52 is 2 SD below: 2.5%. 61 is 1 SD above: (100 − 68)/2 = 16%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q4/5

Scores on a test are approximately normal with mean 500 and standard deviation 100. Use the 68-95-99.7 rule. What percent are between 300 and 600, and what percent are below 400?

300 to 600: _____

Below 400: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q5/5

271

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

81.5%; 16%

GRADING NOTE Form required: both. 47.5% + 34% = 81.5%; 400 is 1 SD below: 16%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSS-ID.A.4

BACKWARD 1

Count standard deviations from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The mean is 50 and the standard deviation is 5. How many standard deviations above the mean is 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q1/5

273

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE (60 − 50)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q1/5

The mean is 12 and the standard deviation is 3. How many standard deviations below the mean is 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q2/5

274

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE (12 − 9)/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q2/5

True or false?

With mean 100 and standard deviation 15, the value 70 is 2 standard deviations below the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q3/5

275

QUESTION ID

S8-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $100 - 2 \cdot 15 = 70$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q3/5

With mean 64 and standard deviation 2.5, which value is 1 standard deviation above the mean?

A. 62.5

B. 66.5

C. 65

D. 67

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q4/5

276

QUESTION ID

S8-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 66.5

GRADING NOTE 64 + 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q4/5

The mean is 30 and the standard deviation is 5. What value is 2 standard deviations below the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q5/5

277

QUESTION ID

S8-B1-Q5

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE 30 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

BACKWARD 2

Find a percent of a population • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 68% of 500?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

279

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

340

GRADING NOTE 0.68 · 500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Half of 5% is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

280

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5%

GRADING NOTE 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

True or false?

34% + 13.5% = 47.5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

281

QUESTION ID

S8-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Addition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

If 95% of values are inside an interval, what percent are outside it?

A. 5%

B. 95%

C. 0%

D. 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

282

QUESTION ID

S8-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5%

GRADING NOTE 100 − 95; 2.5% is one side only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 100% minus 68%, divided by 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

283

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

16%

GRADING NOTE 32/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSS-ID.A.4

BACKWARD 3

Use the symmetry of the normal curve • Grade 6 • Statistics & Probability • 6.SP.A.2

True or false?

In a normal distribution, half of the values are above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q1/5

285

QUESTION ID

S8-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetric about the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q1/5

True or false?

In a normal distribution, more values are between the mean and 1 SD above it than between the mean and 1 SD below it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q2/5

286

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Symmetry makes them equal (34% each).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent of a normal distribution lies between the mean and 2 standard deviations above it?

A. 95%

B. 50%

C. 47.5%

D. 34%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q3/5

287

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 47.5%

GRADING NOTE Half of 95%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent of a normal distribution lies between 1 SD below the mean and the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q4/5

288

QUESTION ID

S8-B3-Q4

TYPE

Short answer

CORRECT ANSWER

34%

GRADING NOTE Half of 68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q4/5

What percent of a normal distribution is more than 1 standard deviation below the mean?

A. 16%

B. 34%

C. 2.5%

D. 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q5/5

289

QUESTION ID

S8-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 16%

GRADING NOTE (100 − 68)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSS-ID.A.4

FORWARD 1

Find a cutoff from a percent • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights are approximately normal with mean 64 in and standard deviation 2.5 in. Use the 68-95-99.7 rule. Above what height are the tallest 2.5%?

Height (in): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q1/5

291

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

69

GRADING NOTE Mean + 2 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q1/5

Times are approximately normal with mean 30 min and standard deviation 5 min. Use the 68-95-99.7 rule. Below what time are the fastest 16%?

Time (min): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q2/5

292

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Mean − 1 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q2/5

True or false?

In a normal distribution with mean 200 and standard deviation 20, the middle 95% is from 160 to 240.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q3/5

293

QUESTION ID

S8-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ±2 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q3/5

Test scores are approximately normal with mean 70 and SD 8. Which score separates the top 16%?

A. 78

B. 74

C. 62

D. 86

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q4/5

294

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 78

GRADING NOTE Mean + 1 SD. 86 marks the top 2.5%; 62 the bottom 16%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lengths are approximately normal with mean 12 cm and standard deviation 0.5 cm. Use the 68-95-99.7 rule. Give the interval that holds the middle 68%.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q5/5

295

QUESTION ID

S8-F1-Q5

TYPE

Short answer

CORRECT ANSWER

11.5; 12.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

FORWARD 2

Compare positions in two populations with standardized values • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Ana scored 85 on a test with mean 75 and SD 5; Ben scored 90 on a test with mean 80 and SD 10. How many standard deviations above the mean is each score?

Ana: _____

Ben: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q1/5

297

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2; 1

GRADING NOTE Form required: both. 10/5 and 10/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q1/5

Using the scores above (Ana 2 SDs above, Ben 1 SD above, both tests normal), which is correct?

A. Ana's score is relatively higher: about 2.5% scored above her, but about 16% scored above Ben.

B. Ben's score is relatively higher, because 90 is a larger number than 85.

C. The scores are equally good, because both are 10 points above their means.

D. They cannot be compared, because the two tests had different means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q2/5

298

QUESTION ID

S8-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Ana's score is relatively higher: about 2.5% scored above her, but about 16% scored above Ben.

GRADING NOTE Standardized position decides. The others compare raw scores or raw gaps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

A value 1 SD below the mean in one normal population and a value 1 SD below the mean in another are at the same percentile.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q3/5

299

QUESTION ID
S8-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both have about 16% below them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q3/5

A 100-m time of 12.0 s comes from a population with mean 13.0 s and SD 0.5 s. How many standard deviations below the mean is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q4/5

300

QUESTION ID

S8-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE (13.0 − 12.0)/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q4/5

Two normal populations: in A, mean 50, SD 5; in B, mean 50, SD 10. Which value is more unusual?

A. they are equally unusual

B. 60 in population A

C. 55 in population B

D. 60 in population B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q5/5

301

QUESTION ID

S8-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 60 in population A

GRADING NOTE 60 is 2 SD above in A, 1 SD in B; 55 is 0.5 SD in B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q5/5

Estimate normal areas with a table or calculator

MAIN	Probability & Statistics main question	**HSS-ID.A.4**
BACKWARD 1	Compute a standardized value	**7.NS.A.3**
BACKWARD 2	Read areas as proportions	**7.SP.C.5**
BACKWARD 3	Find right tails and areas between from left areas	**HSS-ID.A.4**
FORWARD 1	Find the cutoff for a percentile	**HSS-ID.A.4**
FORWARD 2	Make decisions with normal areas	**HSS-ID.A.4**

SET 9 • HSS-ID.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are approximately normal with mean 100 and standard deviation 15. What proportion of people score below 118? Use the table values printed below (areas to the left of z). Give four decimal places.

z	1.0	1.2	1.5	2.0
Area to the left	0.8413	0.8849	0.9332	0.9772

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q1/5

304

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.8849

GRADING NOTE $z = (118 - 100)/15 = 1.2$; area to the left 0.8849.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q1/5

Lengths of calls are approximately normal with mean 50 seconds and standard deviation 8 seconds. What proportion of calls last more than 60 seconds? Use the table values printed below (areas to the left of z). Give four decimal places.

z	1.0	1.25	1.5	2.0
Area to the left	0.8413	0.8944	0.9332	0.9772

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q2/5

305

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.1056

GRADING NOTE $z = 10/8 = 1.25$; $1 - 0.8944 = 0.1056$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q2/5

Daily water use is approximately normal with mean 20 gallons and standard deviation 2.5 gallons. What proportion of days use between 18 and 23 gallons? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−1.0	−0.8	0.8	1.2
Area to the left	0.1587	0.2119	0.7881	0.8849

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q3/5

306

QUESTION ID

S9-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.6730

GRADING NOTE $z = -0.8$ and 1.2; $0.8849 - 0.2119 = 0.6730$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q3/5

Grade-point averages at a school are approximately normal with mean 3.0 and standard deviation 0.4. What proportion are below 2.5? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−1.5	−1.25	−1.0	−0.5
Area to the left	0.0668	0.1056	0.1587	0.3085

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q4/5

307

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.1056

GRADING NOTE z = −0.5/0.4 = −1.25; area 0.1056.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q4/5

Heights of adult men are approximately normal with mean 175 cm and standard deviation 7 cm. What proportion are between 168 and 189 cm? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−2.0	−1.0	1.0	2.0
Area to the left	0.0228	0.1587	0.8413	0.9772

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q5/5

308

QUESTION ID

S9-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.8185

GRADING NOTE $z = -1$ and 2; $0.9772 - 0.1587 = 0.8185$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q5/5

SET 9 • HSS-ID.A.4

BACKWARD 1

Compute a standardized value • Grade 7 • The Number System • 7.NS.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compute (130 − 100)/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q1/5

310

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 30/15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q1/5

Compute (42 − 50)/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q2/5

311

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE −8/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q2/5

True or false?

$(9.5 - 10)/0.5 = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q3/5

312

QUESTION ID

S9-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-0.5/0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q3/5

Compute (61 − 58)/2.

A. 0.67

B. 3

C. 1.5

D. −1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q4/5

313

QUESTION ID

S9-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5

GRADING NOTE 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (2.2 − 3.0)/0.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q5/5

314

QUESTION ID

S9-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE −0.8/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSS-ID.A.4

BACKWARD 2

Read areas as proportions • Grade 7 • Statistics & Probability • 7.SP.C.5

True or false?

An area of 0.8413 to the left of z = 1 means about 84% of the population is below 1 SD above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q1/5

316

QUESTION ID

S9-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Area is proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q1/5

The table gives area 0.0228 to the left of $z = -2$. What does this mean?

A. Exactly 2 values are below the mean.

B. About 22.8% of values are below the mean.

C. About 2.28% of values are above the mean.

D. About 2.28% of values are below 2 SDs below the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q2/5

317

QUESTION ID

S9-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. About 2.28% of values are below 2 SDs below the mean.

GRADING NOTE Read as a proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q2/5

Write 0.1587 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q3/5

318

QUESTION ID

S9-B2-Q3

TYPE

Short answer

CORRECT ANSWER

15.87%

GRADING NOTE × 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q3/5

True or false?

A proportion of a population can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q4/5

319

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Proportions are between 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the largest proportion?

A. 0.5

B. 0.8413

C. 0.9772

D. 0.0228

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q5/5

320

QUESTION ID

S9-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 0.9772

GRADING NOTE Compare decimals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSS-ID.A.4

BACKWARD 3

Find right tails and areas between from left areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The area to the left of z = 1.5 is 0.9332. What is the area to the right of z = 1.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q1/5

322

QUESTION ID
S9-B3-Q1

TYPE
Short answer

CORRECT ANSWER
0.0668

GRADING NOTE 1 − 0.9332.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q1/5

Areas to the left: $z = -1$ gives 0.1587; $z = 1$ gives 0.8413. Find the area between $z = -1$ and $z = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q2/5

323

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0.6826

GRADING NOTE 0.8413 − 0.1587.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q2/5

True or false?

The area to the right of z = 0 is 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q3/5

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q3/5

The area to the left of z = 0.5 is 0.6915. The area between z = 0 and z = 0.5 is

A. 0.3085

B. 0.6915

C. 0.1915

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q4/5

325

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.1915

GRADING NOTE 0.6915 − 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q4/5

The area to the left of $z = 2$ is 0.9772. What is the area to the left of $z = -2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q5/5

326

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.0228

GRADING NOTE By symmetry it equals the area to the right of $z = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSS-ID.A.4

FORWARD 1

Find the cutoff for a percentile • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are normal with mean 100 and SD 15. The table gives area 0.9332 to the left of z = 1.5. What score is at the 93.32nd percentile?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q1/5

328

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

122.5

GRADING NOTE 100 + 1.5·15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q1/5

Call lengths are normal with mean 50 s and SD 8 s. The table gives area 0.1056 to the left of z = −1.25. What length is at the 10.56th percentile?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q2/5

329

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

40

GRADING NOTE 50 − 1.25·8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q2/5

True or false?

A value with z = 0 is at the 50th percentile of a normal distribution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q3/5

330

QUESTION ID

S9-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q3/5

A factory wants only the longest 2.28% of rods to be rejected. Rod lengths are normal with mean 10 cm and SD 0.2 cm; the table gives area 0.9772 to the left of $z = 2$. What cutoff should it use?

A. 10.2 cm

B. 12 cm

C. 10.4 cm

D. 9.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q4/5

331

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10.4 cm

GRADING NOTE 10 + 2·0.2. 9.6 is the lower cutoff; 12 uses z as centimeters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q4/5

Heights are normal with mean 175 cm and SD 7 cm; area 0.8413 is to the left of z = 1. What height separates the tallest 15.87%?

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q5/5

332

QUESTION ID

S9-F1-Q5

TYPE

Short answer

CORRECT ANSWER

182

GRADING NOTE 175 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSS-ID.A.4

FORWARD 2

Make decisions with normal areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A bolt fits if its diameter is between 9.8 and 10.2 mm. Diameters are normal with mean 10.0 mm and SD 0.1 mm. Table: area 0.9772 left of z = 2 and 0.0228 left of z = −2. What proportion of bolts fit?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q1/5

334

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.9544

GRADING NOTE 0.9772 − 0.0228.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q1/5

Fill amounts from both machines are approximately normal. Machine A fills bottles with mean 500 mL, SD 4 mL; machine B with mean 502 mL, SD 1 mL. A bottle is underfilled below 498 mL. Table: area 0.3085 to the left of z = −0.5; the area to the left of z = −4 is less than 0.0001. Which machine underfills fewer bottles?

A. A, because its standard deviation is larger

B. B: under 0.01% versus about 31% for A

C. A, because its mean is closer to 498 mL

D. They underfill the same proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q2/5

335

QUESTION ID

S9-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. B: under 0.01% versus about 31% for A

GRADING NOTE A: z = (498 − 500)/4 = −0.5, area 0.3085. B: z = (498 − 502)/1 = −4, area very close to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q2/5

True or false?

A process makes items whose lengths are approximately normal. If its standard deviation is reduced while its mean stays the same and the process stays normal, the proportion of items more than 1 mm from the mean goes down.

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q3/5

336

QUESTION ID
S9-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The same 1 mm cutoff is now more standard deviations away, so each normal tail is smaller. For other shapes a smaller SD alone would not guarantee this.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q3/5

Scores are normal with mean 70 and SD 10. A prize goes to scores of 85 or more. Table: area 0.9332 left of $z = 1.5$. What proportion of students win?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q4/5

337

QUESTION ID

S9-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.0668

GRADING NOTE $1 - 0.9332$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q4/5

Two normal processes have the same mean. Process 1 has SD 2; process 2 has SD 5. Which has more items within 4 units of the mean?

A. process 2

B. the same

C. cannot tell

D. process 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q5/5

338

QUESTION ID

S9-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. process 1

GRADING NOTE 4 units is 2 SDs for process 1 (95%) and 0.8 SD for process 2 (about 58%).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q5/5

Recognize when a normal model is not appropriate

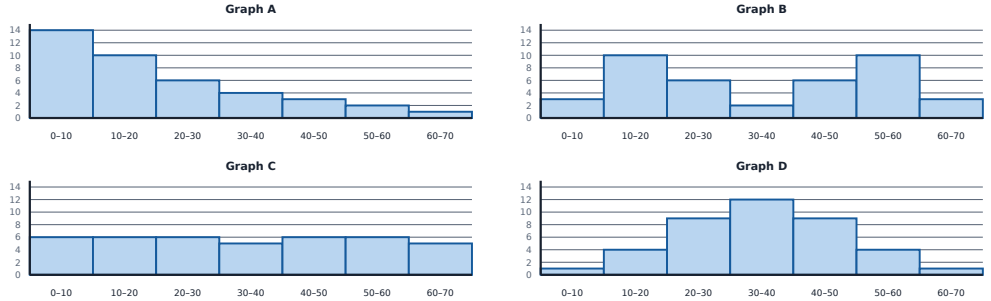
MAIN	Probability & Statistics main question	**HSS-ID.A.4**
BACKWARD 1	Describe the shape of a distribution	**6.SP.A.2**
BACKWARD 2	Relate the mean and median to the shape	**6.SP.B.5.d**
BACKWARD 3	Test a normal model against the data	**HSS-ID.A.4**
FORWARD 1	Compare actual percents with the 68-95-99.7 rule	**HSS-ID.A.4**
FORWARD 2	Choose summaries for data that are not normal	**HSS-ID.A.2**

SET 10 • HSS-ID.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Each histogram shows 40 measurements on the same scale. For which data set is a normal model most reasonable? In each histogram, the horizontal axis shows value and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q1/5

341

QUESTION ID

S10-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE The single symmetric mound fits a normal model. The others are skewed right, have two peaks, or are flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q1/5

Waiting times at a clinic can never be negative. Their mean is 3 minutes and their standard deviation is 4 minutes. If a normal model with this mean and standard deviation were used, what proportion of waiting times would it place below 0 minutes? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−1.0	−0.75	−0.5	0.5
Area to the left	0.1587	0.2266	0.3085	0.6915

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q2/5

342

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.2266

GRADING NOTE $z = (0 - 3)/4 = -0.75$; area 0.2266.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q2/5

Waiting times can never be negative, yet a normal model with their mean (3 minutes) and standard deviation (4 minutes) places about 23% of times below 0. What does this show?

A. The times are normal, but 23% of them were recorded wrong.

B. The standard deviation must have been computed incorrectly.

C. A normal model fits, because the mean is a positive number.

D. A normal model is not appropriate for these waiting times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q3/5

343

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A normal model is not appropriate for these waiting times.

GRADING NOTE A model that predicts impossible values for a large share of the data does not fit. A large SD relative to the mean is typical of right-skewed, bounded data; nothing says it was computed wrongly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q3/5

Which data set is least likely to be approximately normal?

A. the masses of eggs from one large farm

B. repeated careful measurements of one object's length

C. the yearly incomes of all households in a large city

D. the heights of adult women in one country

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q4/5

344

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the yearly incomes of all households in a large city

GRADING NOTE Incomes are strongly skewed right (a few very large values). The others are classic near-normal examples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q4/5

A box plot of 60 values shows minimum 2, Q1 4, median 5, Q3 9 and maximum 30. Which conclusion is correct?



A. The data are roughly symmetric, so a normal model fits well.

B. The data are skewed left, so a normal model is not a good fit.

C. The data are skewed right, so a normal model is not a good fit.

D. A median of 5 shows that the data must be normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q5/5

345

QUESTION ID

S10-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The data are skewed right, so a normal model is not a good fit.

GRADING NOTE The upper whisker (21) and upper box half (4) are far longer than the lower ones (2 and 1): skewed right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q5/5

SET 10 • HSS-ID.A.4

BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

True or false?

A histogram with a long tail to the right is called skewed right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q1/5

347

QUESTION ID

S10-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The tail names the skew.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q1/5

What is the shape of the data 1, 1, 2, 2, 2, 3, 3, 4, 6, 9?

A. skewed right

B. skewed left

C. symmetric

D. uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q2/5

348

QUESTION ID

S10-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. skewed right

GRADING NOTE Most values are low, with a tail toward 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

A distribution with two separate peaks is called unimodal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q3/5

349

QUESTION ID

S10-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two peaks: bimodal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which list of values is roughly symmetric?

A. 1, 10, 18, 19, 19, 20

B. 3, 3, 3, 3, 3, 30

C. 1, 1, 1, 2, 9, 20

D. 2, 4, 5, 5, 6, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q4/5

350

QUESTION ID

S10-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2, 4, 5, 5, 6, 8

GRADING NOTE Centered at 5 with equal spread on each side. The others have a long tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q4/5

For the data 4, 5, 5, 6, 6, 6, 7, 7, 8, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q5/5

351

QUESTION ID

S10-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6; 6

GRADING NOTE Form required: both. Symmetric data: mean = median = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSS-ID.A.4

BACKWARD 2

Relate the mean and median to the shape • Grade 6 • Statistics & Probability • 6.SP.B.5.d

True or false?

In a distribution skewed right, the mean is usually greater than the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q1/5

353

QUESTION ID

S10-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The tail pulls the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q1/5

A data set has mean 52 and median 40. Which shape is most likely?

A. skewed right

B. skewed left

C. uniform

D. symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q2/5

354

QUESTION ID

S10-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. skewed right

GRADING NOTE Mean well above the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For the data 1, 2, 2, 3, 22, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q3/5

355

QUESTION ID

S10-B2-Q3

TYPE

Short answer

CORRECT ANSWER

6; 2

GRADING NOTE Form required: both. 30/5 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q3/5

True or false?

For strongly skewed data, the mean and standard deviation are the best summaries of center and spread.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q4/5

356

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median and IQR resist the tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For data skewed right, which measure best describes a typical value?

A. median

B. mean

C. midrange

D. maximum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q5/5

357

QUESTION ID

S10-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. median

GRADING NOTE The median is resistant to the tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSS-ID.A.4

BACKWARD 3

Test a normal model against the data • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Counts of errors per page (never negative) have mean 8 and standard deviation 4. How many standard deviations below the mean is 0, and what percent would a normal model place below 0 by the 68-95-99.7 rule?

Standard deviations: _____

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q1/5

359

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2; 2.5%

GRADING NOTE Form required: both. (8 − 0)/4 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B3-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A data set with any mean and any standard deviation is approximately normal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q2/5

360

QUESTION ID

S10-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every data set has a mean and SD; the shape decides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B3-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q2/5

Of 20 values with mean 50 and standard deviation 10, 19 lie between 40 and 60. Is this consistent with a normal model?

A. Yes: a normal model predicts about 95% within 1 SD.

B. Yes: every value is within 3 SDs of the mean.

C. No: a normal model predicts about 68% within 1 SD, but 95% are.

D. No: a normal model predicts all values within 1 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q3/5

361

QUESTION ID

S10-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. No: a normal model predicts about 68% within 1 SD, but 95% are.

GRADING NOTE 19/20 = 95% within 1 SD, far above 68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q3/5

A normal model is used for 200 values. About how many values does it predict within 2 standard deviations of the mean?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q4/5

362

QUESTION ID
S10-B3-Q4

TYPE
Short answer

CORRECT ANSWER
190

GRADING NOTE 0.95 · 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q4/5

True or false?

A normal model with mean 3 and SD 4 places about 16% of its values below −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q5/5

363

QUESTION ID

S10-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −1 is 1 SD below the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSS-ID.A.4

FORWARD 1

Compare actual percents with the 68-95-99.7 rule • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Of 40 values, 27 lie within 1 standard deviation of the mean and 38 lie within 2. What percent is each?

Within 1 SD: _____

Within 2 SD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q1/5

365

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

67.5%; 95%

GRADING NOTE Form required: both. 27/40 and 38/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q1/5

Of 40 values, 67.5% are within 1 SD of the mean and 95% within 2 SDs, and a histogram is one symmetric mound. Which conclusion is reasonable?

A. A normal model is not reasonable, because 67.5% is not 68%.

B. The data must be skewed right.

C. A normal model is reasonable for these data.

D. No conclusion is possible without the raw values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q2/5

366

QUESTION ID

S10-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. A normal model is reasonable for these data.

GRADING NOTE Percents match the rule closely and the shape fits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q2/5

True or false?

If 92% of a data set lies within 1 SD of its mean, the data fit a normal model well.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q3/5

367

QUESTION ID

S10-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A normal model gives about 68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q3/5

Of 50 values, 46 lie within 1 standard deviation of the mean. What percent is that?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q4/5

368

QUESTION ID

S10-F1-Q4

TYPE

Short answer

CORRECT ANSWER

92%

GRADING NOTE 46/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q4/5

Which result most strongly suggests a data set is not normal?

A. About half of the values lie above the mean.

B. 96% of the values lie within 2 SDs of the mean.

C. 92% of the values lie within 1 SD of the mean.

D. 70% of the values lie within 1 SD of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q5/5

369

QUESTION ID

S10-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 92% of the values lie within 1 SD of the mean.

GRADING NOTE 92% is far from 68%; the others are close to the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSS-ID.A.4

FORWARD 2

Choose summaries for data that are not normal • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Home prices in a town are strongly skewed right. Which pair of summaries is most appropriate?

A. mode and maximum

B. mean and range

C. mean and standard deviation

D. median and IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q1/5

371

QUESTION ID

S10-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. median and IQR

GRADING NOTE Resistant summaries for skewed data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Home prices (thousands of dollars) are 150, 160, 170, 180, 190, 200, 900. Find the median and the IQR. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q2/5

372

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

180; 40

GRADING NOTE Form required: both. Q1 = 160, Q3 = 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Using a normal model for strongly skewed data can give misleading percentages.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q3/5

373

QUESTION ID
S10-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The model's tails do not match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q3/5

For the prices 150, 160, 170, 180, 190, 200, 900 (thousands), which statement is correct?

A. The value 900 pulls the median above the mean.

B. The mean and the median are equal.

C. The value 900 pulls the mean above the median.

D. The value 900 changes the IQR more than the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q4/5

374

QUESTION ID

S10-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The value 900 pulls the mean above the median.

GRADING NOTE Mean = 1950/7 ≈ 278.6 > 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q4/5

True or false?

The standard deviation is resistant to one extreme value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q5/5

375

QUESTION ID
S10-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It uses every deviation, squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q5/5

Joint and marginal frequencies in two-way tables

MAIN	Probability & Statistics main question	**HSS-ID.B.5**
BACKWARD 1	Complete and read a two-way frequency table	**8.SP.A.4**
BACKWARD 2	Write a part of a whole as a fraction, decimal and percent	**7.NS.A.2.d**
FORWARD 1	Build a joint relative-frequency table	**HSS-ID.B.5**
FORWARD 2	Look for an association by comparing rows	**HSS-ID.B.5**

SET 11 • HSS-ID.B.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were asked how they prefer to take a class. What fraction of all students are 10th graders who chose online? Give a fraction in lowest terms.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q1/5

378

QUESTION ID
S11-M-Q1

TYPE
Short answer

CORRECT ANSWER
9/32

GRADING NOTE 45/160 = 9/32 (joint).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q1/5

160 students were asked how they prefer to take a class. What is the marginal relative frequency of choosing online? Give a fraction in lowest terms.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q2/5

379

QUESTION ID
S11-M-Q2

TYPE
Short answer

CORRECT ANSWER
15/32

GRADING NOTE 75/160 = 15/32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q2/5

A survey of 120 students: 50 are 7th graders and 70 are 8th graders. 18 of the 7th graders play an instrument, and 20 of the 8th graders do not. How many 7th graders do not play, and how many 8th graders do play?

7th graders who do not play: _____

8th graders who play: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q3/5

380

QUESTION ID
S11-M-Q3

TYPE
Short answer

CORRECT ANSWER
32; 50

GRADING NOTE Form required: both. 50 − 18 and 70 − 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q3/5

160 students were surveyed (table). Which statement gives a marginal relative frequency?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

A. 28.125% of all students are 10th graders who chose online.

B. 53.125% of all students chose in person.

C. 56.25% of the 10th graders chose online.

D. 37.5% of the 9th graders chose online.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q4/5

381

QUESTION ID
S11-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 53.125% of all students chose in person.

GRADING NOTE 85/160 uses a total over the grand total. The others are a joint frequency and two conditional ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q4/5

A dealer has 100 cars in two lots (table). What percent of all the cars are gray, and what percent of all the cars are red cars in lot B?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Gray: _____

Red in lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q5/5

382

QUESTION ID

S11-M-Q5

TYPE

Short answer

CORRECT ANSWER

40%; 8%

GRADING NOTE Form required: both. 40/100 and 8/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-M-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q5/5

SET 11 • HSS-ID.B.5

BACKWARD 1

Complete and read a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

120 adults were asked whether they own a dog and whether they walk daily (table). How many walk daily?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q1/5

384

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

51

GRADING NOTE 36 + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q1/5

From the dog-and-walking table, how many adults have no dog and do not walk daily?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q2/5

385

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

GRADING NOTE Read the cell.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q2/5

Of 50 students, 20 are in band and 12 of the band members are 9th graders. 30 of the 50 students are 9th graders. How many 9th graders are not in band?

A. 30

B. 8

C. 18

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q3/5

386

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 18

GRADING NOTE 30 − 12. 8 is band members not in 9th grade.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q3/5

True or false?

In a two-way table, the row totals add up to the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q4/5

387

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every count is in one row.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q4/5

True or false?

In a two-way table, a cell count can be greater than its row total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q5/5

388

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Cells add to the row total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSS-ID.B.5

BACKWARD 2

Write a part of a whole as a fraction, decimal and percent • Grade 7 • The Number System • 7.NS.A.2.d

ANSWER KEY Reduced copy of the question page, with its answer

Write 45/160 as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q1/5

390

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.28125

GRADING NOTE 45 ÷ 160.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 24/120 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q2/5

391

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20%

GRADING NOTE 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q2/5

True or false?

3/8 = 0.375.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q3/5

392

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3 ÷ 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q3/5

Write 15/32 as a percent.

A. 15.32%

B. 47%

C. 2.13%

D. 46.875%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q4/5

393

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 46.875%

GRADING NOTE 15 ÷ 32 = 0.46875.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 30/160 in lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q5/5

394

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/16

GRADING NOTE Divide by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSS-ID.B.5

FORWARD 1

Build a joint relative-frequency table • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

Convert the online-class table to relative frequencies of the grand total (160). What are the joint relative frequencies for 9th grade online and 9th grade in person? Give decimals.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

9th online: _____

9th in person: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q1/5

396

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.1875; 0.3125

GRADING NOTE Form required: both. 30/160 and 50/160.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q1/5

True or false?

In a table of joint relative frequencies of the grand total, all the cells add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q2/5

397

QUESTION ID

S11-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every individual is in one cell.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q2/5

In a joint relative-frequency table, the 9th-grade cells are 0.1875 and 0.3125. What is the 9th-grade marginal relative frequency?

A. 0.5

B. 0.1875

C. 0.6

D. 0.125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q3/5

398

QUESTION ID
S11-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. 0.5

GRADING NOTE Add the row.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q3**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the car table (100 cars), what are the joint relative frequencies for white cars in lot A and gray cars in lot B? Give decimals.

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

White, lot A: _____

Gray, lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q4/5

399

QUESTION ID

S11-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.18; 0.3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q4/5

True or false?

A marginal relative frequency is always smaller than every joint relative frequency in its row.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q5/5

400

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the sum of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSS-ID.B.5

FORWARD 2

Look for an association by comparing rows • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the online-class table, what percent of 9th graders and what percent of 10th graders chose online?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

9th: _____

10th: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q1/5

402

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

37.5%; 56.25%

GRADING NOTE Form required: both. 30/80 and 45/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q1/5

If grade and choice of class were not associated, what would be true?

A. Every cell of the table would contain the same count.

B. The percent choosing online would be about the same in each grade.

C. The number choosing online would be 0 in one grade.

D. The grade totals would have to be unequal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q2/5

403

QUESTION ID
S11-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. The percent choosing online would be about the same in each grade.

GRADING NOTE No association: equal conditional rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q2/5

True or false?

Because more 10th graders (45) than 9th graders (30) chose online, the counts alone prove an association, even if the grades had very different sizes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q3/5

404

QUESTION ID
S11-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Compare rates, not counts, when totals differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q3/5

In the car table, what percent of lot A cars and what percent of lot B cars are gray?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Lot A: _____

Lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q4/5

405

QUESTION ID

S11-F2-Q4

TYPE

Short answer

CORRECT ANSWER

25%; 50%

GRADING NOTE Form required: both. 10/40 and 30/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q4/5

Gray makes up 25% of lot A and 50% of lot B. Which is correct?

A. Lot B has exactly twice as many cars as lot A.

B. Gray is the most common color in lot A.

C. Lot and color are associated: gray is more common in lot B.

D. Lot and color are not associated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q5/5

406

QUESTION ID
S11-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Lot and color are associated: gray is more common in lot B.

GRADING NOTE Rates differ. Lot B has 60 cars, A has 40; white (18) is most common in A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q5/5

Conditional relative frequencies and association

MAIN	Probability & Statistics main question	**HSS-ID.B.5**
BACKWARD 1	Choose the denominator for a conditional relative frequency	**8.SP.A.4**
BACKWARD 2	Find a percent of a group	**6.RP.A.3.c**
BACKWARD 3	Compare two fractions	**4.NF.A.2**
FORWARD 1	Distinguish the two directions of a conditional rate	**HSS-ID.B.5**
FORWARD 2	Recognize when combining groups reverses a comparison	**HSS-ID.B.5**

SET 12 • HSS-ID.B.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were surveyed (table). What fraction of the 9th graders chose online, and what fraction of the 10th graders chose online? Give fractions in lowest terms.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

9th graders: _____

10th graders: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q1/5

409

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

3/8; 9/16

GRADING NOTE Form required: both. 30/80 and 45/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q1/5

In the survey, 37.5% of 9th graders and 56.25% of 10th graders chose online. Which statement is correct?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

A. 9th graders chose online at a higher rate, so grade and choice are associated.

B. The rates are equal, so grade and choice are not associated.

C. 10th graders chose online at a higher rate, so grade and choice are associated.

D. Each grade has 80 students, so grade and choice cannot be associated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S12-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q2/5

410

QUESTION ID

S12-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 10th graders chose online at a higher rate, so grade and choice are associated.

GRADING NOTE Compare the conditional rates. Equal group sizes say nothing about association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S12-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q2/5

160 students were surveyed (table). Of the students who chose online, what percent were 10th graders?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q3/5

411

QUESTION ID
S12-M-Q3

TYPE
Short answer

CORRECT ANSWER
60%

GRADING NOTE 45/75 (the online column).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q3/5

120 adults were surveyed (table). What percent of dog owners walk daily, and what percent of adults without a dog walk daily?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

Dog owners: _____

No dog: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q4/5

412

QUESTION ID
S12-M-Q4

TYPE
Short answer

CORRECT ANSWER
60%; 25%

GRADING NOTE Form required: both. 36/60 and 15/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q4/5

In the survey, 60% of dog owners and 25% of adults without a dog walk daily. Which conclusion is supported?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

A. Dog ownership and walking daily are not associated.

B. Owning a dog causes people to walk daily.

C. Owning a dog and walking daily are associated in this sample.

D. Adults without a dog walk daily more often.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q5/5

413

QUESTION ID
S12-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Owning a dog and walking daily are associated in this sample.

GRADING NOTE Different rates show association; a survey cannot show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q5/5

SET 12 • HSS-ID.B.5

BACKWARD 1

Choose the denominator for a conditional relative frequency • Grade 8 • Statistics & Probability • 8.SP.A.4

Of the 60 dog owners, 36 walk daily. What fraction of dog owners walk daily? Give lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q1/5

415

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE 36/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q1/5

True or false?

To find the percent of 9th graders who chose online, divide the 9th-grade online count by the grand total of all students.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q2/5

416

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by the 9th-grade total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q2/5

In the online-class table, which number is the denominator for the percent of online-choosers who are 9th graders?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

A. 80, the 9th-grade total

B. 75, the online total

C. 30, the cell count

D. 160, the grand total

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q3/5

417

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 75, the online total

GRADING NOTE "Of online-choosers" names the column.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 30 out of 80 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q4/5

418

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

37.5%

GRADING NOTE 30/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q4/5

True or false?

A row relative frequency uses that row's total as the denominator.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q5/5

419

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSS-ID.B.5

BACKWARD 2

Find a percent of a group • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

45 is what percent of 80?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q1/5

421

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

56.25%

GRADING NOTE 45/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q1/5

What is 25% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q2/5

422

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE 0.25 · 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

18 out of 24 is what percent?

A. 18%

B. 75%

C. 133%

D. 24%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q3/5

423

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 75%

GRADING NOTE 18/24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q3/5

True or false?

9/16 = 56.25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q4/5

424

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 ÷ 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q4/5

True or false?

45 is 60% of 80.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q5/5

425

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 45/80 = 56.25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSS-ID.B.5

BACKWARD 3

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is larger, 3/8 or 9/16?

A. 3/8

B. they cannot be compared

C. they are equal

D. 9/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q1/5

427

QUESTION ID

S12-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. 9/16

GRADING NOTE 3/8 = 6/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q1/5

True or false?

3/5 > 1/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q2/5

428

QUESTION ID

S12-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0.6 > 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q2/5

Write 3/8 as an equivalent fraction with denominator 16. What is the numerator?

Numerator: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q3/5

429

QUESTION ID

S12-B3-Q3

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Multiply by 2/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q3/5

True or false?

2/5 > 3/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q4/5

430

QUESTION ID

S12-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.4 < 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Order 1/4, 3/5 and 3/8 from least to greatest.

A. 1/4, 3/5, 3/8

B. 3/8, 1/4, 3/5

C. 1/4, 3/8, 3/5

D. 3/5, 3/8, 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q5/5

431

QUESTION ID

S12-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4, 3/8, 3/5

GRADING NOTE 0.25, 0.375, 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSS-ID.B.5

FORWARD 1

Distinguish the two directions of a conditional rate • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In the car table, what percent of the gray cars are in lot B, and what percent of the lot B cars are gray?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Gray cars in lot B: _____

Lot B cars that are gray: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q1/5

433

QUESTION ID
S12-F1-Q1

TYPE
Short answer

CORRECT ANSWER
75%; 50%

GRADING NOTE Form required: both. 30/40 and 30/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q1/5

True or false?

The percent of lot B cars that are gray equals the percent of gray cars that are in lot B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q2/5

434

QUESTION ID

S12-F1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 50% versus 75%: different denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q2/5

In the car table, what percent of the white cars are in lot A?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

A. 30%

B. 45%

C. 18%

D. 40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q3/5

435

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 45%

GRADING NOTE 18/40. 18% is of all cars; 40% is lot A's share; 30% is 12/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q3/5

In the car table, what percent of the lot A cars are red?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q4/5

436

QUESTION ID
S12-F1-Q4

TYPE
Short answer

CORRECT ANSWER
30%

GRADING NOTE 12/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q4/5

Which statement uses a column total of the car table as its denominator?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

A. 12% of all cars are red cars in lot A.

B. Of the lot A cars, 30% are red.

C. Of the red cars, 60% are in lot A.

D. 40% of all cars are in lot A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q5/5

437

QUESTION ID
S12-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Of the red cars, 60% are in lot A.

GRADING NOTE 12/20. The others divide by a row total or the grand total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSS-ID.B.5

FORWARD 2

Recognize when combining groups reverses a comparison • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Use the hospital table. What is the survival rate for mild cases at hospital A and at hospital B?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q1/5

439

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

95%; 90%

GRADING NOTE Form required: both. 95/100 and 360/400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the hospital table. What is the survival rate for severe cases at hospital A and at hospital B?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q2/5

440

QUESTION ID
S12-F2-Q2

TYPE
Short answer

CORRECT ANSWER
30%; 25%

GRADING NOTE Form required: both. 120/400 and 25/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the hospital table. What is the overall survival rate (all 500 patients) at hospital A and at hospital B?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q3/5

441

QUESTION ID

S12-F2-Q3

TYPE

Short answer

CORRECT ANSWER

43%; 77%

GRADING NOTE Form required: both. 215/500 and 385/500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q3/5

Hospital A has the higher survival rate for mild cases and for severe cases, but the lower rate overall. Why?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A. The two hospitals treated different total numbers of patients.

B. The overall rates were computed with the wrong denominators.

C. A treats many more mild cases, and mild cases survive less often.

D. A treats many more severe cases, and severe cases survive less often.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q4/5

442

QUESTION ID

S12-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A treats many more severe cases, and severe cases survive less often.

GRADING NOTE A: 400 of 500 severe; B: 100 of 500. Both treated 500 patients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q4/5

True or false?

If one group has the higher rate within every category, it must have the higher rate overall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q5/5

443

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hospital data are a counterexample (Simpson's paradox).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q5/5

Represent two quantitative variables on a scatter plot

MAIN	Probability & Statistics main question	**HSS-ID.B.6**
BACKWARD 1	Plot and read points in the first quadrant	**5.G.A.2**
BACKWARD 2	Describe patterns of association	**8.SP.A.1**
FORWARD 1	Compare the direction, form and strength of associations	**HSS-ID.B.6**
FORWARD 2	Distinguish association from causation	**HSS-ID.C.9**

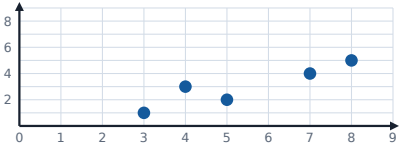
SET 13 • HSS-ID.B.6

MAIN

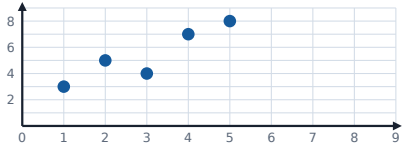
Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

A seedling was measured weekly: (week, height in cm) = (1, 3), (2, 5), (3, 4), (4, 7), (5, 8). Which scatter plot shows the data?

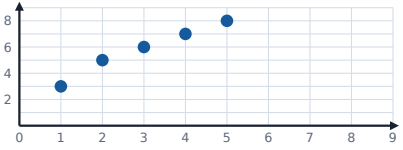
A



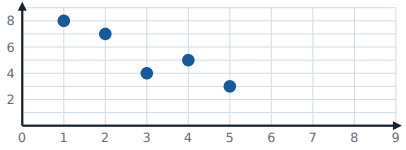
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q1/5

446

QUESTION ID

S13-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE The others swap the coordinates, move the week-3 point to 6, or reverse the heights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q1/5

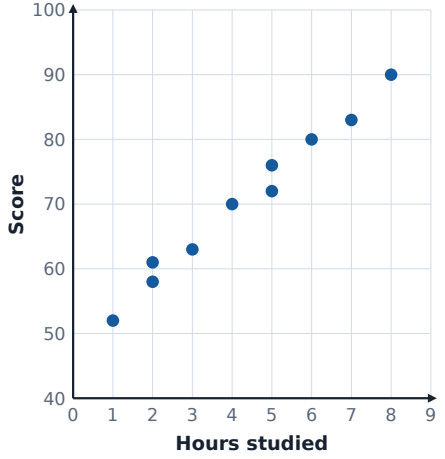
The scatter plot shows hours studied and test scores. Which describes the association?

A. curved, not linear

B. no clear association

C. strong, positive, linear

D. strong, negative, linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q2/5

447

QUESTION ID

S13-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. strong, positive, linear

GRADING NOTE Points rise in a narrow band along a line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q2/5

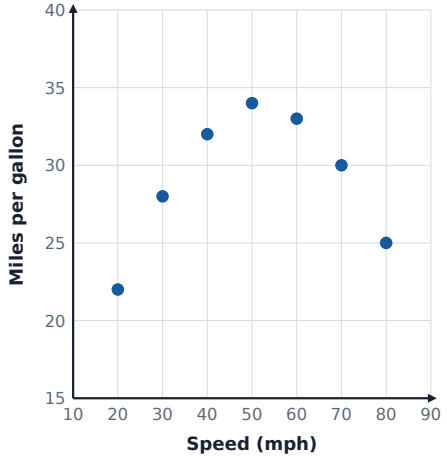
The scatter plot shows a car's speed and fuel efficiency. Which describes the relationship?

A. curved: rises, then falls

B. no association at all

C. strong, negative, linear

D. strong, positive, linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q3/5

448

QUESTION ID
S13-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. curved: rises, then falls

GRADING NOTE Peaks near 50 mph; a line does not describe it.

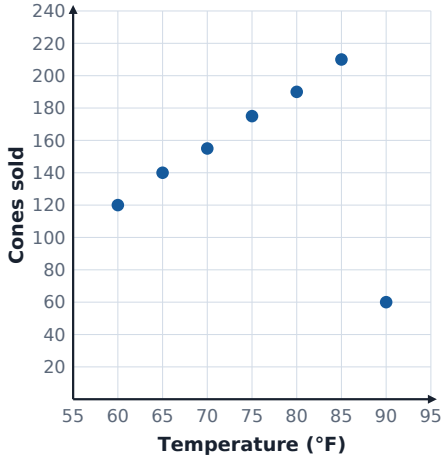
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q3/5

The scatter plot shows daily high temperature and ice-cream cones sold. One day does not follow the pattern. Give its temperature and its cones sold.

Temperature (°F): _____

Cones sold: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q4/5

449

QUESTION ID

S13-M-Q4

TYPE

Short answer

CORRECT ANSWER

90; 60

GRADING NOTE Form required: both. The other six days lie close to a rising line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q4/5

Which pair of variables would most likely show a negative association?

A. the height and arm span of adults

B. hours studied and test score

C. the age of a car and its resale value

D. shoe size and quiz score of adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q5/5

450

QUESTION ID

S13-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. the age of a car and its resale value

GRADING NOTE Value falls with age. Height/arm span and study/score are positive; shoe size and quiz score are unrelated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q5/5

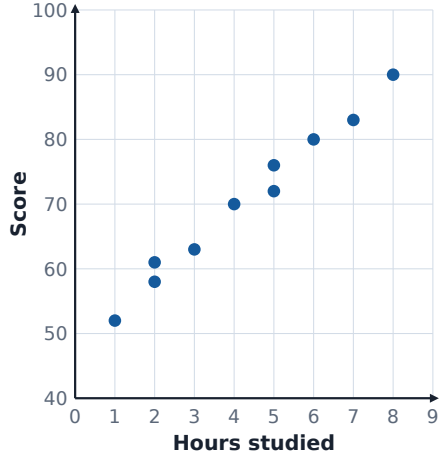
SET 13 • HSS-ID.B.6

BACKWARD 1

Plot and read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

In the study scatter plot, what score did the student who studied 7 hours get?

Score: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

452

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

83

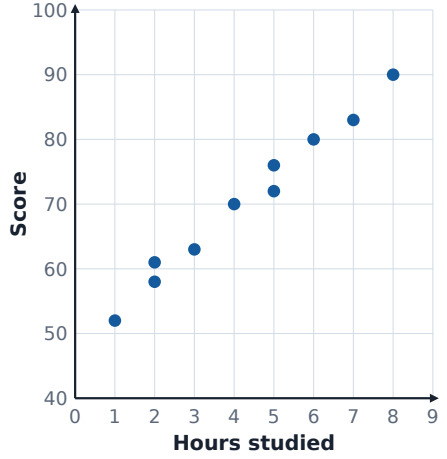
GRADING NOTE Point (7, 83).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

In the study scatter plot, how many students studied 2 hours?

Number: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

453

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE (2, 58) and (2, 61).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

In the study data, what does the point (4, 70) mean?

A. a student studied 4 hours and scored 70

B. a student studied 70 hours and scored 4

C. 4 students scored 70

D. the average score was 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

454

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. a student studied 4 hours and scored 70

GRADING NOTE (x, y) = (hours, score).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

In the ordered pair (3, 63), 63 is the x-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

455

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 63 is the y-coordinate.

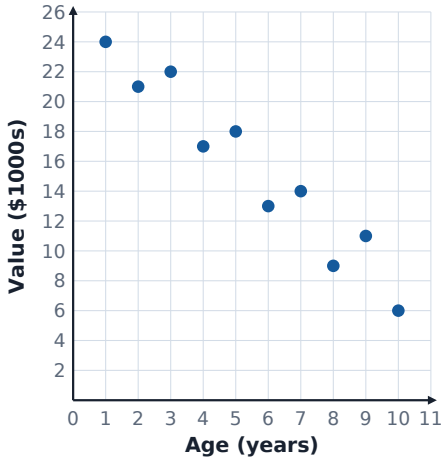
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the car scatter plot, what is the resale value of the 8-year-old car, in thousands of dollars?

Value: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

456

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Point (8, 9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSS-ID.B.6

BACKWARD 2

Describe patterns of association • Grade 8 • Statistics & Probability • 8.SP.A.1

True or false?

When y tends to increase as x increases, the association is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q1/5

458

QUESTION ID

S13-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q1/5

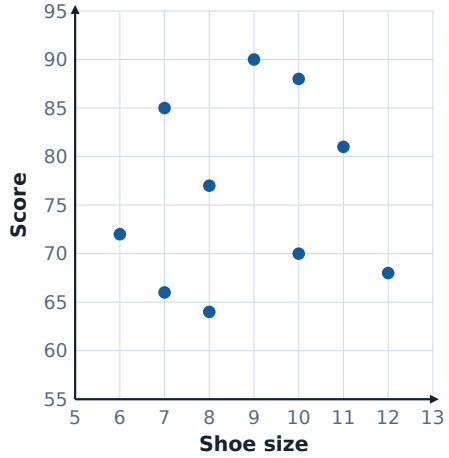
The scatter plot shows shoe size and quiz score. Which describes it?

A. curved

B. strong, positive, linear

C. strong, negative, linear

D. no clear association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q2/5

459

QUESTION ID
S13-B2-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. no clear association

GRADING NOTE The points form a shapeless cloud.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q2/5

True or false?

A scatter plot whose points lie close to a U-shaped curve shows a linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q3/5

460

QUESTION ID

S13-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a nonlinear association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q3/5

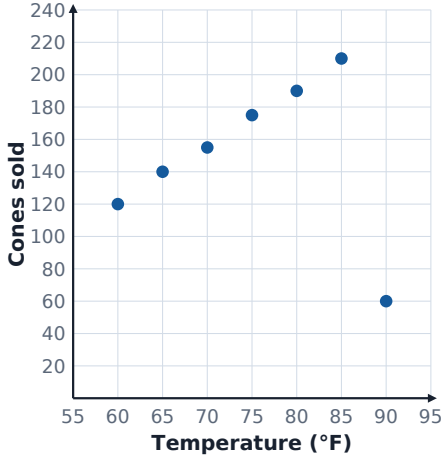
In the ice-cream scatter plot, the point (90, 60) is best described as

A. the trend line

B. an outlier

C. a cluster

D. the y-intercept



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q4/5

461

QUESTION ID
S13-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. an outlier

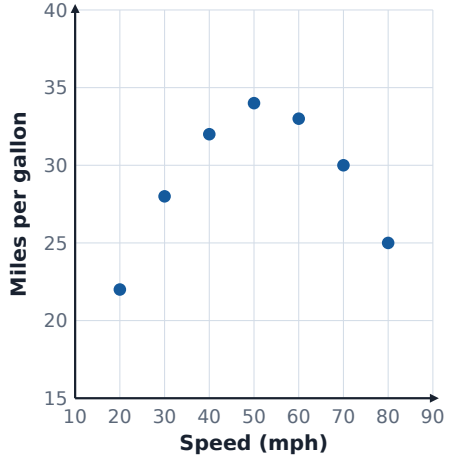
GRADING NOTE It lies far from the pattern of the other points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q4/5

In the fuel-efficiency scatter plot, at what speed is the fuel efficiency greatest?

Speed (mph): _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q5/5

462

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

50

GRADING NOTE Highest point (50, 34).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q5/5

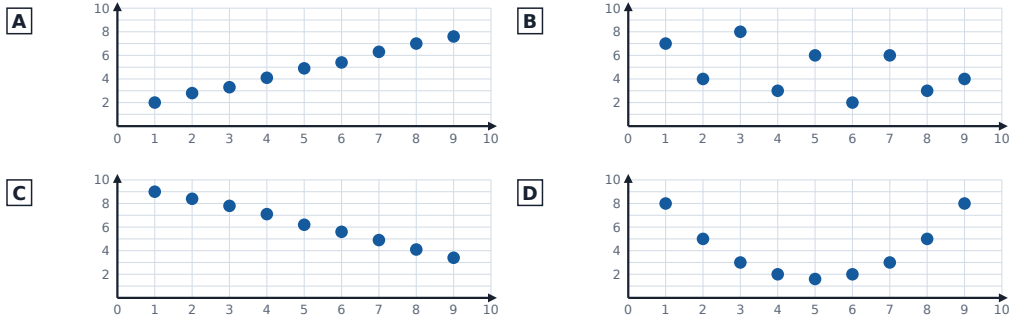
SET 13 • HSS-ID.B.6

FORWARD 1

Compare the direction, form and strength of associations • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Which scatter plot shows a strong, negative, linear association?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q1/5

464

QUESTION ID

S13-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE The others are positive, weak, or curved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q1/5

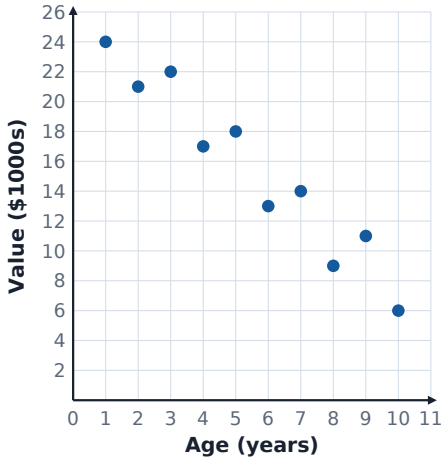
The scatter plot shows the age and resale value of cars. Which describes it?

A. curved, then level

B. strong, negative, linear

C. weak, positive, linear

D. strong, positive, linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q2/5

465

QUESTION ID

S13-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. strong, negative, linear

GRADING NOTE Values fall steadily with age.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q2/5

True or false?

A pattern can show a strong association even when it is not linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q3/5

466

QUESTION ID

S13-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The fuel data are an example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q3/5

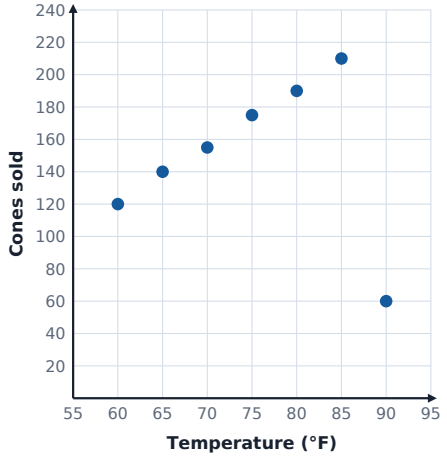
In the ice-cream scatter plot, if the outlier (90, 60) is ignored, the other six days show

A. no association

B. a strong, negative, linear association

C. a U-shaped association

D. a strong, positive, linear association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q4/5

467

QUESTION ID

S13-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a strong, positive, linear association

GRADING NOTE The six points rise along a line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

Removing an outlier can never change how strong an association looks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q5/5

468

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ice-cream data look weak with it and strong without it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSS-ID.B.6

FORWARD 2

Distinguish association from causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Cities with more fire stations tend to have more fires. Which is the best explanation?

A. Fires cause cities to build fewer stations.

B. Fire stations cause fires to start.

C. Larger cities have both more stations and more fires.

D. The association must be a mistake in the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q1/5

470

QUESTION ID
S13-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Larger cities have both more stations and more fires.

GRADING NOTE City size is a lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q1/5

True or false?

A strong association between two variables proves that one causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q2/5

471

QUESTION ID
S13-F2-Q2

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Association alone does not show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q2/5

To show that a fertilizer causes taller plants, which study is needed?

A. a survey of gardeners about their plants

B. a scatter plot of plant heights found in gardens

C. a randomized experiment with fertilized and unfertilized plants

D. measuring only the plants that received fertilizer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q3/5

472

QUESTION ID
S13-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. a randomized experiment with fertilized and unfertilized plants

GRADING NOTE Random assignment controls other causes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q3**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q3/5

Ice-cream sales and swimming accidents both rise during the year. What best explains the association?

A. The two variables are unrelated.

B. Swimming accidents cause ice-cream sales.

C. Eating ice cream causes swimming accidents.

D. Hot weather increases both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q4/5

473

QUESTION ID

S13-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Hot weather increases both.

GRADING NOTE A lurking variable (temperature).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q4/5

True or false?

In an observational study, a lurking variable can create an association between two variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q5/5

474

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both depend on it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q5/5

Use a function fitted to data to solve problems

MAIN	Probability & Statistics main question	**HSS-ID.B.6.a**
BACKWARD 1	Evaluate a function at an input	**HSF-IF.A.2**
BACKWARD 2	Evaluate expressions with exponents	**6.EE.A.2.c**
BACKWARD 3	Solve a linear equation for the input	**HSA-REI.B.3**
FORWARD 1	Judge whether a prediction is reasonable	**HSS-ID.B.6.a**
FORWARD 2	Compare predictions from two fitted models	**HSS-ID.B.6.a**

SET 14 • HSS-ID.B.6.a

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

For a group of students, a line fitted to the data is $y = 2.5x + 12$, where x is hours of practice and y is the predicted number of free throws made out of 50. Predict y for 8 hours of practice.

Predicted y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q1/5

477

QUESTION ID
S14-M-Q1

TYPE
Short answer

CORRECT ANSWER
32

GRADING NOTE 2.5·8 + 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q1/5

A bacteria culture is modeled by $P = 200(1.15)^t$, where t is hours. Predict the population after 6 hours, to the nearest whole number. A calculator or spreadsheet may be used.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q2/5

478

QUESTION ID
S14-M-Q2

TYPE
Short answer

CORRECT ANSWER
463

GRADING NOTE $200 \cdot 1.15^6 = 462.61...$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q2/5

A ball's height is modeled by $h = -4.9t^2 + 20t + 1.5$ (meters, t in seconds). Predict the height at t = 2 seconds.

Height (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q3/5

479

QUESTION ID
S14-M-Q3

TYPE
Short answer

CORRECT ANSWER
21.9

GRADING NOTE −19.6 + 40 + 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q3/5

A line fitted to data is y = 0.8x + 5, where x is the temperature (°C) and y is the predicted number of visitors (hundreds). For what temperature does the model predict 29 hundred visitors?

Temperature (°C): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q4/5

480

QUESTION ID
S14-M-Q4

TYPE
Short answer

CORRECT ANSWER
30

GRADING NOTE 0.8x = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q4/5

A car's value is modeled by $V = 18000(0.85)^t$, where t is years. Predict the value after 4 years, to the nearest dollar. A calculator or spreadsheet may be used.

Value ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q5/5

481

QUESTION ID
S14-M-Q5

TYPE
Short answer

CORRECT ANSWER
9396

GRADING NOTE $0.85^4 = 0.52200625$; $18000 \cdot 0.52200625 = 9396.11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q5/5

SET 14 • HSS-ID.B.6.a

BACKWARD 1

Evaluate a function at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

f(x) = 3x − 4. Find f(6).

f(6): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q1/5

483

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE 18 − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q1/5

$g(t) = 100(2)^t$. Find g(3).

g(3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q2/5

484

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

800

GRADING NOTE 100 · 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q2/5

True or false?

If $h(t) = -5t^2 + 10t$, then $h(1) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q3/5

485

QUESTION ID

S14-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-5 + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q3/5

f(x) = 0.5x + 7. What is f(10)?

A. 7.5

B. 17

C. 57

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q4/5

486

QUESTION ID

S14-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE 5 + 7. 57 treats 0.5x as 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

If $p(t) = 50(1.1)^t$, then $p(0) = 55$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q5/5

487

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $p(0) = 50$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

BACKWARD 2

Evaluate expressions with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 1.5^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q1/5

489

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2.25

GRADING NOTE 1.5 · 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $4 \cdot 3^2 - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q2/5

490

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Exponent first: $36 - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q2/5

True or false?

$2 \cdot 5^2 = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q3/5

491

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2 \cdot 25 = 50$; 100 squares 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q3/5

Evaluate $1000(0.9)^2$.

A. 1800

B. 900

C. 81

D. 810

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q4/5

492

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 810

GRADING NOTE $0.81 \cdot 1000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q4/5

True or false?

$(-3)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q5/5

493

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-3)(-3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSS-ID.B.6.a

BACKWARD 3

Solve a linear equation for the input • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 0.8x + 5 = 21.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q1/5

495

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE 0.8x = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q1/5

Solve 2.5x + 12 = 47.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q2/5

496

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE 2.5x = 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q2/5

True or false?

The solution of $3x - 4 = 11$ is $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q3/5

497

QUESTION ID

S14-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3x = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q3/5

Solve 1.2x + 6 = 30.

A. 28.8

B. 25

C. 20

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q4/5

498

QUESTION ID

S14-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 20

GRADING NOTE 1.2x = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q4/5

True or false?

The solution of $0.5x + 3 = 8$ is $x = 5.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q5/5

499

QUESTION ID

S14-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $0.5x = 5$, $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSS-ID.B.6.a

FORWARD 1

Judge whether a prediction is reasonable • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The line y = 2.5x + 12 was fitted to students who practiced 0 to 12 hours (y = free throws made out of 50). It predicts y = 87 for 30 hours. Which is correct?

A. Reasonable, because 30 hours is a whole number of hours.

B. Unreasonable, because a fitted line must have a negative slope.

C. Unreasonable: 30 hours is far outside the data, and 87 exceeds 50.

D. Reasonable, because the line fits the data from 0 to 12 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q1/5

501

QUESTION ID
S14-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Unreasonable: 30 hours is far outside the data, and 87 exceeds 50.

GRADING NOTE Extrapolation; the value is impossible. A positive slope is not the problem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q1/5

A line fitted to data for x from 10 to 40 is y = 3x − 5. Predict y at x = 25, and state whether x = 25 is inside the data range by entering yes or no.

Predicted y: _____

Inside the range (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q2/5

502

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

70; yes

GRADING NOTE Form required: both. 75 − 5 = 70; 25 is between 10 and 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q2/5

True or false?

Predicting far outside the range of the data (extrapolation) is usually less reliable than predicting inside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q3/5

503

QUESTION ID

S14-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The pattern may not continue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q3/5

The model $V = 18000(0.85)^t$ fits a car's value for $t = 0$ to 8 years. What value does it predict at $t = 0$?

V ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q4/5

504

QUESTION ID

S14-F1-Q4

TYPE

Short answer

CORRECT ANSWER

18000

GRADING NOTE $0.85^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q4/5

True or false?

The model $h = -4.9t^2 + 20t + 1.5$ gives sensible heights for every value of t, including t = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q5/5

505

QUESTION ID
S14-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE h(10) = −288.5, below the ground: the ball has landed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

FORWARD 2

Compare predictions from two fitted models • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Two models fit a town's population (thousands), t years after 2020: linear L = 2t + 40 and exponential $E = 40(1.05)^t$. Predict each for t = 2. Round E to two decimal places.

L: _____

E: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q1/5

507

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

44; 44.10

GRADING NOTE Form required: both. 40 · 1.1025 = 44.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q1/5

For the models $L = 2t + 40$ and $E = 40(1.05)^t$, which statement about large t is correct?

A. E eventually predicts less, because 1.05 is close to 1.

B. The two models always predict the same population.

C. L always predicts more, because its slope is 2.

D. E eventually predicts more, because it grows by a constant percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q2/5

508

QUESTION ID
S14-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. E eventually predicts more, because it grows by a constant percent.

GRADING NOTE Exponential growth eventually exceeds linear growth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q2/5

Using L = 2t + 40 (thousands), in what year t after 2020 does L predict 60 thousand?

t: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q3/5

509

QUESTION ID
S14-F2-Q3

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 2t = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q3/5

True or false?

If two models fit the data equally well inside the data range, they must give the same prediction far outside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q4/5

510

QUESTION ID
S14-F2-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They can diverge, as L and E do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q4/5

A residual check shows the exponential model leaves random scatter and the linear model leaves a curved pattern. Which model should be used for predictions?

A. the exponential model

B. neither, because residuals exist

C. the linear model

D. either, equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q5/5

511

QUESTION ID
S14-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. the exponential model

GRADING NOTE A curved residual pattern shows the linear form is wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q5/5

Choose and fit a linear, quadratic or exponential function

MAIN	Probability & Statistics main question	**HSS-ID.B.6.a**
BACKWARD 1	Tell equal differences from equal factors	**HSF-LE.A.1.a**
BACKWARD 2	Write a linear function from two points	**8.F.B.4**
BACKWARD 3	Write an exponential function from two points	**HSF-LE.A.2**
FORWARD 1	Fit a model with regression technology	**HSS-ID.B.6.a**
FORWARD 2	Check a fitted model with residuals	**HSS-ID.B.6.b**

SET 15 • HSS-ID.B.6.a

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The data fit an exponential function $y = a \cdot b^x$ exactly. Find a and b.

x	0	1	2	3	4
y	3	6	12	24	48

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S15-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q1/5

514

QUESTION ID
S15-M-Q1

TYPE
Short answer

CORRECT ANSWER
3; 2

GRADING NOTE Form required: both. y(0) = 3; each value doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q1/5

The data fit a linear function y = mx + b exactly. Find m and b.

x	0	1	2	3	4
y	5	8	11	14	17

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q2/5

515

QUESTION ID
S15-M-Q2

TYPE
Short answer

CORRECT ANSWER
3; 5

GRADING NOTE Form required: both. Constant difference 3; y(0) = 5.

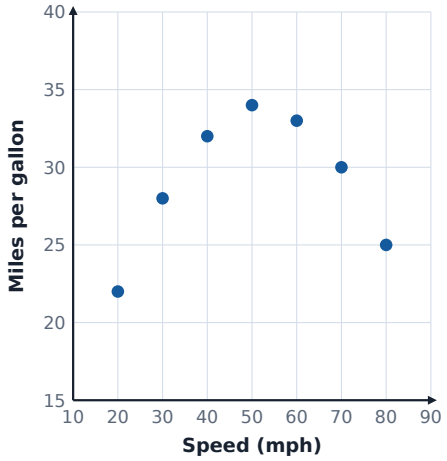
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q2/5

The scatter plot shows a car's speed and fuel efficiency. Which family of functions best fits the data?

A. quadratic **B.** exponential

C. linear **D.** constant



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q3/5

516

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. quadratic

GRADING NOTE The values rise to a peak and fall: a parabola opening downward.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q3/5

A town has 12,000 people and its population grows by about 8% each year. Which model fits, with t in years?

A. $P = 12000 + 0.08t$

B. $P = 12000(1.08)^t$

C. $P = 12000 + 1.08t$

D. $P = 12000(0.08)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q4/5

517

QUESTION ID
S15-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $P = 12000(1.08)^t$

GRADING NOTE Constant percent growth: factor 1.08.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q4/5

An exponential function $y = a \cdot b^x$ passes through (0, 500) and (1, 400). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q5/5

518

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

500; 0.8

GRADING NOTE Form required: both. a = 500; b = 400/500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q5/5

SET 15 • HSS-ID.B.6.a

BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

True or false?

In y = 3x + 5, y changes by equal differences over equal intervals of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q1/5

520

QUESTION ID

S15-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q1/5

For x = 0, 1, 2, 3 the values of y are 2, 6, 18, 54. What is the pattern?

A. squared each step

B. add 3 each step

C. add 4 each step

D. multiplied by 3 each step

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q2/5

521

QUESTION ID

S15-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. multiplied by 3 each step

GRADING NOTE 6/2 = 18/6 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q2/5

True or false?

The values 4, 7, 12, 19 (for x = 0 to 3) have constant first differences.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q3/5

522

QUESTION ID
S15-B1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Differences 3, 5, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q3/5

For x = 0 to 4, y = 4, 7, 12, 19, 28. What is the constant second difference?

Second difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q4/5

523

QUESTION ID
S15-B1-Q4

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE First differences 3, 5, 7, 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q4/5

Which list of y-values (for x = 0, 1, 2, 3) is linear?

A. 1, 2, 4, 8

B. 0, 1, 4, 9

C. 5, 5.5, 6.5, 8

D. 10, 7, 4, 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q5/5

524

QUESTION ID

S15-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 10, 7, 4, 1

GRADING NOTE Constant difference −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

BACKWARD 2

Write a linear function from two points • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q1/5

526

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3; 5

GRADING NOTE Form required: both. 6/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q1/5

Find the slope of the line through (1, 4) and (5, 12).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q2/5

527

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 8/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q2/5

True or false?

The table x = 0, 1, 2 and y = 7, 4, 1 fits y = −3x + 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q3/5

528

QUESTION ID

S15-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Check each point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q3/5

Which function fits x = 0, 1, 2 and y = 2, 4.5, 7?

A. y = 2x + 2.5

B. y = 2.5x + 2

C. y = 4.5x

D. y = 2.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q4/5

529

QUESTION ID

S15-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2.5x + 2

GRADING NOTE Difference 2.5, start 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q4/5

True or false?

The line through (0, 4) and (1, 10) has slope 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q5/5

530

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Slope 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSS-ID.B.6.a

BACKWARD 3

Write an exponential function from two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

An exponential function $y = a \cdot b^x$ passes through (0, 200) and (1, 300). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q1/5

532

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

200; 1.5

GRADING NOTE Form required: both. 300/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-B3-Q1**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q1/5

An exponential function $y = a \cdot b^x$ with $b > 0$ passes through (0, 80) and (2, 20). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q2/5

533

QUESTION ID

S15-B3-Q2

TYPE

Short answer

CORRECT ANSWER

80; 0.5

GRADING NOTE Form required: both. $b^2 = 1/4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-B3-Q2**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q2/5

True or false?

$y = 50(1.2)^x$ passes through (1, 60).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q3/5

534

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $50 \cdot 1.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q3/5

Which exponential function passes through (0, 4) and (1, 12)?

A. $y = 3(4)^x$

B. $y = 4(3)^x$

C. $y = 4 + 8x$

D. $y = 12(4)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q4/5

535

QUESTION ID

S15-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 4(3)^x$

GRADING NOTE a = 4, b = 3. 4 + 8x is linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-B3-Q4**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q4/5

True or false?

$y = 10(0.5)^x$ passes through (2, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q5/5

536

QUESTION ID

S15-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10 \cdot 0.25 = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSS-ID.B.6.a

FORWARD 1

Fit a model with regression technology • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Find the least-squares line y = mx + b for the spring data. Round m and b to two decimal places. A calculator or spreadsheet may be used.

Mass (g)	2	4	5	7	8	10
Length (cm)	15	19	24	26	31	35

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q1/5

538

QUESTION ID
S15-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2.52; 9.86

GRADING NOTE Form required: both. Linear regression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q1/5

The points (0, 2), (1, 5), (2, 10), (3, 17) lie exactly on a quadratic $y = ax^2 + bx + c$. Find a, b and c. A calculator or spreadsheet may be used.

a: _____

b: _____

c: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q2/5

539

QUESTION ID
S15-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1; 2; 2

GRADING NOTE Form required: all three. Second difference 2, so a = 1; c = 2; b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q2/5

Data on y rise faster and faster as x increases, y is always positive, and each step multiplies y by about 1.3. Which regression should be used?

A. linear

B. exponential

C. none of these

D. quadratic

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q3/5

540

QUESTION ID
S15-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. exponential

GRADING NOTE Nearly constant factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q3/5

Using the fitted line y = 2.52x + 9.86, predict y when x = 6.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q4/5

541

QUESTION ID

S15-F1-Q4

TYPE

Short answer

CORRECT ANSWER

24.98

GRADING NOTE 15.12 + 9.86.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q4/5

True or false?

Height-versus-time data for a thrown ball are better fitted by a quadratic than by a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q5/5

542

QUESTION ID
S15-F1-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Projectile heights follow a parabola.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

FORWARD 2

Check a fitted model with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

For the model y = 2.52x + 9.86, find the residual (observed minus predicted) for the data point (5, 24).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q1/5

544

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.54

GRADING NOTE 24 − 22.46.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q1/5

For the model y = 2.52x + 9.86, find the residual (observed minus predicted) for the data point (4, 19).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q2/5

545

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−0.94

GRADING NOTE 19 − 19.94.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q2/5

True or false?

A positive residual means the data point lies above the fitted line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q3/5

546

QUESTION ID
S15-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Observed > predicted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q3/5

A line is fitted to data that actually grow exponentially. What pattern do the residuals usually show?

A. every residual positive

B. a curved (U-shaped) pattern

C. random scatter around 0

D. all residuals equal to 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q4/5

547

QUESTION ID

S15-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a curved (U-shaped) pattern

GRADING NOTE The line misses the curvature systematically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q4/5

True or false?

If the residuals of a linear model are small but form a clear curve, a linear model is still the best family.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q5/5

548

QUESTION ID
S15-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A pattern means the family is wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q5/5

Assess the fit of a function with residuals

MAIN	Probability & Statistics main question	**HSS-ID.B.6.b**
BACKWARD 1	Subtract with signed decimals	**7.NS.A.1.c**
BACKWARD 2	Find a predicted value from a model	**HSF-IF.A.2**
BACKWARD 3	Compare an observed value with a prediction	**HSS-ID.B.6.a**
FORWARD 1	Compute least-squares residuals	**HSS-ID.B.6.b**
FORWARD 2	Compare two models with residuals	**HSS-ID.B.6.b**

SET 16 • HSS-ID.B.6.b

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The model y = 2x + 1 is used for the data in the table. Find the residuals (observed minus predicted) at x = 1 and at x = 2.

x	1	2	3	4	5
y	3.5	4.6	7.2	8.8	11.1

Residual at x = 1: _____

Residual at x = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q1/5

551

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.5; −0.4

GRADING NOTE Form required: both. 3.5 − 3 and 4.6 − 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q1/5

A model predicts y = 3x + 10. Find the residuals (observed minus predicted) for the observed points (4, 20) and (6, 30).

Residual for (4, 20): _____

Residual for (6, 30): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q2/5

552

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

−2; 2

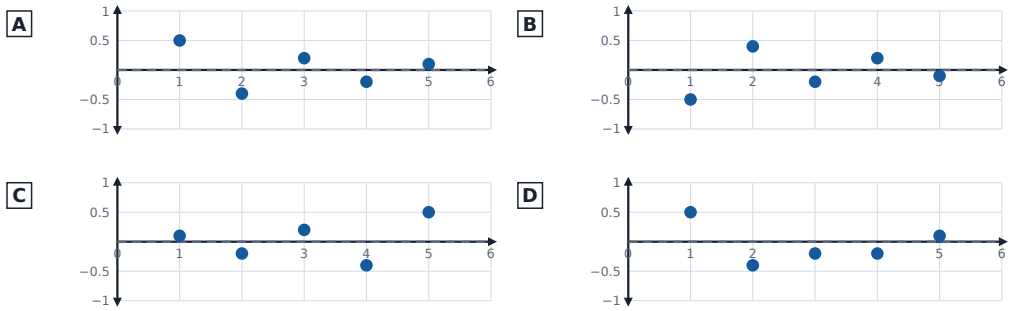
GRADING NOTE Form required: both. 20 − 22 and 30 − 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The model y = 2x + 1 is used for the data in the table (x = 1 to 5; y = 3.5, 4.6, 7.2, 8.8, 11.1). Which plot is the residual plot (observed minus predicted)?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q3/5

553

QUESTION ID
S16-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. Graph A

GRADING NOTE Residuals 0.5, −0.4, 0.2, −0.2, 0.1. The others use predicted minus observed, reverse the order, or flip the sign at x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q3/5

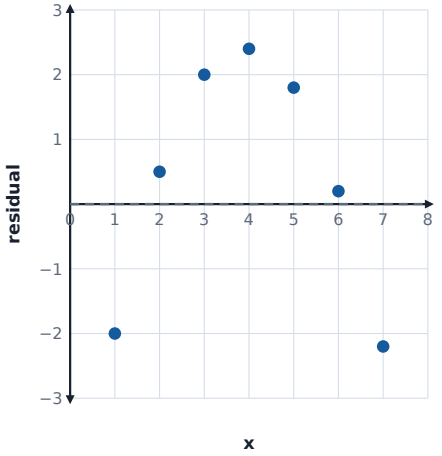
A line was fitted to a data set. The residual plot is shown. Which conclusion is correct?

A. The linear model is appropriate, because some residuals are near 0.

B. The linear model is appropriate, because residuals are both signs.

C. The data must contain an error, because residuals are not all 0.

D. A linear model is not appropriate; the curve suggests a curved model.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q4/5

554

QUESTION ID
S16-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. A linear model is not appropriate; the curve suggests a curved model.

GRADING NOTE A systematic curve in the residuals shows the line misses the form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q4/5

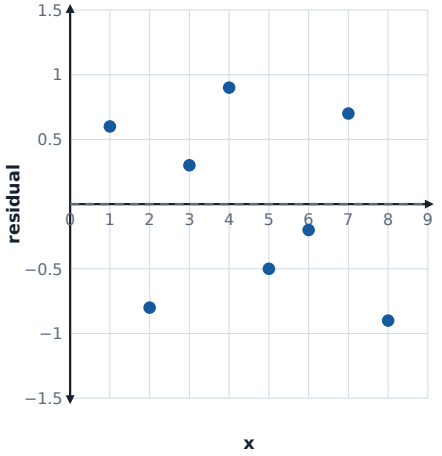
A line was fitted to a different data set, and its residual plot is shown. What does the plot show about the linear model?

A. The linear model is not appropriate, because residuals are not 0.

B. The linear model is appropriate: the residuals show no pattern.

C. A quadratic model is needed, because residuals change sign.

D. The model is perfect, because the residuals add to about 0.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q5/5

555

QUESTION ID
S16-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. The linear model is appropriate: the residuals show no pattern.

GRADING NOTE Random scatter around 0 supports the linear form; residuals are never all 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q5/5

SET 16 • HSS-ID.B.6.b

BACKWARD 1

Subtract with signed decimals • Grade 7 • The Number System • 7.NS.A.1.c

Compute 3.5 − 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q1/5

557

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE Difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 4.6 − 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q2/5

558

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−0.4

GRADING NOTE Below zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q2/5

True or false?

$20 - 22 = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q3/5

559

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Signed result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q3/5

Compute 7.2 − 7.5.

A. −0.3

B. −14.7

C. 14.7

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q4/5

560

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −0.3

GRADING NOTE Smaller minus larger is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q4/5

True or false?

$11 - 11.4 = 0.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q5/5

561

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE -0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSS-ID.B.6.b

BACKWARD 2

Find a predicted value from a model • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

For y = 2x + 1, find the predicted y at x = 3.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q1/5

563

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 6 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q1/5

For y = 3x + 10, find the predicted y at x = 4.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q2/5

564

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE 12 + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B2-Q2**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q2/5

True or false?

For y = −1.5x + 20, the predicted y at x = 4 is 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q3/5

565

QUESTION ID

S16-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −6 + 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q3/5

For y = 0.5x + 3, what is the predicted y at x = 8?

A. 8.5

B. 4

C. 7

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q4/5

566

QUESTION ID

S16-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 7

GRADING NOTE 4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q4/5

True or false?

For y = 4x − 2, the predicted y at x = 0 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q5/5

567

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSS-ID.B.6.b

BACKWARD 3

Compare an observed value with a prediction • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A model predicts y = 2.5x + 12 free throws for x hours of practice. A student practiced 8 hours and made 35. What is the predicted number, and how many more did the student make than predicted?

Predicted: _____

More by: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q1/5

569

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

32; 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q1/5

True or false?

If an observed value is less than the predicted value, the point lies below the fitted line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q2/5

570

QUESTION ID

S16-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Observed < predicted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q2/5

The observed value is 50 and the predicted value is 46. What is observed minus predicted?

A. 4

B. 96

C. −4

D. 46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q3/5

571

QUESTION ID

S16-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE 50 − 46.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q3/5

True or false?

In this collection, a residual is defined as predicted minus observed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q4/5

572

QUESTION ID

S16-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Residual = observed − predicted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q4/5

The observed value is 18 and the predicted value is 20.5. What is observed minus predicted?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q5/5

573

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−2.5

GRADING NOTE 18 − 20.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSS-ID.B.6.b

FORWARD 1

Compute least-squares residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The least-squares line for (1, 2), (2, 5), (3, 4), (4, 7) is y = 1.4x + 1. Find the residuals (observed minus predicted) at x = 1 and x = 2.

At x = 1: _____

At x = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q1/5

575

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−0.4; 1.2

GRADING NOTE Form required: both. 2 − 2.4 and 5 − 3.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q1/5

For y = 1.4x + 1 and the points (1, 2), (2, 5), (3, 4), (4, 7), what is the sum of all four residuals?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q2/5

576

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE −0.4 + 1.2 − 1.2 + 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q2/5

True or false?

For a least-squares line, the residuals always add to 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q3/5

577

QUESTION ID
S16-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A property of least squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q3**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q3/5

For y = 1.4x + 1 and the points (1, 2), (2, 5), (3, 4), (4, 7), which points have the largest residuals in absolute value?

A. (1, 2) only

B. (1, 2) and (4, 7)

C. (4, 7) only

D. (2, 5) and (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q4/5

578

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5) and (3, 4)

GRADING NOTE |1.2| > |0.4|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q4/5

True or false?

A residual of −1.2 means the point lies 1.2 units above the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q5/5

579

QUESTION ID
S16-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Below the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSS-ID.B.6.b

FORWARD 2

Compare two models with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

Two models were fitted to the same data. Linear residuals: 3, −1, −3, −2, 1, 4. Exponential residuals: 0.2, −0.3, 0.1, 0.3, −0.2, −0.1. Which model fits better?

A. the exponential model

B. the linear model

C. neither model can be judged

D. they fit equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q1/5

581

QUESTION ID
S16-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. the exponential model

GRADING NOTE The exponential residuals are small and patternless; the linear ones form a U.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of squared residuals for each model: linear 3, −1, −3, −2, 1, 4 and exponential 0.2, −0.3, 0.1, 0.3, −0.2, −0.1.

Linear: _____

Exponential: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q2/5

582

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

40; 0.28

GRADING NOTE Form required: both. 9+1+9+4+1+16; 0.04+0.09+0.01+0.09+0.04+0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q2/5

True or false?

Of two models fitted to the same data, the one with the smaller sum of squared residuals fits those data more closely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q3/5

583

QUESTION ID
S16-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Smaller total squared error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q3/5

A residual plot fans out: the residuals get larger as x increases. What does this show?

A. the slope is negative

B. prediction errors grow as x increases

C. the model is exact

D. the data have no outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q4/5

584

QUESTION ID
S16-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. prediction errors grow as x increases

GRADING NOTE Spread of errors depends on x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q4/5

True or false?

A residual plot with a clear U shape shows that the model is a good fit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q5/5

585

QUESTION ID
S16-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A pattern shows a poor fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q5/5

Fit a linear function to a linear scatter plot

MAIN	Probability & Statistics main question	**HSS-ID.B.6.c**
BACKWARD 1	Fit a line by eye	**8.SP.A.2**
BACKWARD 2	Find slope and intercept from points	**8.F.B.4**
FORWARD 1	Use the properties of the least-squares line	**HSS-ID.B.6.c**
FORWARD 2	Interpret the fitted line's slope and intercept	**HSS-ID.C.7**

SET 17 • HSS-ID.B.6.c

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the least-squares line y = mx + b for hours studied (x) and score (y). Round m and b to two decimal places. A calculator or spreadsheet may be used.

Hours	1	2	2	3	4	5	5	6	7	8
Score	52	58	61	63	70	72	76	80	83	90

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q1/5

588

QUESTION ID
S17-M-Q1

TYPE
Short answer

CORRECT ANSWER
5.17; 48.28

GRADING NOTE Form required: both. m = 2485/481 ≈ 5.166; b = 23225/481 ≈ 48.285.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q1/5

Find the least-squares line y = mx + b for car age (x) and value in $1000s (y). Round m and b to two decimal places. A calculator or spreadsheet may be used.

Age (years)	1	2	3	4	5	6	7	8	9	10
Value ($1000s)	24	21	22	17	18	13	14	9	11	6

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q2/5

589

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1.88; 25.87

GRADING NOTE Form required: both. m = −311/165 ≈ −1.885; b = 388/15 ≈ 25.867.

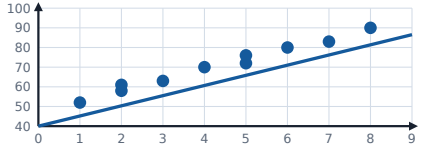
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q2/5

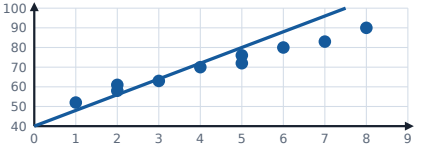
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows the line that best fits the study data (hours, score)?

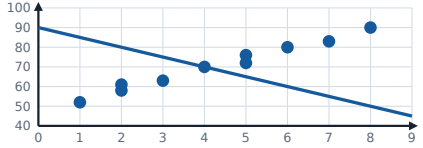
A



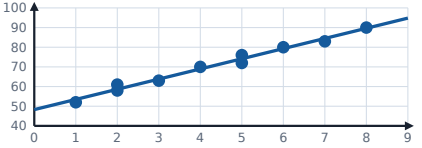
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q3/5

590

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE The least-squares line y = 5.17x + 48.28 runs through the middle. The others are too low, too steep, or falling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q3/5

Using the least-squares line y = 5.17x + 48.28 for hours studied and score, predict the score for 3 hours.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q4/5

591

QUESTION ID
S17-M-Q4

TYPE
Short answer

CORRECT ANSWER
63.79

GRADING NOTE 15.51 + 48.28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q4/5

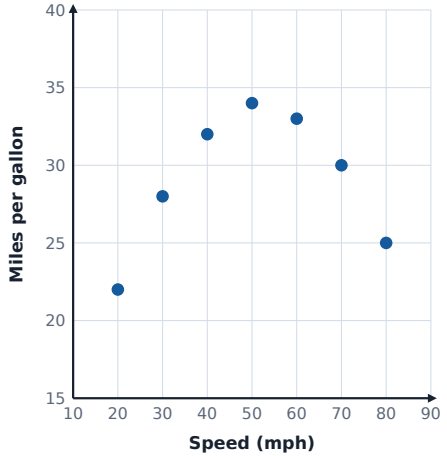
A student wants to fit a least-squares line to the data in this scatter plot. Which statement is correct?

A. A line is appropriate, because r can always be computed.

B. A line is appropriate, because the data have no outliers.

C. A line is not appropriate: the pattern is curved.

D. A line fits well, because the points rise at first.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q5/5

592

QUESTION ID
S17-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. A line is not appropriate: the pattern is curved.

GRADING NOTE The standard applies to scatter plots that suggest a linear association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q5/5

SET 17 • HSS-ID.B.6.c

BACKWARD 1

Fit a line by eye • Grade 8 • Statistics & Probability • 8.SP.A.2

True or false?

A least-squares line must have exactly as many data points above it as below it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q1/5

QUESTION ID

S17-B1-Q1

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Least squares (with an intercept) makes the residuals add to 0; the numbers of points above and below can differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q1/5

Which line passes through (0, 2), (2, 6) and (4, 10)?

A. y = x + 2

B. y = 2x

C. y = 2x + 2

D. y = 4x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q2/5

595

QUESTION ID

S17-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2x + 2

GRADING NOTE Check each point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q2/5

A line drawn by eye passes through (0, 50) and (5, 75). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q3/5

596

QUESTION ID

S17-B1-Q3

TYPE

Short answer

CORRECT ANSWER

5; 50

GRADING NOTE Form required: both. 25/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q3/5

True or false?

A fitted line must pass through every data point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q4/5

597

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It summarizes the trend.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q4/5

Use the line y = 5x + 50 to predict y at x = 4.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q5/5

598

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

70

GRADING NOTE 20 + 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSS-ID.B.6.c

BACKWARD 2

Find slope and intercept from points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (2, 58) and (6, 80).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q1/5

600

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5.5

GRADING NOTE 22/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q1/5

A line has slope 5 and passes through (2, 60). Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q2/5

601

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

50

GRADING NOTE 60 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q2/5

True or false?

The line through (0, 26) with slope −2 has y = 16 at x = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q3/5

602

QUESTION ID

S17-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 26 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q3/5

Which equation is the line through (0, 10) and (4, 2)?

A. y = 2x + 10

B. y = −2x + 2

C. y = −0.5x + 10

D. y = −2x + 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q4/5

603

QUESTION ID

S17-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = −2x + 10

GRADING NOTE Slope −8/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q4/5

True or false?

The slope of the line through (1, 24) and (10, 6) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q5/5

604

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-18/9 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSS-ID.B.6.c

FORWARD 1

Use the properties of the least-squares line • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Find the mean hours $\bar{x}$ and the mean score $\bar{y}$ for the study data.

Hours	1	2	2	3	4	5	5	6	7	8
Score	52	58	61	63	70	72	76	80	83	90

$\bar{x}$: _____

$\bar{y}$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q1/5

606

QUESTION ID
S17-F1-Q1

TYPE
Short answer

CORRECT ANSWER
4.3; 70.5

GRADING NOTE Form required: both. 43/10 and 705/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q1/5

True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q2/5

607

QUESTION ID

S17-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A property of least squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q2/5

The least-squares line is the line that makes which quantity as small as possible?

A. the number of points off the line

B. the largest single residual

C. the sum of the residuals

D. the sum of the squared residuals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q3/5

608

QUESTION ID
S17-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. the sum of the squared residuals

GRADING NOTE Definition. The plain sum is 0 for many lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q3/5

A least-squares line has slope 2 and passes through $(\bar{x}, \bar{y}) = (5, 20)$. Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q4/5

609

QUESTION ID
S17-F1-Q4

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 20 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q4/5

True or false?

Adding one point far from the others can never change the least-squares line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q5/5

610

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An influential point can move it a lot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSS-ID.B.6.c

FORWARD 2

Interpret the fitted line's slope and intercept • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

For hours studied (x) and score (y), the least-squares line is y = 5.17x + 48.28. What does 5.17 mean?

A. A student who studied 0 hours scored exactly 5.17.

B. Each extra hour goes with about 5.17 more points, on average.

C. The score rises 48.28 points for each hour.

D. Each student scored 5.17 points per test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q1/5

612

QUESTION ID
S17-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Each extra hour goes with about 5.17 more points, on average.

GRADING NOTE Slope: predicted change in y per unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q1/5

Using y = 5.17x + 48.28, how much higher is the predicted score for a student who studies 2 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q2/5

613

QUESTION ID
S17-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10.34

GRADING NOTE 2 · 5.17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q2/5

For car age (x, years) and value (y, $1000s), $y = -1.88x + 25.87$. What does 25.87 mean?

A. The oldest car is worth about $25,870.

B. A car is predicted to be worth $0 after 25.87 years.

C. Cars lose about $25,870 each year.

D. A new car (age 0) has a predicted value of about $25,870.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q3/5

614

QUESTION ID
S17-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. A new car (age 0) has a predicted value of about $25,870.

GRADING NOTE Intercept: predicted y at x = 0 (age 0 is near the data).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q3/5

True or false?

In $y = -1.88x + 25.87$, the slope means every car loses exactly $1,880 each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S17-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q4/5

615

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is an average rate for the model, not exact for each car.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S17-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q4/5

Using $y = -1.88x + 25.87$, how much greater (in \$1000s) is the predicted value of a 3-year-old car than a 7-year-old car?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q5/5

616

QUESTION ID
S17-F2-Q5

TYPE
Short answer

CORRECT ANSWER
7.52

GRADING NOTE $4 \cdot 1.88$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q5/5

Interpret the slope and intercept of a linear model

MAIN	Probability & Statistics main question	**HSS-ID.C.7**
BACKWARD 1	Interpret the rate and the initial value of a linear function	**8.F.B.4**
BACKWARD 2	Use a linear model's equation to answer questions	**8.SP.A.3**
FORWARD 1	Interpret parameters with units and judge meaning	**HSS-ID.C.7**
FORWARD 2	Relate the slope to the correlation coefficient	**HSS-ID.C.8**

SET 18 • HSS-ID.C.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for a family's savings is y = 85x + 1200, where x is months and y is dollars. What does 85 mean?

A. The family started with $85 in savings.

B. The family saves $1200 each month.

C. The savings will reach $85 after 1200 months.

D. The savings are predicted to grow by $85 each month.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q1/5

619

QUESTION ID
S18-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. The savings are predicted to grow by $85 each month.

GRADING NOTE Slope: dollars per month.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q1/5

A model for weekly hours of exercise is $y = 6 - 0.2x$, where x is weekly hours of TV. What does 6 mean?

A. A person who watches 0 hours of TV is predicted to exercise 6 hours.

B. Each hour of TV is predicted to add 6 hours of exercise.

C. Each hour of TV is predicted to cut exercise by 6 hours.

D. A person exercises at most 0.2 hours per week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q2/5

620

QUESTION ID
S18-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. A person who watches 0 hours of TV is predicted to exercise 6 hours.

GRADING NOTE Intercept: predicted y at x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A tree's height is modeled by y = 0.4x + 1.5, with y in meters and x in years since planting. What is the predicted growth per year, and what is the predicted height at planting?

Growth per year (m): _____

Height at planting (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q3/5

621

QUESTION ID
S18-M-Q3

TYPE
Short answer

CORRECT ANSWER
0.4; 1.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q3/5

For adults with heights from 150 to 190 cm, arm span is modeled by $y = 1.1x - 15$ (both in cm). Which statement about the intercept -15 is correct?

A. It means a person 0 cm tall has an arm span of -15 cm.

B. It means arm spans are 15 cm less than heights.

C. It is not meaningful: a height of 0 cm is far outside the data.

D. It means the slope of the model is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q4/5

622

QUESTION ID
S18-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. It is not meaningful: a height of 0 cm is far outside the data.

GRADING NOTE Extrapolating to x = 0 makes no sense here.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q4/5

Weekly gasoline use is modeled by y = 0.35x + 4.2 gallons, where x is miles driven per day. By how many gallons do the predictions differ for two drivers whose daily miles differ by 20?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q5/5

623

QUESTION ID
S18-M-Q5

TYPE
Short answer

CORRECT ANSWER
7

GRADING NOTE 20 · 0.35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q5/5

SET 18 • HSS-ID.C.7

BACKWARD 1

Interpret the rate and the initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A plumber charges y = 50x + 80 dollars for x hours. What does 80 mean?

A. the charge per hour

B. a fixed fee charged before any hours

C. the number of hours

D. the total for 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q1/5

625

QUESTION ID

S18-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. a fixed fee charged before any hours

GRADING NOTE Initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q1/5

A phone plan costs y = 0.1x + 25 dollars for x minutes. What is the cost per minute, and what is the fixed cost?

Per minute ($): _____

Fixed ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q2/5

626

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.1; 25

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q2/5

True or false?

In y = 3x + 12, when x increases by 1, y increases by 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q3/5

627

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE By 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q3/5

A candle's height is h = 20 − 1.5t centimeters after t hours. What does 1.5 mean?

A. the candle burns 1.5 cm per hour

B. the candle starts 1.5 cm tall

C. the candle burns 20 cm per hour

D. the candle lasts 1.5 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q4/5

628

QUESTION ID

S18-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the candle burns 1.5 cm per hour

GRADING NOTE Rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q4/5

True or false?

In y = 3x + 12, the value of y when x = 0 is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q5/5

629

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSS-ID.C.7

BACKWARD 2

Use a linear model's equation to answer questions • Grade 8 • Statistics & Probability • 8.SP.A.3

Plant height is modeled by y = 1.5x + 5 (cm after x weeks). Predict the height at 6 weeks.

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q1/5

631

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE 9 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q1/5

In the plant model y = 1.5x + 5, one more week of growth goes with

A. 1.5 more weeks

B. 5 cm more height

C. 1.5 cm more height

D. 6.5 cm more height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q2/5

632

QUESTION ID

S18-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5 cm more height

GRADING NOTE Slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q2/5

True or false?

In the plant model y = 1.5x + 5, the slope means the plants were 1.5 cm tall at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q3/5

633

QUESTION ID

S18-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is the intercept, 5 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q3/5

In the plant model y = 1.5x + 5, how much greater is the predicted height at week 10 than at week 2?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q4/5

634

QUESTION ID

S18-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 8 · 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

A negative slope means the predicted y decreases as x increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q5/5

635

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSS-ID.C.7

FORWARD 1

Interpret parameters with units and judge meaning • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

Using y = 5.17x + 48.28 (score versus hours studied), what is the predicted score increase for 3 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q1/5

637

QUESTION ID
S18-F1-Q1

TYPE
Short answer

CORRECT ANSWER
15.51

GRADING NOTE 3 · 5.17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q1/5

A shipping cost is modeled by $y = 0.02x + 3.5$ dollars for x grams. What does 0.02 mean?

A. Each package costs $0.02 no matter what it weighs.

B. Each additional gram adds about $0.02 to the predicted cost.

C. The predicted cost rises by $3.50 for each gram.

D. A package weighing 0 grams is predicted to cost $0.02.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q2/5

638

QUESTION ID
S18-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. Each additional gram adds about $0.02 to the predicted cost.

GRADING NOTE Dollars per gram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q2/5

True or false?

The intercept of a linear model always has a meaningful interpretation in context.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q3/5

639

QUESTION ID
S18-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Not when x = 0 is outside the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q3/5

For ages 20 to 60, maximum heart rate is modeled by $y = 208 - 0.7x$ beats per minute, x in years. Which statement about 208 is correct?

A. It is the predicted heart rate for a 20-year-old person.

B. It is the prediction at age 0, outside the data: not meaningful.

C. It is the decrease in maximum heart rate for each year of age.

D. It is the largest heart rate that any person can have.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q4/5

640

QUESTION ID
S18-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. It is the prediction at age 0, outside the data: not meaningful.

GRADING NOTE Age 0 is outside 20 to 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q4/5

Using y = 208 − 0.7x, predict the maximum heart rate at age 40.

Beats per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q5/5

641

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

180

GRADING NOTE 208 − 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSS-ID.C.7

FORWARD 2

Relate the slope to the correlation coefficient • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

True or false?

The sign of the correlation coefficient matches the sign of the slope of the least-squares line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q1/5

643

QUESTION ID
S18-F2-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both measure direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q1/5

The least-squares line is $y = -1.88x + 25.87$. Which could be the correlation coefficient?

A. 0.97

B. 1.88

C. −1.88

D. −0.97

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q2/5

644

QUESTION ID

S18-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −0.97

GRADING NOTE Negative, and $|r| \le 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q2/5

True or false?

A larger slope always means a stronger correlation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q3/5

645

QUESTION ID
S18-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Slope depends on units; r does not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q3/5

The least-squares slope equals r·(s_y/s_x). If r = 0.8, s_y = 10 and s_x = 4, what is the slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q4/5

646

QUESTION ID
S18-F2-Q4

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE 0.8 · 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q4/5

The x-values vary and the y-values vary (neither is constant). If the least-squares line has slope 0, what is r?

A. 1

B. −1

C. it cannot be found

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q5/5

647

QUESTION ID
S18-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. 0

GRADING NOTE b = r·s_y/s_x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q5/5

Compute the correlation coefficient with technology

MAIN	Probability & Statistics main question	**HSS-ID.C.8**
BACKWARD 1	Find means and deviations	**6.SP.B.5.c**
BACKWARD 2	Read the direction of an association	**HSS-ID.B.6**
FORWARD 1	Compute r from deviations	**HSS-ID.C.8**
FORWARD 2	See how changes to the data affect r	**HSS-ID.C.8**

SET 19 • HSS-ID.C.8

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Find the correlation coefficient r for hours studied and score. Round to two decimal places. A calculator or spreadsheet may be used.

Hours	1	2	2	3	4	5	5	6	7	8
Score	52	58	61	63	70	72	76	80	83	90

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q1/5

650

QUESTION ID
S19-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.99

GRADING NOTE r = 0.9920.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q1/5

Find the correlation coefficient r for car age and value. Round to two decimal places. A calculator or spreadsheet may be used.

Age (years)	1	2	3	4	5	6	7	8	9	10
Value ($1000s)	24	21	22	17	18	13	14	9	11	6

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q2/5

651

QUESTION ID
S19-M-Q2

TYPE
Short answer

CORRECT ANSWER
−0.97

GRADING NOTE r = −0.9654.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q2/5

Find the correlation coefficient r for hours after sunrise (x) and ice thickness in cm (y). Round to two decimal places. A calculator or spreadsheet may be used.

Hours	1	2	3	4	5	6
Thickness (cm)	48	44	41	35	33	28

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q3/5

652

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

−0.99

GRADING NOTE r = −0.9949.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q3/5

Find r for the temperature and cones-sold data for all 7 days, and again without the 90-degree day. Round each to three decimal places. A calculator or spreadsheet may be used.

Temperature (°F)	60	65	70	75	80	85	90
Cones sold	120	140	155	175	190	210	60

All 7 days: _____

Without the 90-degree day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q4/5

653

QUESTION ID
S19-M-Q4

TYPE
Short answer

CORRECT ANSWER
−0.008; 0.999

GRADING NOTE Form required: both. −0.0077 and 0.9992.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q4/5

Find the correlation coefficient r for the spring data (mass and length). Round to two decimal places. A calculator or spreadsheet may be used.

Mass (g)	2	4	5	7	8	10
Length (cm)	15	19	24	26	31	35

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q5/5

654

QUESTION ID
S19-M-Q5

TYPE
Short answer

CORRECT ANSWER
0.99

GRADING NOTE r = 0.9881.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q5/5

SET 19 • HSS-ID.C.8

BACKWARD 1

Find means and deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 1, 2, 3, 4, 5.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q1/5

656

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 15/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For the values 2, 5, 4, 7 (mean 4.5), find the deviations of 2 and 7 from the mean.

2: _____

7: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q2/5

657

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2.5; 2.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q2/5

True or false?

The deviations of a data set from its mean always add to 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q3/5

658

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Balance property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the mean of 48, 44, 41, 35, 33, 28, rounded to two decimal places?

A. 35

B. 38.17

C. 41

D. 38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q4/5

659

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 38.17

GRADING NOTE 229/6 = 38.1666... ≈ 38.17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q4/5

True or false?

A value below the mean has a positive deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q5/5

660

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSS-ID.C.8

BACKWARD 2

Read the direction of an association • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

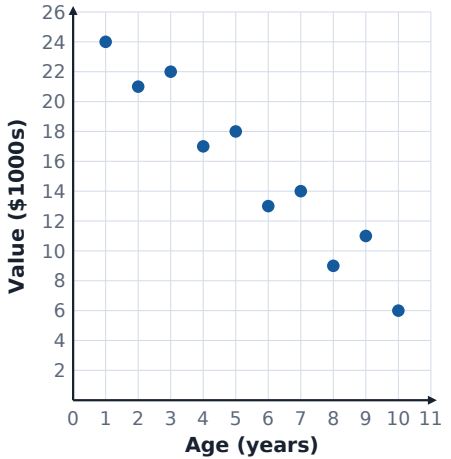
The scatter plot shows car age and value. What is the sign of r?

A. positive

B. cannot be found

C. zero

D. negative



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q1/5

662

QUESTION ID

S19-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. negative

GRADING NOTE Values fall as age rises.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B2-Q1**

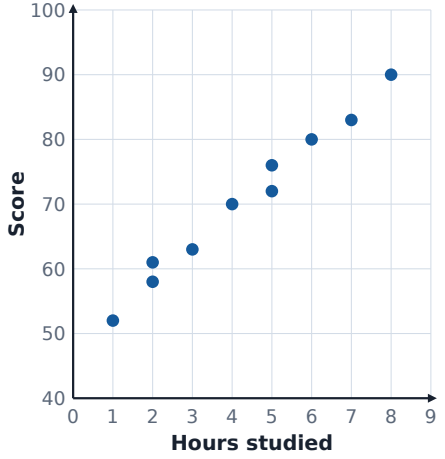
High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q1/5

True or false?

For the hours-studied scatter plot, r is positive.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q2/5

QUESTION ID

S19-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rising pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q2/5

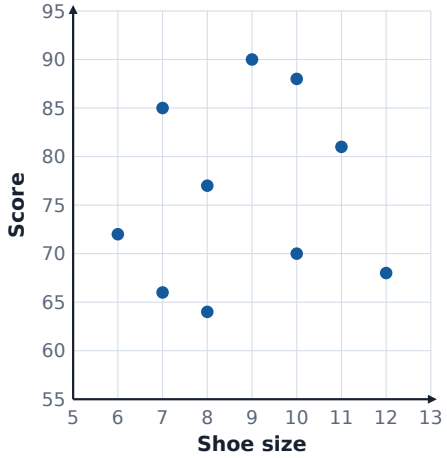
For the shoe-size scatter plot, r is closest to

A. 1

B. 0

C. −1

D. 0.9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q3/5

664

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE No linear pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q3/5

True or false?

For a cloud of points with no pattern, r is close to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q4/5

665

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Close to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q4/5

Which scatter plot would have r close to −1?

A. points along a U-shaped curve

B. a shapeless cloud

C. points falling closely along a line

D. points rising closely along a line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q5/5

666

QUESTION ID

S19-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. points falling closely along a line

GRADING NOTE Strong negative linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSS-ID.C.8

FORWARD 1

Compute r from deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

ANSWER KEY Reduced copy of the question page, with its answer

For (1, 2), (2, 5), (3, 4), (4, 7): $\Sigma(x - \bar{x})(y - \bar{y}) = 7$, $\Sigma(x - \bar{x})^2 = 5$ and $\Sigma(y - \bar{y})^2 = 13$. Compute $r = 7/\sqrt{(5 \cdot 13)}$, rounded to two decimal places.

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q1/5

668

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.87

GRADING NOTE $7/\sqrt{65} = 0.868$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q1/5

True or false?

If all points lie exactly on a line with positive slope, r = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q2/5

669

QUESTION ID
S19-F1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Perfect positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q2/5

For the points (1, 2), (2, 4), (3, 6), what is r?

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q3/5

670

QUESTION ID
S19-F1-Q3

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE Exactly on y = 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q3/5

A calculator reports r = 1.4. What should you conclude?

A. The association is very strong and positive.

B. The data have one outlier.

C. A mistake was made: r is always between −1 and 1.

D. The slope of the line is 1.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q4/5

671

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A mistake was made: r is always between −1 and 1.

GRADING NOTE $|r| \le 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q4/5

True or false?

For the points (1, 6), (2, 4), (3, 2), r = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q5/5

672

QUESTION ID
S19-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE r = −1 (falling line).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSS-ID.C.8

FORWARD 2

See how changes to the data affect r • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

True or false?

Changing x from hours to minutes does not change r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q1/5

674

QUESTION ID
S19-F2-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE r has no units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q1/5

True or false?

Swapping the roles of x and y changes the value of r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q2/5

675

QUESTION ID
S19-F2-Q2

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE r is symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q2/5

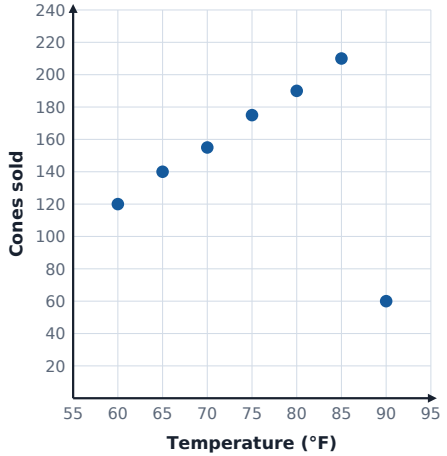
For the cones data, r = 0.999 for six days and −0.008 with the 90-degree day included. What does this show?

A. Temperature causes cone sales to fall.

B. r is not affected by outliers.

C. The six days have no association.

D. One outlier can change r dramatically.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q3/5

676

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. One outlier can change r dramatically.

GRADING NOTE The single point erases the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find r for (1, 3), (2, 5), (3, 7), and for the same points with every y increased by 10. Give both.

Original: _____

After adding 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q4/5

677

QUESTION ID
S19-F2-Q4

TYPE
Short answer

CORRECT ANSWER
1; 1

GRADING NOTE Form required: both. Adding a constant does not change r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q4/5

Which change to a data set could change r?

A. adding 5 to every x-value

B. converting y from meters to centimeters

C. adding a new point far from the others

D. doubling every y-value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q5/5

678

QUESTION ID

S19-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. adding a new point far from the others

GRADING NOTE Shifts and positive rescalings leave r unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q5/5

Interpret the correlation coefficient

MAIN	Probability & Statistics main question	**HSS-ID.C.8**
BACKWARD 1	Describe strength and direction from a plot	**HSS-ID.B.6**
BACKWARD 2	Compare numbers on a scale from -1 to 1	**6.NS.C.7.c**
FORWARD 1	Match correlation coefficients to scatter plots	**HSS-ID.C.8**
FORWARD 2	Correlation is not causation	**HSS-ID.C.9**

SET 20 • HSS-ID.C.8

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

For car age and value, r = −0.97. Which interpretation is correct?

A. weak negative linear association

B. strong positive linear association

C. strong negative linear association

D. no association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q1/5

681

QUESTION ID

S20-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. strong negative linear association

GRADING NOTE |r| near 1, negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q1/5

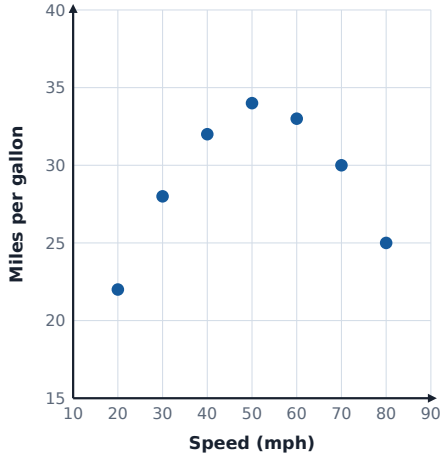
For speed and fuel efficiency, r ≈ 0.245 but the scatter plot shows a clear curve. Which statement is correct?

A. r must have been computed incorrectly.

B. There is a weak linear relationship and nothing more.

C. r measures only linear association, so it misses the curve.

D. There is no relationship between the variables.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q2/5

682

QUESTION ID
S20-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. r measures only linear association, so it misses the curve.

GRADING NOTE A strong nonlinear relationship can have r near 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q2/5

Which correlation coefficient shows the strongest linear association?

A. −0.91

B. 0.85

C. −0.40

D. 0.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q3/5

683

QUESTION ID
S20-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. −0.91

GRADING NOTE Largest |r|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q3/5

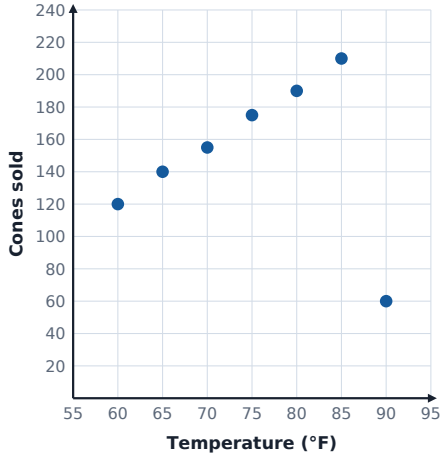
For the cones data, r = −0.008 for all 7 days but 0.999 without the 90-degree day. Which interpretation is best?

A. The six typical days show a strong positive linear association.

B. The 90-degree day is the most typical day.

C. Temperature and cone sales have no relationship on any day.

D. Hotter days have fewer cones sold.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q4/5

684

QUESTION ID

S20-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The six typical days show a strong positive linear association.

GRADING NOTE The outlier hides the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q4/5

Which statement about r is correct?

A. r changes when x is converted from hours to minutes.

B. r is negative whenever the data contain an outlier.

C. r can be as large as 5 for very strong data.

D. r has no units and stays the same if the units change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S20-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q5/5

685

QUESTION ID
S20-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. r has no units and stays the same if the units change.

GRADING NOTE r is unit-free and between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q5/5

SET 20 • HSS-ID.C.8

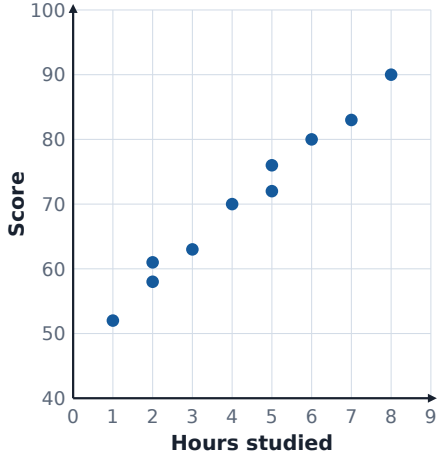
BACKWARD 1

Describe strength and direction from a plot • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

The scatter plot shows hours studied and score. How strong is the linear association?

A. strong **B.** weak

C. cannot be judged **D.** none



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q1/5

687

QUESTION ID

S20-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. strong

GRADING NOTE Points hug a line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q1/5

True or false?

In a strong association, the points lie close to a line or curve.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q2/5

688

QUESTION ID

S20-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Strength.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q2/5

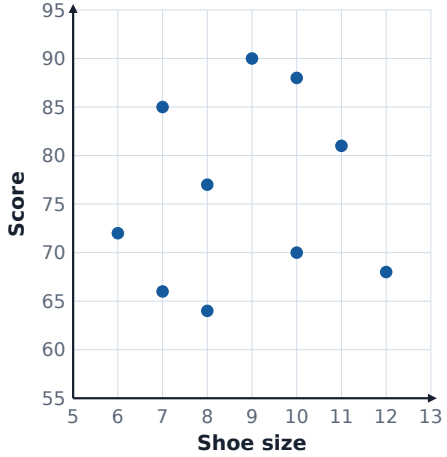
The scatter plot shows shoe size and quiz score. How strong is the association?

A. very weak or none

B. strong negative

C. strong positive

D. perfect



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q3/5

689

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. very weak or none

GRADING NOTE No pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q3/5

True or false?

A scatter plot that falls from left to right shows a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q4/5

690

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

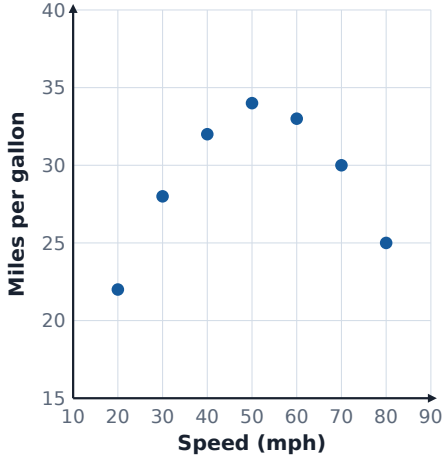
The scatter plot of speed and fuel efficiency has what form?

A. exponential growth

B. linear

C. curved

D. no form



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q5/5

691

QUESTION ID

S20-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. curved

GRADING NOTE Rises then falls.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

BACKWARD 2

Compare numbers on a scale from −1 to 1 • Grade 6 • The Number System • 6.NS.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

What is $|-0.91|$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q1/5

693

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.91

GRADING NOTE Distance from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q1/5

True or false?

$|-0.8| > |0.6|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q2/5

694

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.8 > 0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q2/5

Which is closest to 0?

A. −0.1

B. −0.6

C. 0.5

D. 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q3/5

695

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −0.1

GRADING NOTE Smallest absolute value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q3/5

True or false?

-0.9 is greater than 0.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q4/5

696

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-0.9 < 0.2$, though $|-0.9|$ is larger.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q4/5

Which has the greater absolute value, −0.75 or 0.7?
Enter the number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q5/5

697

QUESTION ID
S20-B2-Q5

TYPE
Short answer

CORRECT ANSWER
−0.75

GRADING NOTE 0.75 > 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

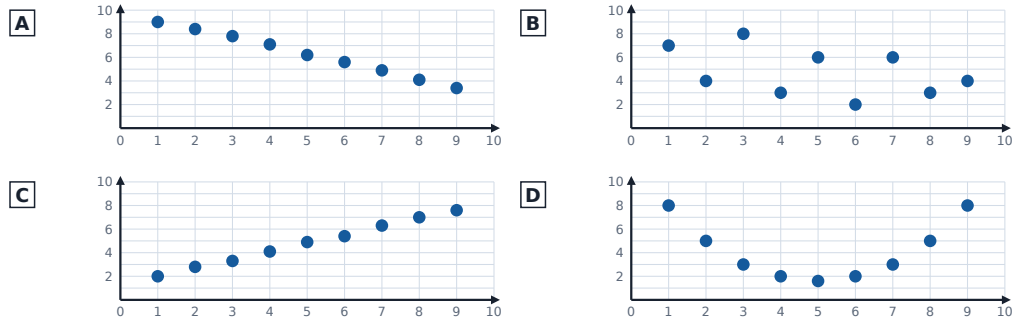
Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSS-ID.C.8

FORWARD 1

Match correlation coefficients to scatter plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Which scatter plot has r ≈ −0.45?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q1/5

699

QUESTION ID

S20-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Weak negative. The others have r ≈ −1.00, 1.00 and 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q1/5

A U-shaped scatter plot that is perfectly symmetric has r equal to

A. 0.5

B. 0

C. −1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q2/5

700

QUESTION ID
S20-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. 0

GRADING NOTE The rising and falling halves cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q2/5

True or false?

r = −0.45 indicates a stronger linear association than r = 0.30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q3/5

701

QUESTION ID
S20-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 0.45 > 0.30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q3/5

Which value of r goes with points falling steadily in a narrow band?

A. 0.99

B. −0.99

C. 0

D. −0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q4/5

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −0.99

GRADING NOTE Strong negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q4/5

True or false?

Two data sets with the same r must have scatter plots of the same shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q5/5

703

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Very different plots can share an r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

FORWARD 2

Correlation is not causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Across towns, the number of libraries and the number of crimes have r = 0.9. Which is the best explanation?

A. Libraries cause crime.

B. r = 0.9 proves a cause in one direction.

C. Crime causes towns to build libraries.

D. Larger towns tend to have more of both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q1/5

705

QUESTION ID
S20-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. Larger towns tend to have more of both.

GRADING NOTE Population is a lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q1/5

True or false?

A correlation of r = 0.95 shows that changes in x cause changes in y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q2/5

706

QUESTION ID

S20-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Strong correlation is not evidence of cause by itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q2/5

Which kind of study can show that x causes y?

A. a scatter plot of past data

B. a study with r close to 1

C. a randomized experiment

D. a survey with many people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q3/5

707

QUESTION ID
S20-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. a randomized experiment

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q3/5

True or false?

A lurking variable can produce a strong correlation between two variables that do not affect each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q4/5

708

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both follow it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q4/5

Students who sleep more have higher grades (r = 0.6) in an observational study. Which conclusion is justified?

A. Higher grades cause these students to sleep more.

B. Sleep and grades are positively associated in this group.

C. Sleep and grades are unrelated for these students.

D. More sleep causes higher grades for these students.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S20-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q5/5

709

QUESTION ID
S20-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Sleep and grades are positively associated in this group.

GRADING NOTE Only association is shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S20-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q5/5

Distinguish between correlation and causation

MAIN	Probability & Statistics main question	**HSS-ID.C.9**
BACKWARD 1	Recognize a strong correlation	**HSS-ID.C.8**
BACKWARD 2	Describe the association in a scatter plot	**8.SP.A.1**
FORWARD 1	Tell experiments from observational studies	**HSS-IC.B.3**
FORWARD 2	Judge causal claims from experiment results	**HSS-IC.B.5**

SET 21 • HSS-ID.C.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Over several years, a city's sunscreen sales and its number of pool visits are strongly correlated. What best explains this?

A. Swimming causes people to buy sunscreen.

B. Sunny, hot weather raises both.

C. Sunscreen causes people to swim.

D. The data must be wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q1/5

712

QUESTION ID
S21-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Sunny, hot weather raises both.

GRADING NOTE A lurking variable drives both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q1/5

Countries with more doctors per person have longer life spans. Which conclusion is justified?

A. Longer life spans cause countries to train more doctors.

B. Adding more doctors in a country will lengthen life spans.

C. Doctors per person and life span are positively associated.

D. Doctors per person and life span are unrelated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q2/5

713

QUESTION ID
S21-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Doctors per person and life span are positively associated.

GRADING NOTE Observational data show association only; wealth may drive both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q2/5

Researchers want to know whether a new study app raises test scores. Which design can show cause?

A. Find the correlation of app use and scores.

B. Survey students who already chose to use the app.

C. Compare the scores of last year's users and nonusers.

D. Randomly assign students to use the app or not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q3/5

714

QUESTION ID
S21-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Randomly assign students to use the app or not.

GRADING NOTE Random assignment balances other causes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q3/5

Children with larger shoe sizes have higher reading scores. What is the most likely lurking variable?

A. age

B. reading-book color

C. school lunch menu

D. shoe brand

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q4/5

715

QUESTION ID
S21-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. age

GRADING NOTE Older children have bigger feet and read better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q4/5

Which statement is correct?

A. Association implies causation when r is above 0.9.

B. If two variables are not correlated, neither can cause anything.

C. An observed association alone does not establish a causal relationship.

D. Causation can only be shown with a scatter plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q5/5

716

QUESTION ID
S21-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. An observed association alone does not establish a causal relationship.

GRADING NOTE Association is not enough by itself to show cause. The others are false: no r value proves cause, scatter plots cannot show cause, and a lack of correlation does not rule out every causal effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q5/5

SET 21 • HSS-ID.C.9

BACKWARD 1

Recognize a strong correlation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

True or false?

r = −0.92 shows a strong linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q1/5

718

QUESTION ID

S21-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE |r| near 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which r shows the weakest linear association?

A. −0.7

B. 0.05

C. 0.6

D. −0.95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q2/5

719

QUESTION ID

S21-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 0.05

GRADING NOTE Smallest |r|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A strong correlation means the two variables must be causally related.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q3/5

720

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not necessarily.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q3/5

A data set has r = 0.88. Is the linear association positive or negative? Enter positive or negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q4/5

721

QUESTION ID

S21-B1-Q4

TYPE

Short answer

CORRECT ANSWER

positive

GRADING NOTE Sign of r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q4/5

r = 0.88 for study time and grades in a survey. What does it describe?

A. a curved association

B. a weak association

C. a strong positive linear association

D. proof that study time causes grades

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q5/5

722

QUESTION ID

S21-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. a strong positive linear association

GRADING NOTE Association only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSS-ID.C.9

BACKWARD 2

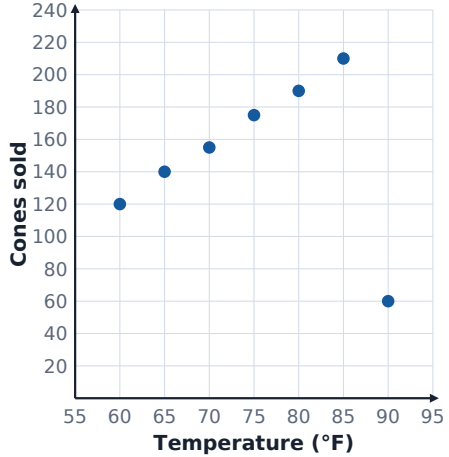
Describe the association in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

True or false?

In the ice-cream scatter plot, the six typical days show a positive association.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q1/5

724

QUESTION ID

S21-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rising pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q1/5

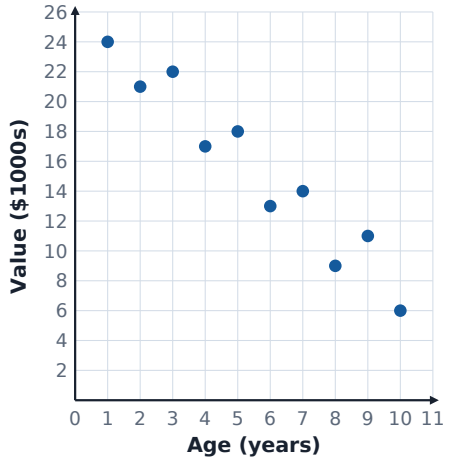
The scatter plot shows car age and value. Which describes it?

A. curved association

B. positive linear association

C. no association

D. negative linear association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q2/5

725

QUESTION ID

S21-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. negative linear association

GRADING NOTE Falling pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

A scatter plot can show an association between two variables that do not cause each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q3/5

726

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Correlation without cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q3/5

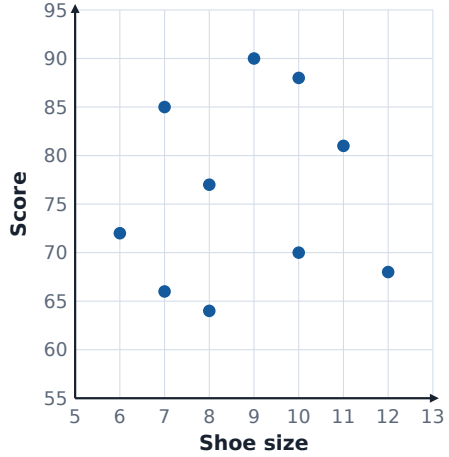
The scatter plot shows shoe size and quiz score of adults. Which describes it?

A. curved

B. no clear association

C. strong negative

D. strong positive



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q4/5

727

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. no clear association

GRADING NOTE Cloud.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

An association seen in a scatter plot shows which variable is the cause.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q5/5

728

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A plot cannot show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSS-ID.C.9

FORWARD 1

Tell experiments from observational studies • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers record the coffee intake and blood pressure of 500 adults without changing anything. What type of study is this?

A. randomized experiment

B. simulation

C. sample survey of opinions

D. observational study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q1/5

730

QUESTION ID

S21-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. observational study

GRADING NOTE Nothing is imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q1/5

True or false?

In a randomized experiment, the researcher assigns the treatments by chance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q2/5

731

QUESTION ID

S21-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q2/5

Which study is a randomized experiment?

A. Researchers compare towns that chose different school lunches.

B. Volunteers are randomly assigned to a new diet or the usual diet.

C. A survey asks people which diet they prefer.

D. Doctors record the diets their patients already follow.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q3/5

732

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Volunteers are randomly assigned to a new diet or the usual diet.

GRADING NOTE Chance assignment of treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q3/5

True or false?

An observational study can establish cause as well as a randomized experiment can.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q4/5

733

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It cannot control lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q4/5

Why does random assignment help support causal conclusions?

A. It removes all variability from the measurements.

B. It guarantees that r will be close to 1.

C. It makes the sample larger than other designs do.

D. It tends to balance other variables between the groups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q5/5

734

QUESTION ID

S21-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. It tends to balance other variables between the groups.

GRADING NOTE Balancing lurking variables. The others are false: random assignment does not remove variability, fix r, or enlarge the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSS-ID.C.9

FORWARD 2

Judge causal claims from experiment results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In a randomized experiment, 40 plants got fertilizer and grew a mean of 18 cm; 40 got none and grew a mean of 12 cm. What is the difference in means (fertilizer minus none)?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q1/5

736

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 18 − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q1/5

(Hypothetical simulation results.) In an experiment, 80 plants were randomly assigned, 40 to fertilizer and 40 to none, and the mean growth difference (fertilizer minus none) was 6 cm. Under the no-effect model, re-randomizing the 80 results many times gave differences of 6 cm or more (fertilizer minus none) in only 1% of the re-randomizations. Using a one-sided rule (fewer than 5% counts as significant), what can be concluded?

A. Fertilizer is unrelated to growth.

B. The fertilizer likely caused more growth.

C. The difference is due to chance alone.

D. No conclusion is possible without r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q2/5

737

QUESTION ID

S21-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. The fertilizer likely caused more growth.

GRADING NOTE 1% < 5%: a difference this large is rare under the no-effect model, and random assignment lets it be credited to the fertilizer. The others ignore the evidence or bring in r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q2/5

True or false?

In a randomized experiment, a difference much larger than chance alone would produce supports a causal conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q3/5

738

QUESTION ID
S21-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Random assignment rules out lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An observational study found a large difference between coffee drinkers and others. Why can't it prove cause?

A. Observational studies do not collect any data.

B. Coffee drinkers may differ in other ways that matter.

C. Coffee intake cannot be measured accurately.

D. The difference was too large to be believable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q4/5

739

QUESTION ID

S21-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Coffee drinkers may differ in other ways that matter.

GRADING NOTE Lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q4/5

True or false?

If an experiment's difference could easily happen by chance, the experiment proves the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q5/5

740

QUESTION ID
S21-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It only fails to show an effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q5/5

Statistics as a process of inference from samples

MAIN	Probability & Statistics main question	**HSS-IC.A.1**
BACKWARD 1	Recognize random and representative samples	**7.SP.A.1**
BACKWARD 2	Use a sample to estimate a population count	**7.SP.A.2**
FORWARD 1	See how sample size affects sampling variability	**HSS-IC.A.1**
FORWARD 2	Attach a margin of error to an estimate	**HSS-IC.B.4**

SET 22 • HSS-IC.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the sample proportion who bike? Give a decimal.

Sample proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q1/5

743

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.31

GRADING NOTE 62/200. This is the statistic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q1/5

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the parameter of interest?

A. the 200 students who were in the sample

B. the sample proportion 0.31 from the survey

C. the proportion of all 1,500 students who bike

D. the 62 sampled students who said they bike

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q2/5

744

QUESTION ID

S22-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. the proportion of all 1,500 students who bike

GRADING NOTE A parameter describes the population. 0.31 is the statistic; 62 and 200 are counts or people.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q2/5

In a random sample of 200 of a school's 1,500 students, the proportion who bike is 0.31. Use it to estimate how many of the 1,500 students bike.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q3/5

745

QUESTION ID

S22-M-Q3

TYPE

Short answer

CORRECT ANSWER

465

GRADING NOTE 0.31 · 1500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q3/5

Which sample of a school's students allows an inference about all of the students?

A. the first 200 students to arrive at school one morning

B. a random sample of 200 students chosen from the school roster

C. all 200 students who belong to the school bike club

D. the 200 students who chose to answer an online poll

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q4/5

746

QUESTION ID

S22-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a random sample of 200 students chosen from the school roster

GRADING NOTE Only random selection makes the sample representative. The others are voluntary or convenience samples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q4/5

Two random samples of 200 students from the same school give sample proportions 0.31 and 0.27. Why do they differ?

A. One of the two samples must have been chosen incorrectly.

B. Sample proportions can never be used to estimate a parameter.

C. Sample statistics vary from one random sample to another by chance.

D. The population proportion changed between the two samples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q5/5

747

QUESTION ID

S22-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Sample statistics vary from one random sample to another by chance.

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q5/5

SET 22 • HSS-IC.A.1

BACKWARD 1

Recognize random and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

True or false?

A random sample tends to be representative of its population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q1/5

749

QUESTION ID

S22-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each member has the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q1/5

A town wants residents' opinion on a new park and has a complete list of all residents. Which method gives a random sample of residents?

A. ask the first 100 people at the park

B. ask the members of the parks committee

C. post an online survey and use the first replies

D. pick 100 names by lottery from the complete resident list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q2/5

750

QUESTION ID

S22-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. pick 100 names by lottery from the complete resident list

GRADING NOTE Chance selection from the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q2/5

True or false?

A sample of volunteers who choose to respond is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q3/5

751

QUESTION ID

S22-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Self-selected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q3/5

A survey of shoppers at a sporting-goods store is used to estimate how much all town residents exercise. What is the main problem?

A. the sample is not representative

B. the sample is a census

C. the sample is too random

D. the sample has no data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q4/5

752

QUESTION ID

S22-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the sample is not representative

GRADING NOTE Shoppers there likely exercise more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The whole group a study wants to learn about is called the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q5/5

753

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSS-IC.A.1

BACKWARD 2

Use a sample to estimate a population count • Grade 7 • Statistics & Probability • 7.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 50 students, 12 play an instrument. Estimate how many of 600 students play.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q1/5

755

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

144

GRADING NOTE 12/50 · 600.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q1/5

In a random sample of 40 light bulbs, 2 are defective. Estimate the number defective in 2,000 bulbs.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q2/5

756

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE 0.05 · 2000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q2/5

True or false?

If 30% of a random sample prefer pizza, about 30% of the population is estimated to prefer pizza.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q3/5

757

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sample proportion estimates the population proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q3/5

In a random sample of 80 voters, 44 support a measure. What is the estimated percent of all voters who support it?

A. 55%

B. 36%

C. 44%

D. 80%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q4/5

758

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 55%

GRADING NOTE 44/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q4/5

True or false?

Estimates from a random sample are always exactly equal to the population value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q5/5

759

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They vary by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSS-IC.A.1

FORWARD 1

See how sample size affects sampling variability • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Many random samples of size 25 and many of size 400 are taken from the same population. Which is true?

A. Larger samples are always biased.

B. Proportions from samples of 25 vary less.

C. Both sizes vary the same amount.

D. Proportions from samples of 400 vary less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q1/5

761

QUESTION ID

S22-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Proportions from samples of 400 vary less.

GRADING NOTE Larger n, less variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q1/5

True or false?

Increasing the size of a random sample reduces bias caused by a poor sampling method.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q2/5

762

QUESTION ID

S22-F1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size reduces variability, not bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q2/5

A random sample of 400 voters has 228 who favor a law. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q3/5

763

QUESTION ID

S22-F1-Q3

TYPE

Short answer

CORRECT ANSWER

0.57

GRADING NOTE 228/400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q3/5

Which estimate of a population proportion is likely to be closest to the true value?

A. one from 1,000 volunteers who called in

B. one from a random sample of 50

C. one from 50 friends of the researcher

D. one from a random sample of 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q4/5

764

QUESTION ID

S22-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. one from a random sample of 1,000

GRADING NOTE Large and random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q4/5

True or false?

Sample proportions from repeated random samples center around the population proportion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q5/5

765

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Random sampling is unbiased.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSS-IC.A.1

FORWARD 2

Attach a margin of error to an estimate • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample gives a sample proportion of 0.57 with a margin of error of 0.05. Give the interval of plausible values for the population proportion.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q1/5

767

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.52; 0.62

GRADING NOTE Form required: both. 0.57 ± 0.05.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q1/5

A poll reports 48% with a margin of error of 3 percentage points. Which value is plausible for the population percent?

A. 52%

B. 55%

C. 44%

D. 50%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q2/5

768

QUESTION ID

S22-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 50%

GRADING NOTE 45% to 51%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q2/5

True or false?

A larger random sample usually gives a smaller margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q3/5

769

QUESTION ID
S22-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Less variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q3/5

(Hypothetical simulation results.) Simulated sample proportions have standard deviation 0.025. Using margin of error = 2 × standard deviation, what is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q4/5

770

QUESTION ID

S22-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.05

GRADING NOTE 2 · 0.025.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q4/5

True or false?

A margin of error accounts for bias from a sample of volunteers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q5/5

771

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It covers only chance variation in random samples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q5/5

Simulate a probability model with random digits

MAIN	Probability & Statistics main question	**HSS-IC.A.2**
BACKWARD 1	Design a simulation for a compound event	**7.SP.C.8.c**
BACKWARD 2	Find a probability as a relative frequency	**7.SP.C.6**
FORWARD 1	Use two-digit groups for probabilities in hundredths	**HSS-IC.A.2**
FORWARD 2	Use simulation results to judge a model	**HSS-IC.A.2**

SET 23 • HSS-IC.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A player makes 70% of free throws. To simulate, digits 0-6 mean "make" and 7-9 mean "miss". Use the first 10 digits of line 1 of the table as 10 shots. How many shots are made?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Shots made: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q1/5

774

QUESTION ID

S23-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Line 1 begins 1588139986; the digits 0-6 among them: 1, 5, 1, 3, 6 → 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A player makes 70% of free throws (digits 0-6 = make, 7-9 = miss). Each line's first 10 digits are one trial of 10 shots, giving 8 trials. In how many trials does the player make 8 or more shots?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q2/5

775

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Makes per line: 5, 8, 7, 4, 4, 3, 7, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q2/5

An event has probability 0.4. Which assignment of single random digits simulates it?

A. 4 = success; all other digits = failure

B. 0-3 and 9 = success; 4-8 = failure

C. 0-4 = success; 5-9 = failure

D. 0-3 = success; 4-9 = failure

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q3/5

776

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0-3 = success; 4-9 = failure

GRADING NOTE 4 of 10 equally likely digits. The others give 0.5, 0.5 and 0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner lands on red with probability 0.25. Read line 5 of the table as ten two-digit numbers; 00-24 means red. How many of the ten spins are red?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Red spins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q4/5

777

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Line 5 pairs: 17, 78, 48, 64, 88, 76, 63, 44, 00, 29.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q4/5

A simulation of 50 families of 4 children (each birth a girl with probability 0.5) gave the results in the table. What proportion of the simulated families had 3 or more girls? Give a decimal.

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q5/5

778

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.32

GRADING NOTE (13 + 3)/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q5/5

SET 23 • HSS-IC.A.2

BACKWARD 1

Design a simulation for a compound event • Grade 7 • Statistics & Probability • 7.SP.C.8.c

To simulate a fair coin with random digits, which assignment works?

A. 0-4 = heads; 4-9 = tails

B. 0 = heads; 1-9 = tails

C. even digits = heads; odd digits = tails

D. 0-6 = heads; 7-9 = tails

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q1/5

780

QUESTION ID

S23-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. even digits = heads; odd digits = tails

GRADING NOTE Even and odd digits split 5 and 5. 0-4 and 4-9 overlap at 4; 0 alone gives 0.1; 0-6 gives 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q1/5

In the digits 4 1 7 0 7 5 3 6 4 5, digits 0-4 mean heads. How many heads?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q2/5

781

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE 4, 1, 0, 3, 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q2/5

True or false?

A trial of a simulation for "number of heads in 10 flips" uses 10 random digits.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q3/5

782

QUESTION ID

S23-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One digit per flip.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q3/5

To simulate rolling a fair six-sided die with digits, what should be done with 0, 7, 8 and 9?

A. skip them and read the next digit

B. stop the simulation

C. count them as a 6

D. count them as a 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q4/5

783

QUESTION ID

S23-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. skip them and read the next digit

GRADING NOTE Only 1-6 are used, so each face stays equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q4/5

True or false?

Running more trials of a simulation makes the estimated probability less reliable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q5/5

784

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE More trials, more reliable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSS-IC.A.2

BACKWARD 2

Find a probability as a relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) In 50 simulated trials, an event happened 16 times. What is its relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q1/5

786

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.32

GRADING NOTE 16/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q1/5

An event happened in 9 of 200 trials. What is its relative frequency as a percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q2/5

787

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4.5%

GRADING NOTE 9/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q2/5

True or false?

A relative frequency of 0.32 means the event happened in 32% of the trials.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q3/5

788

QUESTION ID

S23-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Decimal to percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q3/5

An event happened in 12 of 60 trials. What is its relative frequency?

A. 0.2

B. 0.6

C. 5

D. 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q4/5

789

QUESTION ID

S23-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.2

GRADING NOTE 12/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q4/5

True or false?

The relative frequency from 10 trials is always exactly equal to the true probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q5/5

790

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It varies, especially in few trials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSS-IC.A.2

FORWARD 1

Use two-digit groups for probabilities in hundredths • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

An event has probability 0.35. Which assignment of two-digit numbers simulates it?

A. 35 = success; others = failure

B. 00-35 = success; 36-99 = failure

C. 00-34 = success; 35-99 = failure

D. 01-34 = success; others = failure

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q1/5

792

QUESTION ID

S23-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 00-34 = success; 35-99 = failure

GRADING NOTE 00-34 is 35 of the 100 numbers. 00-35 has 36; 01-34 has 34; 35 alone has probability 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q1/5

Read line 3 as ten two-digit numbers; 00-34 means success. How many successes?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Successes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q2/5

793

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Line 3 pairs: 72, 94, 05, 57, 61, 69, 87, 16, 89, 76.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q2/5

True or false?

Two-digit groups 00-99 give 100 equally likely outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q3/5

794

QUESTION ID

S23-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each has probability 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin model uses 0-4 = heads. Using the first 10 digits of each of the 8 lines as 8 trials of 10 flips, in how many trials are there 3 or fewer heads?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q4/5

795

QUESTION ID

S23-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Heads per line: 3, 5, 4, 3, 3, 2, 5, 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q4/5

True or false?

To simulate a probability of 1/3 with single digits, use 1-3 = success, 4-9 = failure and skip 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q5/5

796

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3 of 9 used digits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSS-IC.A.2

FORWARD 2

Use simulation results to judge a model • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In 100 simulated sets of 10 fair-coin flips (table), in what proportion of the sets were there 9 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q1/5

798

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.02

GRADING NOTE 2/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A coin claimed to be fair gave 9 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: the model is a fair coin. Count the simulated results that are 9 or more heads (a one-sided check: only an excess of heads counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: 9 or more heads happened in only 2% of the trials.

B. Not consistent: a fair coin must give exactly 5 heads.

C. Consistent: a fair coin can land heads 9 times.

D. Consistent: 9 or more heads happened in 2% of the trials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q2/5

799

QUESTION ID

S23-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Not consistent: 9 or more heads happened in only 2% of the trials.

GRADING NOTE 2% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q2/5

True or false?

A result that happens in 25% of the simulated trials is consistent with the model under the 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q3/5

800

QUESTION ID
S23-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 25% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q3/5